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and Philosophy of Science

Stefania Centrone

Pierluigi Minari

Oskar Becker, On the Logic of Modalities (1930): Translation, Commentary and Analysis



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and Philosophy of Science

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Oskar Becker, On the Logic of Modalities (1930): Translation, Commentary and Analysis

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*This book is dedicated to Prof. Dr Volker
Peckhaus*

Preface

The history of twentieth-century modal logic is all too often presented as the American success story that started with the work of Clarence Irving Lewis, while prewar modal logic research in Europe is passed off as a side-show of well-intended failures. This book is intended as a first attempt to drastically correct such a picture. It is related to the research project *Modal Logic and Austro-Polish Philosophy* awarded to the first author by the German Research Association (DFG) in Summer 2017, which started in October 2018. The preparatory work has enormously profited from the first author's 1-year research (2017–2018) within the research program *Modalities and Conditionals: Systematic and Historical studies* at the University of Helsinki, whose principal investigator was Sara Negri.

The core of the project *Modal Logic and Austro-Polish Philosophy* consists in formally exploiting ideas, attempts, and solutions emerging from prewar modal logic with the aim to open up unexpected, radically new paths in various areas of contemporary research in modal logic.

The starting point of the research should have been, and indeed it has been, to make the key works in modal logic that appeared in prewar continental Europe accessible to a wider audience. Three works in particular come to the mind when one thinks of European research in modal logic prior to the war: Jan Łukasiewicz's *O logice trójwartościowej* (Łukasiewicz 1920), Ernst Mally's *Grundgesetze des Sollens* (Mally 1926), and Oskar Becker's *Zur Logik der Modalitäten* (Becker 1930).

The present book focuses on Oskar Becker's essay *On the Logic of Modalities*.

The decision to begin therewith was, in part, due to the fact that Becker seemed to have been deeply influenced by his teacher, Edmund Husserl, and an in-depth knowledge of Husserl's work appeared to be essential to correctly interpret certain choices of Becker about modalities. Such background had been elaborated in several volumes and articles that have appeared mostly in the Springer series *Synthese Library* in recent years, for instance, the monograph *Logic and Philosophy of Mathematics in the Early Husserl* (*Synthese Library* 2010) (Centrone 2010) or the volume *Essays on Husserl's Logic and Philosophy of Mathematics* (*Synthese Library* 2017) (Centrone 2017), which contains our joint work on Husserl and the

algebraists of logic, *Husserl and Boole* (Centrone and Minari 2017a) and *Husserl and Schröder* (Centrone and Minari 2017b). On the other side, two important volumes have paved the way for an adequate understanding of the philosophical significance of Oskar Becker's contribution to the history and philosophy of mathematics, namely *Die Philosophie und die Wissenschaften. Zum Werk Oskar Beckers* edited by Annemarie Gethmann-Siefert and Jürgen Mittelstraß (Gethmann-Siefert and Mittelstraß 2002) and *Oskar Becker und die Philosophie der Mathematik* edited by Volker Peckhaus (Peckhaus 2005). At a variance with the wide spectrum of topics dealt with in these volumes, our choice has been to focus exclusively on Oskar Becker's pioneering contributions to modal logic in 1930, with the aim to evaluate Becker's scientific accomplishments on this topic and possibly to make them, so to say, bear fruit.

We first decided to give an idea of the work that had to be done on this aspect of Oskar Becker's research in a booklet titled *Oskar Becker on Modalities*, which appeared in 2019 within the series *Philosophische Hefte* edited by Axel Gelfert and Thomas Gil for the *Logos Verlag* in Berlin (Centrone and Minari 2019). On this basis, we then turned to a deeper examination and a rigorous proof of all the claims made in the booklet, but not only that.

The present work is the result of two years of intense work on Oskar Becker's *On the Logic of Modalities*. In the first place, we try to give an idea of the reasons that led Becker to deal with modalities, first of all, the birth of intuitionistic logic, two years earlier, with the publication of Arend Heyting's seminal work *The Formal Rules of Intuitionistic Logic* (Heyting 1930). It may perhaps turn out to be surprising that Oskar Becker is the first philosopher ever to have put forward the idea of a translation of intuitionistic logic into modal logic, even if it is Kurt Gödel that actually realized it shortly later.

This book provides the reader with the basic modal logical tools necessary to read *On the Logic of Modalities* and gives a sketch of Lewis's Logic **S3**, which is the target of Becker's essay, in the modal logical symbolism that is current nowadays. Next, two extensions of **S3** proposed by Oskar Becker are presented and all objections raised by Kurt Gödel in his *Review of Becker 1930: On the Logic of Modalities* (Gödel 1931) are considered with the aim of correcting some historical misunderstandings related to them.

We dwell at length on the controversy between Oskar Becker and Ernst Cassirer with respect to the meaning and the legitimacy of the use of *ideal elements* in logic and mathematics. This results in an interesting multi-voice canon whose main actors are none other than Husserl, Weyl, Hilbert, and Brouwer.

We then provide the English translation of Becker's essay, accompanying the whole translation with footnotes that explain step by step, in current symbolism, and sometimes even in words, what is going on in the text, in order to allow even a reader not versed in logic to read the text easily.

The Appendix at the end of the book contains the detailed proofs of our assertions about Becker's logical accomplishments, conjectures, and mistakes as well. It is preceded by a systematic presentation of the Kripke-style semantics for **S3**—of the

completeness theorems and other related results—in a comprehensive way which, to our knowledge, is not yet to be found in the literature.

We are particularly indebted to our teacher, Ettore Casari, who aroused our interest in logic, mathematics, and in Husserl's and Bolzano's writings and made us realize that restriction to one single field of research can be more of a hindrance than a help for original work. Very special thanks go to Klaus Mainzer, emeritus of excellence at the Technical University of Munich, and Niklas Hebing, Head Office of the Section of Humanities and Social Science by the German Research Association (DFG), who both strongly supported this work. The encouragement at a decisive moment and the friendly advice we received from Otavio Bueno, editor in chief of Synthese Library, were truly invaluable.

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Chapter 1

Introduction



Oskar Becker was involved with Nazism—he was indeed, as is written in an official file of the SS Security Service,¹ “not a party member but a loyal to National Socialism, who tries to consolidate the National Socialistic ideology”.

We wish to make fully explicit that we completely distance ourselves from Becker’s political views, and that our book, which deals exclusively with Becker’s accomplishments in modal logic and its philosophy, neither endorses nor defends in any way Becker’s political and moral failings.

* * *

This book contains the first English translation of Oskar Becker’s essay *On the Logic of Modalities* (*Zur Logik der Modalitäten*)² that appeared in 1930 in the *Yearbook for Philosophy and Phenomenological Research*.³ Our commentary aims to present, to contextualize and to evaluate the pioneering contributions to modal logic contained in this work.

¹ Cp. Leaman and Simon (1992).

² Becker (1930). Henceforth cited using the page number(s) of the original pagination and, in brackets, of the English translation contained in the present volume (the original pagination is reproduced in the margin).

³ Becker (1930). The *Yearbook for Philosophy and Phenomenological Research* (*Jahrbuch für Philosophie und phänomenologische Forschung*) was founded by Edmund Husserl in 1912 and served the Husserl’s circle as an important organ during Husserl’s Freiburg period (1916–1938). The first issue of the journal was published in 1913 and contains Husserl’s *Ideas for a Pure Phenomenology and Phenomenological Philosophy* (Husserl 1913). Volume 8 includes Heidegger’s masterpiece *Being and Time* (Heidegger 1927) as well as Oskar Becker’s famous investigation on the logic and ontology of mathematical phenomena “*Mathematical Existence* (*Mathematische Existenz*)” (Becker 1927).

Oskar Becker (Leipzig 1889–Bonn 1964) was a German philosopher, logician, mathematician and historian of mathematics. He is often remembered, together with Martin Heidegger, for being one of the most prominent students of Edmund Husserl (1859–1938). He was, together with Moritz Geiger (1880–1937), Martin Heidegger (1889–1976), Alexander Pfänder (1870–1941), Adolf Reinach (1883–1917) and Max Scheler (1874–1928), one of the members of the editorial board of the *Yearbook*.

Oskar Becker got his PhD in mathematics in 1914 with a work⁴ entitled *On the Decomposition of Polygons in non-intersecting triangles on the Basis of the Axioms of Connection and Order* (*Über die Zerlegung eines Polygons in exclusive Dreiecke auf Grund der ebenen Axiome der Verknüpfung und Anordnung*). In 1922 he wrote under Husserl's supervision his *Habilitationsschrift*, *On Investigations of the Phenomenological Foundation of Geometry and their physical Application* (*Beiträge zur phänomenologischen Begründung der Geometrie und ihrer physikalischen Anwendungen*).⁵ In 1927 Oskar Becker published in the *Yearbook* his masterpiece *Mathematical Existence*,⁶ where he uses the Husserlian phenomenology to clarify the process of counting. In 1952—when the study of modal logic was already well beyond its pioneering era—Becker came back to the subject publishing a monograph, *Investigations on the Modal Calculus* (*Untersuchungen über den Modalkalkül*), perhaps too old-fashioned for the time.⁷

* * *

The essay *On the Logic of Modalities* represents an attempt to treat modal logical issues with a phenomenological method. This enterprise appeared from the outset not to be easy at all, for logic and phenomenology are completely different disciplines. Depending on the way in which it constructs its formal systems, formal logic can be seen as *the theory of the correct inferences*, or alternatively, as the *theory of purely formal truths*, that is, as the theory of those truths that hold without any condition. Phenomenology, instead, deals with the description of lived experiences.

Indeed, we might better say that in his investigations Becker pursued two loosely related goals. The first, more technical in character, was to find axiomatic conditions that reduced to the finite the number of logically non-equivalent combinations arising from the iterated application of the operators “not” and “it is impossible that (...)” in Lewis's modal system, as we will explain in details below. The second,

⁴ Becker (1914). For a complete bibliography of Becker's works see Zimny (1969).

⁵ Becker (1923). For a precise placement of Oskar Becker in the political and philosophical panorama of his time see Wolters (2002).

⁶ Becker (1927). Hereto see at least: Gethmann (2003), Peckhaus (2005), Gethmann-Siefert and Mittelstraß (2002), and Mancosu (1998). A very important contribution for a correct understanding of Oskar Becker's position in the history of modal logic is, in our opinion (Peckhaus 2002).

⁷ Becker (1952). It is worth noticing that Becker proposes here a Leibnizian semantics for the modal operators, which he calls the “statistical interpretation of the modal calculus”, cp. Martin (1969) and, in particular, Peckhaus (2002), 174 and Copeland (2002), 107–108.

more philosophically oriented and in a sense much more ambitious, was to treat the logic of modalities from a phenomenological perspective and to understand, from this perspective, the philosophical and logical-ontological problems underlying, and posed by, Intuitionism.

On the Logic of Modalities consists of two parts, loosely related as the above mentioned corresponding goals are. Part I opens with a general Introduction that shortly reviews the Aristotelian conception of modalities as well as Hugh MacColl's pioneering modal logical investigations in his *Symbolic Logic and its Applications*⁸ of 1906. It then focuses on C. I. Lewis's *Survey of Symbolic Logic*⁹ of 1918. The latter work contains the first presentation of the so-called "Survey system", known since 1932 as "modal system **S3**."¹⁰

De facto, **S3** is the actual object of the investigations contained in Part I of Becker's essay. As pointed out by Emil L. Post, the system Lewis presents in 1918 *collapses* into classical logic. Lewis corrects it in a paper entitled *Strict Implication: An Emendation*¹¹ and published in 1920, where the system effectively becomes the logic we nowadays know as "**S3**."¹² In his essay Becker faithfully reports both that the original version of the "Survey system" proves the *collapse* of modalities, as well as Lewis's amendment thereof. Incidentally, "*collapse of modalities*" is a customary expression in the modal-logical jargon. It means that a modal logical system proves that *necessity* and *truth* are one and the same, or equivalently (as it is the case in the "Survey system") that *impossibility* and *falsity* are one and the same. Obviously, such a system is *trivial* from a modal point of view.

Becker's Introduction touches on the paradoxes of *material* and *strict implication* and sets out to establish a propositional modal logic that is *decidable* as the classical propositional logic is:¹³

The aim of the present essay is now closely related to both MacColl's and Lewis's investigations. The ultimate goal of our investigations is to develop an elementary logical calculus that takes adequately into account the *modalities* of the statement, namely in such a way that the so-called *elementary decision problem* is solvable, as in the ordinary propositional calculus.

Part I, *On the Rank Order and the Reduction of Logical Modalities*, is specifically devoted to the problems of ranking and iteration of modalities. Becker sets out to modify **S3** by means of some additional axioms effecting the reduction of complex modalities to simple ones in order to obtain two new modal systems—he calls them "*the six modalities calculus*" (henceforth denoted here by **S3'**) and "*the ten modalities calculus*" (henceforth **S3''**) with the following properties:

⁸ MacColl (1906).

⁹ Lewis (1918).

¹⁰ The name appears for the first time in Appendix II of Lewis and Langford (1932).

¹¹ Lewis (1920).

¹² Cp. Cresswell et al. (2016), 281 f.

¹³ Becker (1930), 4 (58).

- (i) *the number of irreducible modalities is finite,*
- (ii) *the positive (and by consequence the negative) modalities are arranged in a linear order with respect to logical strength.*

He believes that, since the “System of Strict Implication” has the conjunction, the negation and the impossibility as primitive logical constants, it is possible to generate within it infinitely many non equivalent nested modalities through the iteration of the negation and the impossibility operators. Such modalities, as Kurt Gödel (1906–1978) puts it in his *Review of Becker 1930*, “cannot even be linearly ordered according to their logical strength in the sense that, of any two affirming modalities, one will imply the other, and similarly for negating ones.”¹⁴ Otherwise said, there are modalities that are incomparable in Lewis’s system.

That said, it is worth to be mentioned that Oskar Becker neither shows that the two systems he sets up (and others he tentatively introduces, as we will see below) really differ from one another, nor that his additional axioms cannot be derived from those of Lewis, nor that in his own systems, with *six* and, respectively, with *ten* “irreducible” modalities, such modalities cannot be further reduced.¹⁵

Actually, nine years later, W. T. Parry will show, in a paper entitled *Modalities in the Survey System of Strict Implication*,¹⁶ that, at a variance with what Becker seems to believe, **S3** has a finite number of modalities. More precisely, Parry shows, with the help of a number of suitable theses he is able to derive in the system, that it is possible to reduce all the complex modalities in **S3** to a finite number of irreducible modalities, viz. 42. He also shows that no further reduction is possible.

Part **II** of Becker’s essay explores, more or less independently from Part **I**, the connection between modal and intuitionistic logic both from a formal and from a phenomenological perspective.

From a formal perspective, the particular interest of a (propositional) modal calculus with nested modalities that is *decidable* lies in the fact, so Becker, that Brouwer’s idea to set up a finite logic grounded on *evidence*, or—to put it with Husserl—on *evidence levels* seems to be realizable only within the framework of a modal formal system.

Indeed, Becker is the first logician and philosopher of mathematics to put forward the idea of a *modal interpretation* of intuitionistic logic,¹⁷ more precisely the idea of a possible sound and faithful translation of intuitionistic logic into modal logic. However, the first actual translation is to be found in a one-page celebrated and influential paper entitled *An interpretation of the intuitionistic propositional*

¹⁴ Gödel (1931).

¹⁵ Hereto cp. Gödel (1931).

¹⁶ Parry (1939).

¹⁷ A modal translation of intuitionistic logic was in a sense foreshadowed also by Ivan Orlov in Orlov (1928). The paper, written in Russian, remained however unknown outside the Soviet Union for a very long time. It is only since the 1990’s that Orlov’s pioneering contributions were “rediscovered” and came to be known and discussed, see Chagrov and Zakharyashchev (1992) and Došen (1992).

calculus written in 1933 by Kurt Gödel.¹⁸ The basic idea of Gödel is similar to the one Oskar Becker outlines in *On the Logic of Modalities*.

Becker suggests to add to classical logic the predicates “(...) is provable”, “(...) is such, that its negation is provable” and “(...) is undecided”. Such predicates should express Brouwer’s primitive logical concepts.

Similarly, Gödel’s idea is to extend the language of classical propositional logic with the unary operator “it is provable that (...)”, denoted by “B”, and to add to an axiomatic calculus for propositional classical logic *three* axiom-schemas and *one* rule of inference. The axiom-schemas are the modal schemas *K*, *T* and 4 that characterize modal logics that are nowadays standard, the rule of inference is the necessitation rule that is contained in all *normal* modal systems. We will introduce both the schemas and the rule of inference in detail later on.

Notice, incidentally, that both Becker and Gödel seem to take the predicate “(...) is provable” and the operator “it is provable that (...)” as conveying the same piece of information. Actually, the predicate “(...) is provable” denotes the property of a proposition to be provable, while the operator “it is provable that (...)” takes a proposition as input and gives a different proposition as output. (Unfortunately, such practice of systematically neglecting the difference between predicate and operator is, even nowadays, quite widespread among logicians.)

Gödel writes:¹⁹

One can interpret Heyting’s propositional calculus by means of the notions of the ordinary propositional calculus and the notion “*p* is provable” (written “B*p*”), if one adopts for that notion the following system $\mathfrak{S}$ of axioms:

1. $Bp \rightarrow p$
if it is provable that *p*, then it is true that *p*
2. $Bp \rightarrow ((B(p \rightarrow q) \rightarrow Bq)$
if it is provable that *p* and it is provable that *p* implies *q*, then it is provable that *q*
3. $Bp \rightarrow BBp$
if it is provable that *p*, then it is provable that it is provable that *p*

In addition, [...] the new rule of inference is to be added

$$\frac{A}{BA}$$

From *A*, it is provable that *A* may be inferred

By substituting throughout the operator “B” (“it is provable that (...)”) by the operator “□” (“it is necessary that (...)”) one obtains one of the modal logical systems that are nowadays standard, namely Lewis’s system **S4**.

* * *

Becker’s more logically oriented investigations, mainly contained in Part I of *On the Logic of Modalities*, will be carefully analyzed in Chap. 2, while his more

¹⁸ Gödel (1933a).

¹⁹ Gödel (1933a), 301.

philosophically oriented discussions, to be found in Part **II**, will be examined at length in Chap. 3. Chapter 4 contains the English translation of *On the Logic of Modalities*, together with an apparatus of explanatory footnotes. The final Chap. 5 (Appendix) provides a self-contained and comprehensive survey of the most relevant “technical” issues related to Lewis’s system **S3** and other non normal modal systems in the neighbourhood.

Chapter 2

Part I of *On the Logic of Modalities*



2.1 The Conditional, or *The Crows on the Roofs*

Since Becker as well as MacColl and Lewis all refer to the old controversy about the right interpretation of conditional sentences, let us briefly dwell on this issue.

The controversy traces back to the Megarians and the Stoics. As Józef Maria Bocheński puts it in his *A History of Formal Logic*:¹

The definition of implication was a matter much debated among the Megarians and Stoics: All dialecticians say that a connected (proposition) is sound, when its consequent follows from its antecedent — but they dispute about when and how it follows, and propound rival criteria.

Even so Callimachus, librarian at Alexandria in the 2nd century B.C., said: ‘the very crows on the roofs croak about which implications are sound’.

In ancient times the quarrel was, above all, between a truth-functional and a modal interpretation of the conditional. Philo (of Megara) said that an implication is true when it is not the case that it begins with the true and ends with the false.² This conception of the conditional was later adopted by Gottlob Frege (1848–1925) and by the American logician and founder of American Pragmatism Charles Sanders Peirce (1839–1914).

In his *Gedankengefüge* (1923) Frege calls conditional sentences “*hypothetische Satzgefüge*” and what is expressed by them “*hypothetische Gedankengefüge*”. He writes:³

[A] hypothetical compound thought is true if its consequent is true; it is also true if its antecedent is false, regardless of whether the consequent is true or false. The consequent

¹ Bocheński (1956), 116.

² *Ibid.*, 117.

³ Frege (1923), 46.

must always be a thought. Given [...] that “A” and “B” are sentences proper, then “not (not A and B)” expresses a hypothetical compound with the sense (thought-content) of “A” as consequent and the sense of “B” as antecedent. We may also write instead: “if B, then A.” But here, indeed, doubts may arise. It may perhaps be maintained that this does not square with linguistic usage. I reply, it must once again be emphasized that science has to be allowed its own terminology, that it cannot always bow to ordinary language. Just here I see the greatest difficulty for philosophy: the instrument it finds available for its work, namely ordinary language, is little suited to the purpose, for its formation was governed by requirements wholly different from those of philosophy. So also logic is first obliged to fashion a usable instrument from those already to hand. And for this purpose it initially finds but little in the way of usable instruments available. [...] The thought expressed by the compound sentence “If I own a cock which has laid eggs today, then Cologne Cathedral will collapse tomorrow morning” is [...] true. Someone will perhaps say: “But here the antecedent has no inner connection at all with the consequent.” In my account, however, I required no such connection, and I ask that “if B, then A” should be understood solely in terms of what I have said and expressed in the form “not (not A and B).” It must be admitted that this conception of a hypothetical compound thought will at first be thought strange. But my account is not designed to square with ordinary linguistic usage, which is generally too vague and ambiguous for the purposes of logic.

Thus, as far as the truth-conditions of conditional propositions are concerned, Frege is, whether he knew it or not, a follower of Philo.

In turn, Peirce is overtly a follower of Philo:⁴

As far as I am concerned, I am a follower of Philo [...]. It is completely irrelevant, whether this conception is in accordance with ordinary language.

Diodorus of Sicily and Chrysippus of Soli raised objections against Philo’s truth-functional conception of implication. While Diodorus saw the conditional as a temporal quantification—namely by regarding a conditional true if and only if, for every instant of time *t*, it is not the case that at the instant *t* the antecedent is true and the consequent is false—Chrysippus had a virtually⁵ modal conception of the conditional: according to him, a conditional is true if and only if the premise is *incompatible* with the negation of the consequence.

As we saw, Frege and Peirce shared Philo’s conception. In Hugh MacColl’s essay published in 1880 in the journal *Mind* and in his *Symbolic Logic* of 1906 a concept similar to that of Chrysippus can be found.⁶

Clarence Irving Lewis, on the basis of MacColl’s investigations, pleaded in favor of the Chrysippean modal reading as an appropriate interpretation of the conditional, which he called “strict implication”. In 1918 Lewis published his first modal system, eventually called **S3**. Later on he developed his modal systems **S1–S5**,⁷ which are stepwise based on one another. They are conceived as alternatives to the non modal logic presented in Russell’s and Whitehead’s *Principia Mathematica* (1910–1913).⁸

⁴ Peirce (1992), 125 f.

⁵ If we assume that “incompatibility” and “impossibility of the conjunction” mean the same.

⁶ MacColl (1880), MacColl (1906).

⁷ Lewis and Langford (1932).

⁸ Whitehead and Russell (1910).

Lewis formalized the strict implication (in symbols: “ $\rightarrow$ ”) in terms of negation, possibility and conjunction⁹ as follows:¹⁰

$$p \rightarrow q := \neg \Diamond (p \wedge \neg q)$$

Due to the interdefinability of $\Box$ and $\Diamond$, this is logically equivalent to

$$\Box (p \rightarrow q)$$

Similarities and differences in comparison with the “material implication”, as Russell called the Philonian conditional, are obvious:

$$p \rightarrow q := \neg (p \wedge \neg q)$$

Lewis’s comment on this was the following:¹¹

“ p strictly implies q ” is to mean “it is false that it is possible that p should be true and q false” [...]

“ p materially implies q ” is to mean “it is not the case that p is true and q is false.”

Lewis’s main reason for formally introducing the Chrysippean conditional were the paradoxes of material implication, in particular:

- | | |
|--|-----------------------------------|
| (1) $p \rightarrow (q \rightarrow p)$ | (<i>argumentum a fortiori</i>), |
| (2) $\neg p \rightarrow (p \rightarrow q)$ | (<i>ex falso quodlibet</i>). |

Lewis rephrases them as follows:¹²

- (1*) If p is true, then any proposition q materially implies p .
- (2*) If p is false, then p materially implies any proposition q .

However, Lewis admitted that the strict implication also yields the following analogous paradoxical theorems:

- (3) $\neg \Diamond \neg p \rightarrow (q \rightarrow p)$,
- (4) $\neg \Diamond p \rightarrow (p \rightarrow q)$,

⁹ In order to comply with the now current logical notation, we prefer not to adopt Lewis’s symbolic apparatus, except for the symbol “ $\rightarrow$ ” of strict implication. In particular, we use “ $\rightarrow$ ”, “ $\neg$ ”, “ $\wedge$ ”, “ $\vee$ ”, “ $\Box$ ” and “ $\Diamond$ ” to denote, respectively, material implication, conjunction, disjunction, necessity and possibility.

¹⁰ Lewis and Langford (1932), 124.

¹¹ *Ibid.*, 124, 136.

¹² *Ibid.*, 142.

in his words:¹³

- (3*) A proposition which is necessarily true is implied by any proposition [...]
- (4*) A proposition which is impossible implies any proposition.

In the later *Symbolic Logic* he will carefully discuss this issue, arguing that the paradoxicality of these theorems is just illusory: they are “paradoxical only in the sense of expressing logical truths which are easily overlooked”.¹⁴

2.2 The Decision Problem and *Leibniz’s Dream*

Part I of Becker’s essay pursues the goal, as we already said, of finding a propositional modal logic with a finite number of irreducible iterated modalities that is *decidable*. In the present section we focus on the *decision problem* and hint at the problems one encounters when searching for a decidable propositional modal logic *without an adequate semantic* (and a *completeness theorem*) for such a calculus being available.

The *Entscheidungsproblem* (German for “decision problem”) is the question whether the first-order predicate logic is decidable or not. In this form the question is to be found in Hilbert & Ackermann *Outlines of Mathematical Logic* (*Grundzüge der mathematischen Logik*) published in Berlin in 1928.¹⁵ Is there an effective method to decide the set of logically valid formulas of first order logic?

The conceptual issues underlying the problem have a long history and may be traced back at least to Leibniz. In his *Dissertatio de Arte Combinatoria* (1666)¹⁶ the young Leibniz had advanced the hypothesis that all concepts could be reduced, through analysis, to a finite number of simple concepts with a procedure analogous to prime number decomposition. Such simple concepts should have been transposed in a *lingua characteristica*, i.e. in a universal sign-system capable of directly coding simple concepts and, by means of syntactical rules, complex conceptual expressions. Leibniz takes it to be possible to translate all deductive problems in the *lingua characteristica* and to decide them by means of a *calculus ratiocinator*. A typical deductive problem has the following form: “Does the conclusion *C* logically follow from the premises $A_1, A_2, \dots, A_n$?”. In symbols:

$$A_1, A_2, \dots, A_n \models^? C$$

A *calculus ratiocinator* may be thought of as a complex of calculation rules capable of “deciding”, or of answering mechanically in a positive or negative way, any

¹³ *Ibid.*, 174.

¹⁴ Lewis and Langford (1932), 248 ff.

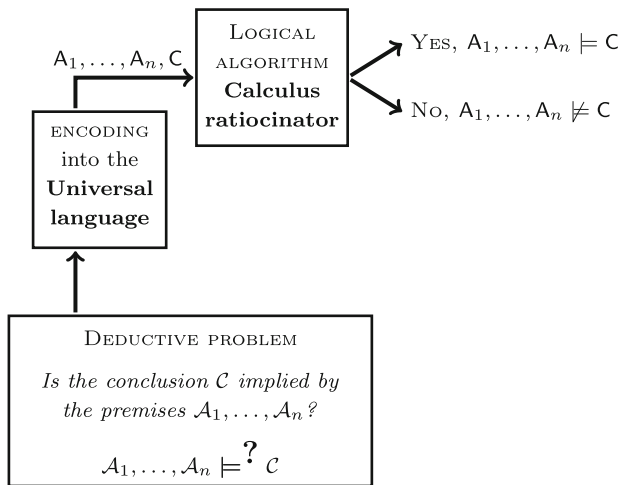
¹⁵ Hilbert and Ackermann (1928), 4; 7–9.

¹⁶ Leibniz (1666).

deductive problem; i.e., applied to our schematization, whether C does follow from $A_1, A_2, \dots, A_n$ or not. We interpret thus Leibniz's famous "*calculemus*". Leibniz writes:¹⁷

I feel that controversies can never be finished, nor silence imposed upon the sects, unless we give up complicated reasonings in favor of simple calculations, words of vague and uncertain meaning in favor of fixed symbols [*characteres*]. Thus, it will appear that every paralogism is nothing but an error of calculation [...] If controversies were to arise, there would be no more need of disputation between two philosophers than between two calculators. For it would suffice for them to take their pencils in their hands and to sit down at the abacus, and say to each other (and if they so wish also to a friend called to help): Let us calculate without further ado!

For the sake of convenience let us depict Leibniz's idea by the following schema:



To prove the decidability of a certain set, for instance of the set of all **S3**-tautologies, it suffices to exhibit a decision algorithm for it; that is, in our example, an algorithm that takes as input a formula A in the language of **S3** and terminates its computation with a (conventionally fixed) output/answer **1/yes**, if A is a **S3**-tautology, and **0/no**, if A is not a **S3**-tautology. However, in order to prove the *undecidability* of a set, one must have a mathematical counterpart for the informal

¹⁷ *De arte characteristic ad perficiendas scientias ratione nitentes* (Leibniz 1999a). The original Latin text reads as follows: "Sed ut redeam ad expressionem cogitationum per characteres, ita sentio nunquam temere *controversias finiri* neque *sectis* silentium imponi posse, nisi a ratiocinationibus complicatis ad *calculos* simplices, a vocabulis vagae incertaeque significationis ad *characteres* determinatos revocemur. Id scilicet efficiendum est, ut omnis *paralogismus* nihil aliud sit quam error *calculi* [...] Quo facto quando orientur controversiae, non magis disputatione opus erit inter duos philosophos, quam inter duos Computistas. Sufficiet enim calamos in manus sumere sedereque ad abacos, et sibi mutuo (accito si placet amico) dicere: *calculemus*".

notion of *algorithm* (on which the informal notion of *decidable set* depends). Oskar Becker neither seems to be aware of the problem of finding an adequate formal counterpart for the concept of *decidability* nor seems to have any inkling of the necessity of making the concept of *algorithm* mathematically precise.

Thanks to the investigations of the Hilbert School, the recognition of distinct logical levels (propositional, first-order, higher-order), as it is nowadays standard, was known since the years 1917–1919, as well as the decidability of classical propositional logic. The first ‘official’ proof thereof is to be found in P. Bernays’s *Habilitationsschrift* (1918); indeed, the result is already implicit, as an easy consequence, in the *normal form theorem* proved by Hilbert in his 1905 Lectures on *Logical Principles of Mathematical Thought*.¹⁸

Now, Becker seems to take **S3** as undecidable on the basis of his mistaken belief that **S3** has an infinite number of iterated modalities. His main aim in *On the Logic of Modalities* is to set up a propositional *modal* logic that is decidable, more precisely, since he seems to assume that a modal logic with a finite number of irreducible modalities has to be decidable, to find extensions of **S3** that have a finite number of irreducible modalities. To this aim, he introduces two systems, which we agreed to call **S3'** and **S3''**, that represent, so Becker, two *new* modal logics with *six* (**S3'**) and, respectively, *ten* (**S3''**) irreducible modalities.

Today we know that **S3** is decidable. Indeed, any (under very general conditions) propositional logic *L* that is finitely axiomatizable and has the finite model property is, by Harrop’s theorem, decidable.¹⁹

A complete semantics for **S3** (as well as for **S3'** and **S3''**) together with the finite model property would have applied to obtain the decidability of theoremhood for each of these logics.²⁰ But neither Lewis provides a semantics for **S3** nor Becker provides a semantics for **S3'** and **S3''**.

Decidability could also be proved syntactically by reducing it to terminating proof-search in an analytic (e.g. Gentzen-style) presentation of **S3**, **S3'** and **S3''**,²¹ or by proving the equivalence (modulo a translation) between **S3**, **S3'** and **S3''** and some other propositional logics already known to be decidable, or by developing methods based on translations of such modal logics into suitable decidable fragments of first-order logic. None of these alternative was available at that time or had even been conjectured by Lewis or Becker.

¹⁸ Hilbert (1905). Cp. Zach (1999), 333–335.

¹⁹ A logic *L* has the finite model property if any non-theorem of *L* is falsified by some finite model of *L*.

²⁰ The Kripke-style semantics for the system **S3**, together with the completeness and the finite model property theorems, is presented in full details in the appendix, Sects. 5.4, 5.5 and 5.6.

²¹ Gentzen-style sequent calculi for Lewis’s non normal modal systems have been introduced in Matsumoto (1960) (for **S2**) and Ohnishi (1961) (for **S2** and **S3**). The recent paper Tesi (2020) provides G3-style labelled sequent calculi with good structural properties for **S2**, **S3** and some neighbour systems.

2.3 Normal Modal Logics: a Quick *Resumé*

As a preliminary to the presentation and discussion of Becker's investigations on Lewis's system **S3**, to be found in the next sections, it is convenient to review here the best known *normal* modal logics, **K**, **D**, **T**, **K4**, **B**, **S4**, **S5**, as they are usually axiomatically characterized as extensions of classical logic. More precisely, the axioms comprehend all classical tautologies (or a "sufficient" selection thereof) as well as one or more additional axioms (in schematic form) that characterize the specific logic in question; the inference rules are the *modus ponens* of classical logic and one specifically modal rule, the necessitation rule

$$\frac{A}{\Box A}$$

introduced, as we already said, by Gödel in his 1933 paper. It says that if a proposition A is provable within the system in question, its necessitation $\Box A$ is also provable.²²

Once an axiom system for the minimal normal modal logic **K** is given, it is simple to give an axiomatization for the other above mentioned modal logics, for they amount to **K** plus a few additional axiom schemas.

Thus, we will recall the standard axiomatization for the logic **K** as well as the modal logical schemas T , 4 , B , 5 or E , and then we will see which schemas give which logic. Doing so turns out to be useful to the aim of proving, later on, to which standard modal systems Becker's systems **S3'** and **S3''** are equivalent.

The formal modal propositional language $\mathcal{L}^\Box$ is defined as usual. The alphabet contains:

- denumerably many propositional variables (or 'atoms'): $p_0, p_1, p_2 \dots$;
- the Boolean connectives: $\neg, \vee, \wedge, \rightarrow$;
- one modal operator: $\Box$, for *necessity*;
- auxiliary symbols: parentheses.

The modal operator $\Diamond$ for *possibility* is conveniently not taken as primitive, and $\Diamond A$ is instead introduced as a metalinguistic abbreviation for $\neg\Box\neg A$.

The set of formulas of $\mathcal{L}^\Box$ is inductively defined as usual: atoms are formulas, if A and B are $\mathcal{L}^\Box$ -formulas then also $(\neg A)$, $(A \vee B)$, $(A \wedge B)$, $(A \rightarrow B)$, $(\Box A)$ are $\mathcal{L}^\Box$ -formulas, and nothing else is a formula.

An axiomatic calculus for the basic axiom system **K** is set up as indicated in Table 2.1.

²² Notice that we are considering here only axiomatic systems which do not allow derivations with open assumptions. The extension of this formulation of the rule of necessitation from pure axiomatic systems to natural deduction and to axiomatic systems allowing derivability from open assumptions has lead to many misunderstandings in the literature on modal logic, as to the alleged failure of the deduction theorem—see Hakli and Negri (2012) for a thorough discussion of this issue.

Table 2.1 The calculus **K**

<i>Axioms and axiom schemas:</i>		
— all classical tautologies		
— $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ (schema <i>K</i>)		
<i>Inference rules</i>		
$\frac{A \quad A \rightarrow B}{B}$	MP	Modus ponens
$\frac{A}{\Box A}$	RN	Necessitation rule

A *formal proof* in **K** is a finite list $A_1, \dots, A_n$ of formulas such that for all i ($1 \leq i \leq n$): A_i is an (instance of) an axiom (schema) of **K**, or A_i follows by the *modus ponens* rule from two previous formulas A_j, A_k ($j, k < i$) in the list, or A_i follows by the *necessitation rule* from a previous formula A_j ($j < i$) in the list. A is a theorem of **K** (in symbols $\vdash_{\mathbf{K}} A$) iff there exists a formal proof $A_1, \dots, A_n$ in **K** such that A_n is A .

A modal logic which extends **K** by one or more extra axiom schemas is called a *normal modal logic*. **K** is thus the minimal normal modal logic.

Let us now recall the modal schemas *D*, *T*, *4*, *B*, *E* and their meaning:

- T*: $\Box A \rightarrow A$
D: $\Box A \rightarrow \Diamond A$
B: $A \rightarrow \Box \Diamond A$
4: $\Box A \rightarrow \Box \Box A$
E: $\Diamond A \rightarrow \Box \Diamond A$

The schema *T* claims that *if a proposition A is necessary, then it is also true* (“ab necesse ad esse valet consequentia”, in the Scholastics’ reading). It is also known as “epistemic schema”, since it is compatible with an *epistemic* interpretation of the modal operators. If we take “ $\Box$ ” to be a placeholder for the operator “it is known that (...)” or, “the agent x knows that (...)” the schema turns out to be in accordance with the Platonic conception of knowledge displayed in *Theaetetus* 201d- 210a, namely:

x **knows that** p if and only if

1. it is true *that* p
2. x believes *that* p
3. x is justified in believing *that* p

or, in other words, the schema *T* is consistent with the platonic conception of knowledge as *true belief with an account (justified)*.

The schema *D* claims that *if a proposition A is necessary, then it is also possible*. It is known as “deontic schema”, since it is consistent with a *deontic* interpretation of the modal operators. If we take “ $\Box$ ” to be a placeholder for the operator “it is obligatory that (...)” (and consequently “ $\Diamond$ ” to be a placeholder for the operator “it is permitted that (...)” the schema says that whatever is obligatory is also permitted.

The schema B is also known (after Becker²³) as *Brouwer schema*. It claims that *if a proposition A is true, then it is necessarily possible that it is true*.

The schemas 4 and E are both consistent with an *epistemic* interpretation of modalities and are also known as “*positive*” and “*negative*” *introspection* principle, respectively. Under an epistemic reading the schema 4 says that *if the agent x knows that p , he also knows that he knows that p* ; while the schema E , in the equivalent reformulation:

$$E' : \neg \Box A \rightarrow \Box \neg \Box A$$

says that *if the agent x does not know that p , then he knows that he does not know that p* .

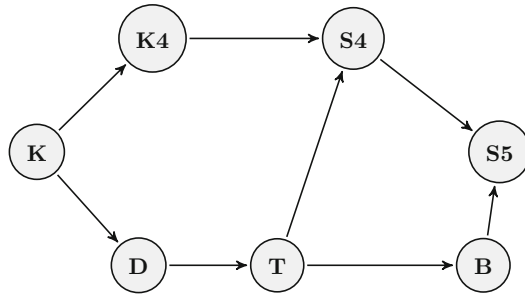
Let us now recall the most interesting axiomatic extensions of $\mathbf{K}$ which the above schemas give rise to:

- $\mathbf{D} := \mathbf{K} + D$
- $\mathbf{T} := \mathbf{K} + T$
- $\mathbf{K4} := \mathbf{K} + 4$
- $\mathbf{B} := \mathbf{K} + T + B$
- $\mathbf{S4} := \mathbf{K} + T + 4$
- $\mathbf{S5} := \mathbf{K} + T + E$

All the systems $\mathbf{K}$, $\mathbf{D}$, $\mathbf{T}$, $\mathbf{K4}$, $\mathbf{B}$, $\mathbf{S4}$, $\mathbf{S5}$ are therefore *normal* modal logics.

The “strength” relations between them can be summarized in the following diagram, where:

- an arrow leading from a system $\mathbf{L}_1$ to a system $\mathbf{L}_2$ means that $\mathbf{L}_1 \subset \mathbf{L}_2$ (that is, any $\mathbf{L}_1$ -theorem is also a $\mathbf{L}_2$ -theorem, but not conversely),
- the *absence* of any arrow between two systems $\mathbf{L}_1$ and $\mathbf{L}_2$ means that they are **incomparable**, that is: there is at least one $\mathbf{L}_1$ -theorem which is not a $\mathbf{L}_2$ -theorem, and there is as well at least one $\mathbf{L}_2$ -theorem which is not a $\mathbf{L}_1$ -theorem.



²³ The reason why Becker uses this name for the schema is explained in the next section.

2.4 Lewis's S3 and Becker's Extensions

As we already said, the formal language and the style of axiomatization employed by Lewis in his *Survey* and by Becker in his *On the Logic of Modalities* are different from the now current ones.

As primitive logical operators, they take:

- the unary operators “ $-$ ” and “ $\sim$ ”, respectively for *negation* and *impossibility*,
- and the binary operators “ $\times$ ” and “ $=$ ”, respectively for *conjunction* and *strict equivalence*.

Thus “ $-A$ ”, “ $\sim A$ ”, “ $A \times B$ ”, “ $A = B$ ” correspond, respectively, to “ $\neg A$ ”, “ $\neg \Diamond A$ ”, “ $A \wedge B$ ”, “ $\Box(A \leftrightarrow B)$ ” in our notation.

Concerning the non-primitive logical operators, both Lewis and Becker use the symbol “ $\subset$ ”, corresponding to the symbol “ $\rightarrow$ ” in the current notation (as well as to Russell’s “ $\supset$ ”), to denote the material implication. As for the strict implication Lewis *fish-hook* symbol “ $\supset$ ” is rendered in Becker’s text (probably for typographical reasons) with the symbol “ $<$ ”, which we retain henceforth in the analysis and translation of Becker’s text. The symbol for material disjunction is “ $+$ ”, corresponding to our “ $\vee$ ”. These non-primitive operators are *defined* in the expected way:

$A \subset B =_{df} -(A \times -B)$, $A < B =_{df} \sim (A \times -B)$ and $A + B =_{df} -(-A \times -B)$. To summarize, see Table 2.2.

In turn, Lewis’s (and Becker’s) axiomatization of the system **S3** is *not* given as an extension of an axiomatic calculus for classical logic by means of additional axioms and inference rules (as the axiomatizations of the normal systems reviewed

Table 2.2 Correspondences to current logical notation

Logical operator	Lewis	Becker	Current
Negation	$-$	$-$	$\neg$
Material conjunction	$\times$	$\times$	$\wedge$
Material disjunction	$+$	$+$	$\vee$
Material implication	$\subset$	$\subset$	$\rightarrow$ (or: $\supset$)
Material equivalence	$\equiv$	$\equiv$	$\leftrightarrow$
Impossibility	$\sim$	$\sim$ (or: U)	$\neg \Diamond$
Possibility	$- \sim$	$- \sim$ (or: M)	$\Diamond$
Necessity	$\sim -$	$\sim -$ (or: N)	$\Box$
Consistency	$\circ$	$\circ$	$\Diamond(\dots \wedge \dots)$
Strict disjunction	$\wedge$	$\wedge$	$\Box(\dots \vee \dots)$
Strict implication	$\supset$	$<$	$\Box(\dots \rightarrow \dots)$
Strict equivalence	$=$	$=$	$\Box(\dots \leftrightarrow \dots)$

in the previous section are). Actually, it is not at all trivial to prove that all classical tautologies are theorems of the original axiomatization of **S3**.²⁴

All in all, for a contemporary reader with a basic knowledge of logic it would be cumbersome to decipher and translate in the current formalism Lewis's and Becker's formulas and investigations.

In this chapter, we therefore present the (emended) system **S3** *not* in its original formulation, but in the *equivalent* standard formulation now current in modal logic.

We then present—again, *not* in the original but in the now standard formulation—the axiomatization of Becker's **S3'** and **S3''**. Becker's claims and accomplishments will be evaluated in the next sections.

The proofs of all our assertions (e.g. as to the equivalence or non-equivalence of the logics in question) in the present and in the next sections are carried out in the Appendix (Chap. 5). Here we try to explain what is going on in *On the Logic of Modalities* in a way accessible to a wider audience.

2.5 Lewis's Survey System S3

An axiomatic calculus for **S3** in the style of the most known modal logics, see Sect. 2.3 above, is found in Table 2.3.

Thus, **S3** contains all classical tautologies, the schema T , the schema K^+ , the rule of *modus ponens* and *only a restricted* version of the necessitation rule RN , that we indicate by " RN^- ".

The schema K^+ is, like the schema K , a distributivity law (of the operator $\Box$ on the connective $\rightarrow$). At a variance with K it contains a further $\Box$ after the principal connective $\rightarrow$.²⁵

Table 2.3 The calculus **S3**

Axioms and axiom schemas

- all classical tautologies
- $\Box A \rightarrow A$ (schema T)
- $\Box(A \rightarrow B) \rightarrow \Box(\Box A \rightarrow \Box B)$ (schema K^+)

Inference rules

$\frac{A \quad A \rightarrow B}{B} \text{ MP}$	<i>Modus ponens</i>
$\frac{A}{\Box A} \text{ RN}^-$	Provided A is a tautology or an instance of T or K^+

²⁴ See Lewis and Langford (1932), 136 ff.

²⁵ Notice that K is provable in **S3**, by using K^+ , T and the transitivity of implication.

The inference rule RN^- says that if a proposition A is either a classical tautology, or an instance of T or K^+ , then its necessitation $\Box A$ is a **S3**-theorem. Thus, if “ $\top$ ” denotes any tautology, say $p \rightarrow p$, we have that $\Box \top$ is provable in **S3**. By contrast, $\Box \Box \top$ cannot be obtained from $\Box \top$ by means of RN^- , since it is *neither* a classical tautology nor an instance of T or K^+ . Indeed, it is possible to prove that $\Box \Box \top$ is *not* a theorem of **S3**, which implies that the *unrestricted* necessitation rule RN is not admissible in Lewis’s system: **S3** is *not* a normal modal system.

As we already said, the original 1918 version of **S3** contained an additional axiom, which was responsible for the modal collapse (as Post pointed out to Lewis) and was therefore dropped by Lewis in his 1920 “*Emendation*”. The axiom schema says: the strict implication $A \rightarrow B$ is strictly implied by the strict implication $\neg \Diamond B \rightarrow \neg \Diamond A$. Formally, in our notation:²⁶

$$(*) \quad (\neg \Diamond B \rightarrow \neg \Diamond A) \rightarrow (A \rightarrow B)$$

It is interesting to notice here how Becker accounts for the implausibility of $(*)$, by providing the following informal, yet convincing and intriguing countermodel:²⁷

Example: One may think of a sequence of numbers generated by drawing arbitrarily many times, one after another, numbered balls from an urn, whereby, each time, the drawn ball is put back in the urn, before drawing the next ball. *It is unknown* how many balls are in the urn and how they are numbered. [...]

[T]he converse

$$(\sim q < \sim p) < (p < q)$$

does not hold, as our example—suitably modified—does show. Namely, let now “ q ” and “ p ” stand for:

q : “19 appears among the first 100 places.”

p : “19 appears among the first 200 places.”

Then

$$\sim q < \sim p$$

does hold.

Actually, the *impossibility* that 19 occurs in the sequence (at a variance with its contingent, i.e., *incidental* not-occurring) can be due exclusively to the fact that no balls with the number 19 are contained in the urn. This impossibility holds for all places of the sequence, if it holds for some.

However, from $\sim q < \sim p$ it does *in no way* follow $p < q$, i.e. the implication: “If 19 appears among the first 200 places, then it necessarily appears within the first 100 places”, since this sentence is trivially false.

²⁶ That is, $(\sim q \rightarrow \sim p) \rightarrow (p \rightarrow q)$ in Lewis’s symbolism. Notice that the converse of $(*)$, $(A \rightarrow B) \rightarrow (\neg \Diamond B \rightarrow \neg \Diamond A)$ is a theorem of **S3** (actually an axiom, $(p \rightarrow q) \rightarrow (\sim q \rightarrow \sim p)$, in Lewis’s presentation).

²⁷ Becker (1930), 8–9 (63–65).

2.6 Becker's Six Modalities System S3'

Let us now consider Becker's system S3'. He writes:²⁸

Before going into the actual meaning of the reduction problem of the infinitely many nested modalities, which arise through the iteration and composition of the symbols “ $\sim$ ” and “ $-$ ”, we present a purely formal investigation, by which Lewis's system becomes a closed system, thanks to the addition of a further axiom. This can be done in several ways. [...] The assumptions introduced by Lewis are (apparently) not sufficient to obtain a closed system of irreducible modalities.

Therefore we add to Lewis's axioms the new axiom 1.9:

$$-(\sim p) < \sim (\sim p)$$

Becker is saying that—as summarized in Table 2.4—this system (once formulated in our notation) is obtained from S3 by the addition of one single axiom schema, namely $\Box(\Diamond A \rightarrow \Box\Diamond A)$, which is a “boxed-version” of the schema E .

Becker gives a detailed proof of the fact that this system has 6 irreducible modalities:

- Positive modalities: $\Box A$, $\Diamond A$, A (“factual” truth);
- Negative modalities: $\neg\Box A$, $\neg\Diamond A$, $\neg A$ (“factual” falsity).

and that they are linearly ordered, as to logical strength, as follows

- Positive modalities: $\Box A < A < \Diamond A$
- Negative modalities: $\neg\Diamond A < \neg A < \neg\Box A$.

Table 2.4 The calculus S3'

Axioms and axiom schemas

- all classical tautologies
- $\Box A \rightarrow A$ (schema T)
- $\Box(A \rightarrow B) \rightarrow \Box(\Box A \rightarrow \Box B)$ (schema K^+)
- $\Box(\Diamond A \rightarrow \Box\Diamond A)$ (schema $\Box E$)

Inference rules

$\frac{A \quad A \rightarrow B}{B} \text{MP}$	<i>Modus ponens</i>
$\frac{A}{\Box A} \text{RN}^-$	Provided A is a tautology or an instance of T or K^+

²⁸ Becker (1930), 11–12 (67–67). Recall that “ $-A$ ” corresponds to our “ $\neg A$ ”.

2.7 Becker's Ten Modalities System S3''

In Brouwer's and Heyting's²⁹ Intuitionistic logic the *double negation principle*, $\neg\neg A \leftrightarrow A$, is not valid. More precisely, the left-to-right direction of the biconditional, $\neg\neg A \rightarrow A$, is not intuitionistically acceptable. The other direction of the biconditional,

$$A \rightarrow \neg\neg A \quad (\text{WDN})$$

is instead intuitionistically valid.

As we know, Becker is also trying to explore the connection between intuitionistic and modal logic. He is thus naturally led, in particular, to interpret the intuitionistic negation (“ $\neg$ ”)—which is *stronger* than classical negation—in modal terms, as *impossibility* (“ $\sim$ ”) or, as he uses to say, *absurdity* (*Absurdität*). By replacing, in the intuitionistic law (WDN), “ $\neg$ ” with “ $\sim$ ” and “ $\rightarrow$ ” with “ $<$ ” one gets

$$A < \sim\sim A$$

(that is $\Box(A \rightarrow \neg\Diamond\neg\Diamond A)$, in our notation)

“Truth—as he puts it³⁰—implies the absurdity of the absurdity (but not conversely!)”.

Such principle is thus equivalent to the “boxed-version” of the schema *B*, $A \rightarrow \Box\Diamond A$, considered in Sect. 2.3. It should be now clear why Becker called it “Brouwer's *axiom*”, a name still current in the literature.

According to Becker, this is a reasonable axiom to consider in order to extend Lewis's **S3**:³¹

One can now add “Brouwer's *axiom*” to this setting (this is the weakest addition we propose):

$$p = \neg\neg p < \sim\sim p \quad (1.91)$$

[...] As an [additional] axiom we choose [...]:

$$\sim\neg p < \sim\sim\neg p \quad (1.92)$$

[...] If one postulates (1.91) $\times$ (1.92) one can thus set up a ten modalities calculus.

²⁹ Becker refers explicitly to Heyting (1930), which contains the first (complete) presentation of intuitionistic logic as a formalized calculus. The paper was published in the same year of *On the Logic of Modalities*, but was circulating since 1928.

³⁰ Becker (1930), 17 (75).

³¹ Becker (1930), 17–18 (75–77).

Table 2.5 The calculus **S3'***Axioms and axiom schemas*

- all classical tautologies
- $\Box A \rightarrow A$ (Schema T)
- $\Box(A \rightarrow B) \rightarrow \Box(\Box B \rightarrow \Box B)$ (Schema K^+)
- $\Box(A \rightarrow \Box \Diamond A)$ (Schema $\Box B$)
- $\Box(\Box A \rightarrow \Box \Box A)$ (Schema $\Box 4$)

Inference rules

$\frac{A \quad A \rightarrow B}{B} \text{MP}$	<i>Modus ponens</i>
$\frac{A}{\Box A} \text{RN}^-$	Provided A is a tautology or an instance of T or K^+

In the current standard form we adopted, Becker's second extension of Lewis's **S3**—summarized in Table 2.5—results from **S3** by adding *two* axiom schemas, namely $\Box(A \rightarrow \Box \Diamond A)$ (1.91), the “boxed-version” of the Brouwer's schema B , and $\Box(\Box A \rightarrow \Box \Box A)$ (1.92)³² which is a “boxed-version” of the schema 4 (see Sect. 2.3).

Becker's claim, supported by a detailed (putative, see below, p. 25) proof, is that this system has 10 irreducible modalities:

- Positive modalities: $\Box A, \Diamond \Box A, \Diamond A, \Box \Diamond A, A$ (“factual” truth);
- Negative modalities: $\neg \Box A, \neg \Diamond \Box A, \neg \Diamond A, \neg \Box \Diamond A, \neg A$ (“factual” falsity).

and that they are linearly ordered, as to logical strength, as follows

- Positive modalities: $\Box A < \Diamond \Box A < A < \Box \Diamond A < \Diamond A$
- Negative modalities: $\neg \Diamond A < \neg \Box \Diamond A < \neg A < \neg \Diamond \Box A < \neg \Box A$.

2.8 Becker's Further “Experiments”

Becker did also tentatively consider other two possible ways to extend Lewis's **S3** in order to get a system with a finite number of irreducible modalities or, at least, a system with a possibly infinite number of irreducible modalities, yet all *pairwise comparable* with respect to logical strength. Here is a sketchy account of these two modal “experiments” (as he calls them³³).

³² In Becker (1930), 22 f. (82 f.) Becker acutely analyzes the “concrete (phenomenological) meaning” of this schema in the light of Husserl's distinction between contingent and formal apriori.

³³ Becker (1930), 2 (56).

2.8.1 A Variant of $S3'$

This variant, let us call it $S3'^*$, is proposed in a short *Observation*³⁴ following the presentation and the investigation of the *six modalities calculus* $S3'$. It is obtained by replacing the characteristic axiom schema $\Box E$

$$\Box(\Diamond A \rightarrow \Box\Diamond A)$$

of $S3'$ (see Table 2.3) with the axiom schema $\Box E^*$

$$\Box(\Box\Diamond A \rightarrow \Box A)$$

Becker's claim is that also this new system $S3'^*$ has 6 irreducible modalities, exactly the same as $S3'$, and that they are ordered with respect to logical strength in the same way as they are ordered in $S3'$. The only remarkable difference between the two systems, according to Becker, is that while in $S3'$ A is stronger than $\Box\Diamond A$ ($\sim\sim A$ in his notation), in $S3'^*$ the other way around is the case: $\Box\Diamond A$ is stronger than A . His “formalist” conclusion is the following:³⁵

Thus, $\sim\sim p$ is stronger than p (in contrast with Brouwer's conception). From a *purely formal point of view* it seems that also this approach can be carried through without contradiction; although it has perhaps no concrete meaning.

2.8.2 A More Abstract Approach

In §5 of Part I,³⁶ entitled “On a Calculus of Modalities with Minimal Requirements, that Still Allows for a Linear Rank Order”, Becker tentatively develops a very interesting, more abstract approach to the problem of “completing” Lewis's calculus in such a way that in the resulting system, independently from the number (finite or infinite) of irreducible modalities, any two modalities (combinations of primitive modalities) be comparable with respect to logical strength:³⁷

One can try to free oneself from all requirements that are imposed by special *material* [*sachliche*] assumptions, and introduce solely the most general *formal* conditions of a calculus of modalities. Firstly, one can drop the requirement that the reduction to a finite number of primitive modalities be possible. [...]

One will, however, keep the general requirement of a linear rank order of modalities, whereby the implicational relation of any two distinct modalities is univocally determined.

³⁴ Becker (1930), 15–16 (73–73).

³⁵ Becker (1930), 16 (73).

³⁶ Becker (1930), 25–30 (85–91).

³⁷ Becker (1930), 25 (85).

Otherwise a modality could no longer be univocally determined by its "rank of logical strength". Such requirement should be in any case the upper bound of our formal freedom.

Now, the question is, on the one hand, whether Lewis's calculus satisfies this requirement and, on the other hand, whether suitable axioms for a theory of the rank order of modalities can be established independently from Lewis's calculus. The answer to the first question is negative, while the answer to the second question is positive.

Becker conveniently uses here as basic, elementary positive modality the operator " $\Box$ " (" N " in his symbolism, corresponding to Lewis's " $\sim -$ "). He then fixes a number of "rules"³⁸ to which composed ("non-elementary") modalities in the calculus of modalities should obey.

By a *composed modality* he means a finite, possibly empty string of $\Box$'s and $\neg$'s (e.g. $\Box\Box$, $\neg\Box\neg$, $\neg\Box\neg\Box\neg\Box$, ...). Capital Greek letters Λ , Π , ... vary over composed modalities. Given two composed modalities Λ and Π , the composed modality arising from their juxtaposition (in the given order) is denoted by " $\Lambda\Pi$ ".³⁹ A composed modality Λ is *positive* (*negative*) if and only if it contains an *even* (*odd*) number of negations. Thus e.g. $\Box$ and $\neg\Box\neg$ (that is: $\Diamond$) are positive, while $\Box\neg$ (equivalent to $\sim$, *impossible*) is negative.

The rules are intended to impose a number of conditions concerning the *preservation within the calculus of relations of logical strength between modalities under composition/juxtaposition*, and can be equivalently rephrased as follows.

For any basic or composed modalities Λ , Λ' and Π ,

if

$$\Lambda\Lambda\rightarrow\Lambda'A$$

is a theorem of the calculus of modalities for every formula A

then also

- $\Lambda\Pi\Lambda\rightarrow\Lambda'\Pi\Lambda$ [rule R_1]
- $\Pi\Lambda\Lambda\rightarrow\Pi\Lambda'A$, where Π is *positive* [rule R_2]
- $\Pi\Lambda'A\rightarrow\Pi\Lambda\Lambda$, Π is *negative* [rule R_3]

shall be theorems of the calculus of modalities for every formula A

In other words, these three rules say that if the modality Λ turns out to be at least as strong as the modality Λ' in the calculus, then also $\Lambda\Pi$ shall be at least as strong as $\Lambda'\Pi$ (R_1), and $\Pi\Lambda$ shall be at least as strong as $\Pi\Lambda'$ when Π is positive (R_2), as well as $\Pi\Lambda'$ shall be at least as strong as $\Pi\Lambda$ when Π is negative (R_3).

Incidentally, it is worth observing that *these rules have not been correctly interpreted and formalized* by Churchman (1938), the first and, as far as we know, unique scholar who has carefully analyzed this "experiment" by Becker. Indeed, the

³⁸ The relevant ones are "Rule II" and "Rule III", which in our presentation, see below, correspond to rule (R_1) and, respectively, rule (R_2) plus rule (R_3).

³⁹ E.g., for $\Lambda = \Box\Box$ and $\Pi = \neg\Box\neg$, we have $\Lambda\Pi = \Box\Box\neg\Box\neg$.

inference rules

$$\frac{A \rightarrow B}{\Box A \rightarrow \Box B} \quad \text{and} \quad \frac{A \rightarrow B}{\Diamond A \rightarrow \Diamond B}$$

still known in the current literature as Becker's *rules*, were given this name by (Churchman 1938)⁴⁰ because *incorrectly* regarded—a fact which apparently has never been noticed in the literature—as specific instances of the above rule (R₂) of Becker, as we detailedly explain in Sect. 5.9.3 of the Appendix.⁴¹

Now, the point is that Lewis's calculus, while being closed under these three rules, contains *incomparable* modalities. So, this is Becker's very interesting idea, one should try to devise the “weakest possible axiomatic conditions” one should add to the above rules R₁–R₃, *in order to obtain a calculus in which all the modalities are linearly ordered*, that is, are pairwise comparable in strength.

At the end of a rather elaborate argument, he arrives at the claim that a stepwise generalization of Brouwer's schema (in the form $\Box B$, see above):

$$B_1 = \Box(A \rightarrow \Box \Diamond A) \quad (\text{that is } \Box B)$$

$$B_2 = \Box(A \rightarrow \Box \Box \Diamond A)$$

$$B_3 = \Box(A \rightarrow \Box \Box \Box \Diamond A)$$

⋮

provides an infinite number of axiomatic conditions which, added to a basis system like Lewis's **S3**⁴² together with the above rules (R₁)–(R₃), produce a calculus—let us call it **SM**, for further reference—whose modalities, although infinite in number, are linearly ordered. He leaves as an open problem the question whether all these infinite modalities of **SM** are irreducible.

2.9 Becker's Logical Accomplishments: An Assessment

The evaluation of Becker's formal investigations, and in particular of the results and claims we illustrated in the previous section, may well start from the questions raised by Gödel in his *Review of On the Logic of Modalities*:⁴³

[T]he author proposes various additional axioms and then seeks to specify a system, with as few assumptions as possible, for which a linear ordering [of modalities with respect to logical strength] still exists. All in all, three different kinds of the calculus of modalities emerge [...]. As far as the purely formal side is concerned, one can hardly take exception to anything here, but there remain essential gaps to be filled in, some of which the author

⁴⁰ Hereto cp. e.g. Hughes and Cresswell (1968), 200; 207 and Peckhaus (2002), 169.

⁴¹ See also Centrone and Minari (2022).

⁴² Becker does not characterize explicitly the “basis system”; this point is discussed at length in the Appendix, Sect. 5.9.3.

⁴³ Gödel (1931), 6.

himself points out. Above all, it is nowhere shown that the three systems set up really differ from one another and from Lewis's system (in other words, that the additional axioms are not in fact equivalent and do not follow from Lewis's); nor, furthermore, that the six, or ten, basic modalities obtained cannot be still further reduced.

As we have seen, Becker actually proposed *four* calculi altogether: **S3'** (the *six modalities calculus*, see Sect. 2.6), **S3''** (the *ten modalities calculus*, see Sect. 2.7), the variant **S3'*** of **S3'** (see Sect. 2.8.1) and finally **SM** (the calculus with infinitely many linearly ordered modalities, see Sect. 2.8.2). It is likely that "the three different kinds of the calculus of modalities" Gödel is hinting at are the extensions **S3'**, **S3''** and **SM** of **S3**. Anyway, including also **S3'***, the questions are:

- (i) Are **S3'**, **S3''**, **S3'*** and **SM** pairwise non-equivalent?
- (ii) Can the additional axioms of these systems be derived from **S3**?
- (iii) Are the claimed "irreducible" modalities of these calculi really irreducible?

The deciding answers to these questions follow from the following three results:

1. **Both S3' and S3'' are equivalent to the system S5.** **S3'** is indeed *one of the two equivalent axiomatizations of the normal modal system S5* (see Chap. 5), as it is explicitly introduced for the first time (and thus named) in 1932 in the *Appendix II* of Lewis & Langford *Symbolic Logic*.⁴⁴ The other equivalent axiomatization of **S5** indicated there is exactly Becker's system **S3''**! So, the six and the ten modalities calculi proposed by Becker are in fact different axiomatizations of one and the same modal system, and while the six "irreducible" modalities of **S3'** are really irreducible, the ten "irreducible" modalities of **S3''** of course boil down to the six of **S3'**, alias **S5**. This said, and given the fact that Becker's axiom schemas $\Box B$, $\Box 4$ and $\Box E$ are indeed *not derivable* from **S3**,⁴⁵ we have a conclusive answer to questions (i)–(iii) above as to **S3'** and **S3''**.
2. **S3'* collapses.** This was first observed by Parry in 1939, while one year earlier Churchman⁴⁶ was still conjecturing that **S3'*** had an infinite number of irreducible modalities. Indeed, using the **S3**-theorem $\Box \Diamond (A \rightarrow \Box A)$, Parry was able to show that $A \rightarrow \Box \Box A$ is a theorem of **S3'*** and thus "reduces to the system of material implication".⁴⁷ Of course, this also shows (see questions (ii) and (iii) above) that the additional axiom $\Box E^*$ of **S3'*** *cannot* be derived from **S3**, and that the six "irreducible" (according to Becker) modalities of **S3'*** boil down to two, that is (actual) truth and falsity.
3. **SM is equivalent to the system S5.** Finally, as far as the fourth "system of modalities", **SM**, tentatively proposed by Becker, Churchman proved in 1938⁴⁸

⁴⁴ Lewis and Langford (1932), 501.

⁴⁵ The irreducibility of the six modalities of **S5**, alias **S3'**, and the other facts and claims we mentioned are proved in Lewis and Langford (1932), 497 ff.

⁴⁶ Churchman (1938), 78.

⁴⁷ Parry (1939), 153 f. The proof is presented in the Appendix, Sect. 5.9.1.

⁴⁸ Churchman (1938), 78 ff.

that it is equivalent to $S3'$ —that is, by what we said above, to $S5$. This claim is indeed correct, *under one out of diverse possible interpretations of this system*,⁴⁹ although Churchman's proof thereof is not, because he did not adequately formalize the system SM as Becker intended it⁵⁰.

All in all, we can say that Becker's claims about $S3'$ *were right*—and so also that *he was indeed the first*, two years before the official birthdate of the system $S5$, to identify this modal system and to investigate some of its properties.

Becker's claims about $S3''$, $S3'^*$ and (perhaps) SM *were instead wrong*. Yet, concerning $S3''$ the proofs he gave to support his claims were correct, except that he did not notice that the characteristic schema $\Box E$ of $S3'$ can be (easily) derived from the two characteristic schemas $\Box B$ and $\Box 4$ of $S3''$, modulo Lewis's system $S3$. Concerning SM , again, his proof that the (supposedly infinite) modalities of this system are linearly ordered is very clever, and correct. Unfortunately he did not notice that, on the basis of $S3$ (of $S2$), already the first schema B_1 (resp. the first two schemas B_1 and B_2) of the infinite sequence of schemas $\{B_n\}_{n \geq 0}$ he postulated, together with the rule R_2 are sufficient to prove the schema $\Box 4$ —and so to make SM equivalent to $S3'$, alias $S5$. Whether the same does happen with a basis *weaker* than $S2$ is an interesting question, open as far as we know and worth to be investigated.

In the light of these results—of the four systems he proposed, three are in fact equivalent ($S3'$, $S3''$ and SM) while the remaining one ($S3'^*$) collapses—one might be tempted to underrate Becker's *formal* contributions in *On the Logic of Modalities*. On the contrary, and notwithstanding these shortcomings, Becker's *pioneering* work, containing sophisticated insights and interesting technical solutions, has played an extremely important role in the *early* development of modal logic in the decade 1930–1940, as witnessed by the scientific contributions of other scholars who, at that time, referred to Becker's investigations and to the problems raised by him, and took them as a basis for further developments and investigations.⁵¹

Last but not least, Becker was the first to advance the idea of a *modal interpretation of intuitionistic logic*—the reader should keep in mind that Becker was writing in 1930, and that the birthdate of intuitionistic logic as a formalized system is 1928, thanks to Heyting's axiomatization.⁵²

The idea is exposed and elaborated in the *Appendix to the Part I*⁵³ of *On the Logic of Modalities*:⁵⁴

⁴⁹ See the detailed analysis in Sect. 5.9.3.

⁵⁰ As we explained on p. 24.

⁵¹ In this regard, Feys (1937) and Feys (1938) should at least be added to the already mentioned works Gödel (1931) and Gödel (1933a), Lewis and Langford (1932), Churchman (1938), Parry (1939) referring to, and discussing, Becker's work.

⁵² Published two years later in Heyting (1930).

⁵³ Becker (1930), 30–35 (91–96).

⁵⁴ Becker (1930), 31 (92).

How is now the Heyting's calculus related to the uncompleted and the completed Lewis's calculus?

First of all, the question concerning an appropriate "translation" of the symbols arises.

More precisely, his idea is to define a vocabulary translation associating to each intuitionistic (H) logical operator ($\rightarrow, \vee, \wedge, \neg$) a corresponding logical operator (L) of Lewis's calculus **S3**, in such a way that every theorem of the intuitionistic propositional calculus **IPC** becomes, once transformed according to the translation, a theorem of **S3**. Becker tentatively considers three candidate translations:⁵⁵

- (T1) $H: \rightarrow, \vee, \wedge, \neg \Rightarrow L: \rightarrow, \vee, \wedge, \neg$
 (T2) $H: \rightarrow, \vee, \wedge, \neg \Rightarrow L: \rightarrow, \vee_s, \wedge, \neg\Diamond$ (where $A \vee_s B =_{df} \Box(A \vee B)$)
 (T3) $H: \rightarrow, \vee, \wedge, \neg \Rightarrow L: \rightarrow, \vee, \wedge, \neg\Diamond$

As to (T1), he observes that the T1-translation of every intuitionistic theorem is *obviously* a **S3**-theorem, because intuitionistic logic is included in classical non modal logic, and the latter in turn is included in the Lewis's system. On the other side, he rightly stresses that⁵⁶

[...] this is a worthless triviality. Indeed, the purpose of a comparison between intuitionistic and modal logic can only be to make the deficits of the former with respect to the latter comprehensible by interpreting the intuitionistic notions by the specific modal-logical notions, that is by Lewis's "strict" notions (*strict implication, strict logical sum, impossibility*).

Concerning the second translation, he again rightly observes that the T2-translation $A \vee_s A \leftrightarrow_s A$ of $A \vee A \leftrightarrow A$, which is one of the axioms of Heyting's calculus, is *not a theorem* of **S3**—the latter would collapse otherwise!

Finally, as far as the third of the proposed translations is concerned, Becker claims (without giving a detailed proof) that the T3-translation of the axiom $(A \rightarrow B) \wedge (A \rightarrow \neg B) \rightarrow \neg A$, that is the formula

$$(*) \quad (A \rightarrow B) \wedge (A \rightarrow \neg\Diamond B) \rightarrow \neg\Diamond A$$

is *not* a theorem of **S3**, and thereby concludes his "translation-experiments" as follows:⁵⁷

At this point a further investigation must begin, with the aim to assess *whether and which additions must be made to the extended Lewis's System (Calculus of 10 Modalities, Calculus of 6 Modalities) in order to validate the translation of Heyting's axiom (11) [i.e. (*) above]*. Further problems can nevertheless arise because of the difference of the undefined notions in Heyting's and Lewis's system. The solution of these tasks and the overcoming of these difficulties shall be left to future work.

Indeed, it is not difficult to prove that Becker was right in claiming that (*) is undervivable in **S3**. Actually, it is possible to prove even more: *every normal modal*

⁵⁵ Here rephrased in the non-Lewisian symbolism so far employed.

⁵⁶ Becker (1930), 31 (92).

⁵⁷ Becker (1930), 33 (94).

system containing the schemas T and $(*)$ collapses. This implies that also with respect to the extended Lewis's systems mentioned by Becker the translation T3 would boil down to the *trivial* translation T1.

To conclude, let us tell the end of the story: only three years later someone else, namely Kurt Gödel, did the “future work” and “found the solution of these tasks” as predicted by Becker. In the already mentioned *An interpretation of the intuitionistic propositional calculus*⁵⁸ Gödel, who in his 1931 *Review of On the Logic of Modalities* had hastily mentioned and too roughly dismissed Becker's idea of a *modal interpretation of intuitionistic logic*, saying⁵⁹

[...] the author discusses, from a formal as well as a phenomenological standpoint, the connections that in his opinion obtain between modal logic and the intuitionistic logic of Brouwer and Heyting. It seems doubtful, however, that the steps here taken to deal with this problem on a formal plane will lead to success,

provided the first *sound and faithful translation* of propositional intuitionistic logic **IPC** into a modal system, namely **S4**.⁶⁰ In the paper, Becker is mentioned for having introduced the axiom $\Box A \rightarrow \Box \Box A$ ($\Box 4$, see Sect. 2.7) but, quite unfairly, *not for having anticipated the very idea of a modal translation of intuitionistic logic*.⁶¹ By the way, as was later proved by Hacking,⁶² it is also possible to define a sound and faithful translation of intuitionistic logic even in the Lewis's modal system **S3**—as Becker had tried to do.

⁵⁸ Gödel (1933a).

⁵⁹ Gödel (1931), 6.

⁶⁰ Gödel's translation $()^G$ from the formulas of the language $\mathcal{L}$ of intuitionistic propositional logic to the formulas of the modal propositional language $\mathcal{L}^\Box$ of **S4** (see Sect. 2.3) is such that, for every $\mathcal{L}$ -formula A , the following does hold: (i) if $\vdash_{\text{IPC}} A$ then $\vdash_{\text{S4}} (A)^G$ (*soundness*), and (ii) if $\vdash_{\text{S4}} (A)^G$ then $\vdash_{\text{IPC}} A$ (*faithfulness*). Point (ii) was only conjectured by Gödel, and proved 15 years later by McKinsey and Tarski (McKinsey and Tarski 1948) through the introduction of a suitable algebraic-semantics characterization for **S4** (an elegant, purely syntactical proof of the faithfulness of Gödel's translation can be found in Dyckhoff and Negri 2012). See the Appendix, Sect. 5.7, for a detailed proof.

⁶¹ Likewise, no mention of Becker is found in A. S. Troelstra's “Introductory note to *An interpretation of the intuitionistic propositional calculus*”, in Feferman et al. (1986), 296–299. Actually, Becker's anticipation of the idea of a modal interpretation of intuitionistic logic is completely neglected in the literature, apart from few cursory mentions, see e.g. Hughes and Cresswell (1972), 305 fn. 364 and Peckhaus (2002), 171–172.

⁶² Hacking (1963).

Chapter 3

Part II of *On the Logic of Modalities*



3.1 Introduction

Part II of Oskar Becker's *On the Logic of Modalities* bears the title "*The Logical Interpretation of the Mathematical Intuitionism from the Standpoint of Modality*." Subtitle: "*With Special Consideration of E. Cassirer's 'Philosophy of Symbolic Forms'*."

The ultimate goal of this part is to show that a proper understanding of the problem of modalities gives insight (i) into the conceptual presuppositions at the basis of the intuitionistic foundation of mathematics¹ as well as (ii) into the relations between classical and intuitionistic logic. And, finally, (iii) to bring to light the relations between the alethic modalities, necessity and possibility, and the temporal modes of the present, on the one side, and on the past and the future of the other side.

This is preceded by a careful reconstruction of Cassirer's thoughts on intuitionism contained in Part III of Cassirer's *Philosophy of Symbolic Forms*, entitled "*Phenomenology of Knowledge*", that had appeared in Hamburg in 1929.

For our aims it is worth noticing that

- (i) Cassirer's manuscript was already completed in 1927, i.e. in the year in which, as we said above,² Oskar Becker's masterpiece *Mathematical Existence* as well as Martin Heidegger's masterpiece *Being and Time* had been published together in volume 8 of the *Yearbook for Philosophy and Phenomenological Research*.
- (ii) The third volume of *The Philosophy of Symbolic Forms* contains a chapter on mathematical concepts-formation that includes a long footnote, in which

¹ Cp. Becker (1930), 35 (96).

² See 1, fn. 3.

Cassirer is critical towards the analysis of the role of temporality in the foundation of the numerical field, that Oskar Becker delivers in *Mathematical Existence*.³ This critical remark by Cassirer is reproduced in the present volume at the end of the English translation of *On the Logic of Modalities*.

- (iii) *On the Logic of Modalities. Part II* constitutes Oskar Becker's answer to Cassirer's criticism.

The discussion between Cassirer and Oskar Becker is actually a multi-part dialogue with many participants, the most important of which are Edmund Husserl, David Hilbert, Luitzen Egbertus Jan Brouwer and Hermann Weyl. Let us first of all ask the question: What is at issue?

3.2 On Edmund Husserl

At issue is, most likely, a double lecture Edmund Husserl presented to the Mathematical Society in Göttingen in the Winter term of 1901/1902.⁴ The lecture is well known in the literature on Husserl as his "*Doppelvortrag*."⁵ All that remains of it is contained in the manuscript with signature KI 26 conserved in the Husserl-Archives Leuven, which bears the title (written by Husserl in pencil) "ad Vortrag Göttingen 1901." We know that Hilbert urged upon Husserl to publish the *Doppelvortrag* and that Husserl was occupied with the elaboration of the definitive text during the Christmas holidays 1901/1902.⁶ Husserl did not publish the *Doppelvortrag* then, nor did he ever reconsider the possibility of a publication, though on several occasions he emphasized its importance.⁷ The themes and problems discussed in the *Doppelvortrag*, in particular the two correlated concepts of *definite theory* and *definite manifold* were later extensively reconsidered in his *Formale und Transzendente Logik*.⁸ In a footnote to §72 of his *Ideas I*, which bears the title "*Eidetic Sciences: Concrete, Abstract, 'Mathematical'*", Husserl bitterly complains that his ideas about definiteness have found their way into the literature without mention of their original sources. He writes:

The concepts introduced here served me already at the beginning of the 1890's in the investigations pertaining to the theory of formal-mathematical disciplines, which I intended as a continuation of my *Philosophy of Arithmetic*, namely to the purpose of finding a fundamental solution to the problem of the imaginary. Since then I have often had occasion

³ Cassirer (1929), 469–471; Engl. transl. 404–405.

⁴ E. Husserl, *Das Imaginäre in der Mathematik: Zu einem Vortrag in der mathematischen Gesellschaft in Göttingen* (1901), in: Husserl (1970), 430–451, Engl. transl. 409–452.

⁵ Cp., among others, Schuhmann and Schuhmann (2001), Da Silva (2000), Hartimo (2007), Centrone (2010), 149 ff., Centrone (2011).

⁶ Cp. Schuhmann and Schuhmann (2001), 87.

⁷ Cp. e.g. Husserl (1913), 72; Husserl (1929), 85.

⁸ Husserl (1929), 78 ff.

to develop the relevant concepts and theories in lectures and seminars, partly in complete detail; and in the Winter semester of 1901/02 I dealt with them in a double lecture to the Göttingen Mathematical Society. Some parts of this train of thoughts have found their way into the literature, without mention of their original sources. The close relationship of the concept of definiteness to the “axiom of completeness” introduced by Hilbert for the foundation of arithmetic will be immediately obvious to every mathematician.

3.2.1 *Existence*

The *Doppelvortrag* can be read as Husserl’s own contribution to the solution of a problem Hilbert was concerned with around 1900: under which conditions is it possible to consider a certain object, or a certain set of objects, say, for instance, the set of real numbers, as existent? At this stage of his foundational research Hilbert was convinced that: “if one succeeds in proving that the constitutive marks of a [certain] concept cannot ever lead to a contradiction using a finite number of logical inferences,”⁹ then the mathematical existence of the object-concept is proven. Otherwise said, if one can prove that the axiom-system for the real numbers is consistent, then *there is* the set of real numbers.¹⁰

Husserl’s problem is slightly different. He asks under which conditions the system of natural numbers can be stepwise expanded up to the system of the real numbers.

At root, the *Doppelvortrag* can be seen as an application of what Hilbert in his paper *On the Concept of Number* calls the “genetic method.”¹¹ the concept of real number is obtained by way of successive expansions of the concept of natural number by dropping certain restrictions on the executability of certain operations.

In this context Husserl uses the expression “imaginary” *à la* Leibniz, that is as a synonym of “empty.”¹² An imaginary concept is a concept to which there corresponds no object within a certain field. For instance, the negative integers turn out to be imaginary number-constructs within the field of natural numbers, while some real numbers turn out to be imaginary number-constructs within the field of rational numbers. To the concept expressed by “ $\sqrt{2}$ ”, for instance, there corresponds no object within the field of rational numbers.

The expansion of the rules of calculation that are valid for the theory of finite cardinal numbers in a purely formal way, without considering the correlative

⁹ Hilbert (1900b).

¹⁰ Sieg (2002), 365.

¹¹ Hilbert (1900a), 180–181.

¹² Cp. Leibniz (1962), II, Ch. 30, §5: “[...] [P]ossible ideas become merely chimerical when the idea of actual existence is groundlessly attached to them [...]”. Hereto cp. Bolzano (1837) I, §70, 323: “There are [...] ideas which have [...] no object, simply because the attributes they would ascribe to their object are contradictory. [...] [i]t has been customary to call them empty, impossible, or imaginary ideas.”

modification of the concrete numerical field is, so Husserl, at the source of the difficulties that concern the arithmetical method, the most important of which is related to the application, in calculations, of “imaginary numbers”, that is, of number-constructs that are devoid of reference in one or more arithmetical fields, whose use, nevertheless, yields correct results.

3.2.2 *Definiteness and Universal Arithmetic*

The idea Husserl develops in the *Doppelvortrag* is that the solution of the problem of the imaginary is closely connected with a specific property that only some formal theories have—the property of *definiteness*. The stepwise expansion of the theory of natural numbers to the theory of real numbers is possible, in the sense that it does not lead to contradictions, if and only if the axiom system to be expanded is *definite* and its expansion is proven to be both *consistent* and *conservative* over the old theory. Husserl believes that the *definiteness* of the narrow system guarantees the *conservativity* of the broader one. The interpretation of Husserl’s definiteness is controversial in the literature,¹³ but the property that, according to us, seems to be more in line with Husserl’s word is that of *syntactic completeness* of a theory. Thus, an axiom system is definite, in our interpretation, if and only if it formally decides each formula of its language (i.e., it either proves the formula or its negation).

The conception of mathematical theories that emerges from Husserl’s analysis is as follows. Each *arithmetic* is a *system of equations* or a *system of operations* that axiomatizes a certain numerical field. Each numerical field is an independent reality with its own mathematical structure. The *system of equations* and the *field of experience* or *structure* that is axiomatized by that system are *one and the same thing*, yet considered from two different points of view, namely that of *the theory* and that of *the structure*, which are terminologically distinguished in Husserl’s later *Formal and Transcendental Logic*¹⁴ as *apophantic* (i.e. concerned with the forms of propositions, of proofs, and of theories as systems of propositions) and *ontological* (i.e. concerned with objects in general, with sets and relations among sets, with any kind of manifold, with any kind of connection, order, magnitude, ...).

By “universal arithmetic” Husserl means a formal theory of operations that is valid in all numerical systems (the wholes, the rationals, etc.). Different numerical fields can be interpreted, under certain restrictions or by suitably dropping restrictions on the executability of certain operations, by the same algorithm; in other words, the “objects” of these fields can stand in the same relations and can be

¹³ Cp., at least, Majer (1997), Da Silva (2000), Hartimo (2007), Centrone (2010), Centrone (2011).

¹⁴ Cp. *Formal and Transcendental Logic* (Husserl 1929), Ch. 1 “*Formal Logic as Apophantic Analytics*”, Ch. 2 “*Formal Apophantics, Formal Mathematics*”, Ch. 3 “*Theory of Deductive Systems and Theory of Multiplicities*”. Cp. above all the whole second part of *Formal and Transcendental Logic*: “*Phenomenological Clarification of the Two-Sidedness of Formal Logic as Formal Apophantics and Formal Ontology*.”

axiomatized by the same theory. Husserl says that these fields are “isomorphic” with one another.

It is worth noticing that in a system of operations the “objects” of the field, which are determined solely through the operations, are themselves operations. They are taken as 0-ary operations. One may think, for the sake of clarification, to the fact that in the nowadays current presentation of formal languages, we can choose to treat individual constants as functions letters, namely as symbols for 0-ary operations. In performing this choice, we are essentially fixing whether we see the elements of the field that are denoted by the individual constants as objects or as elementary operations. This is important for the present analysis of the second part of Becker’s *On the Logic of Modalities*, since talk of “sphere of operations”, “which first lead to the individuals that we designate as “numbers””¹⁵ or “of the primacy of the concept function over the concept of thing” is explicit in Cassirer, as well as talk of other different kinds of “definiteness” (“definiteness of extension”, “*Entscheidungsdefinitheit*”, ...) is explicit both in Cassirer and in Weyl.¹⁶

3.2.3 “To Be Grounded in the Essence of Something”

A last point that we think is essential to understand the discussion between Oskar Becker and Ernst Cassirer is to make clear the sense and meaning of Husserl’s often repeated claim that some laws are “grounded in the essence of this or that.”¹⁷

Consider the concept *cardinal number* (*Anzahl*). This concept “determines a field of knowledge.” This means, according to Husserl, that the field of cardinal numbers is constituted by the objects that *fall under* the concept *cardinal number*. The latter are the *infimae species* of the genus *cardinal number*.¹⁸ They are precisely the cardinal numbers for which there is a uniform and univocal principle of construction (i.e. the passage to the successor: $1, 2 = 1 + 1, 3 = 2 + 1, \dots$).¹⁹ The only *relation* that obtains between the objects of this field is the *order relation* ($\leq$). The *connections* or *combinations* “that have . . . a meaning for the objects of this field”²⁰ are the *arithmetical operations* (*elementary operations*: addition,

¹⁵ Cassirer (1929), 430.

¹⁶ Cassirer (1929), 433; Weyl (1921), 53.

¹⁷ Cp. for instance Husserl (1970), 435, Engl. transl. 414, where Husserl speaks of “operations which are grounded in the idea of the cardinal number”, or Husserl (2001), 241, where Husserl writes: “The general logical laws divide in several groups: in *laws which have their roots in the concept of proposition*, in *laws which have their roots in the concept of concept*, in *laws which have their roots in the concept of object*.”

¹⁸ Cp. Husserl (1970), 434, Engl. transl. 414: “If we understand by “cardinal number” the answer to the question “How many?” then the number series is the closed manifold of *particularizations* that are possible in the sphere of the concept *how many*.”

¹⁹ Husserl (1970), 226 ff., Engl. transl. 238 ff.

²⁰ Husserl (1970), 433, Engl. transl. 412.

subtraction, multiplication, division, and *higher operations*: exponentiation, root extraction, logarithm, etc.).

The sense of Husserl's claim that some laws are "grounded in the essence of *cardinal number*" is that they are analytically included in that concept. Thus, for instance, the laws of commutativity and associativity are grounded in the concept of addition by being analytically included in this concept.

3.3 On Cassirer and Brouwer

Let us first consider Cassirer's position about the issues under consideration independently of Becker's criticism. The third volume of *The Philosophy of Symbolic Forms* considers the problem of knowledge, in particular, the problem of exact knowledge which, according to Cassirer, essentially coincides with the problem of the form of exact science.²¹

Chapter 4 is entitled "*The Object of Mathematics*" and contains four important paragraphs, on which Oskar Becker focuses in his critical discussion:

1. *The Formalistic and Intuitionistic Theories of Mathematics.*
2. *The Growth of the Theory of Sets and the Foundational Crisis of Mathematics.*
3. *The Position of the Sign in the Theory of Mathematics.*
4. *The "Ideal Elements" and Their Meaning for the Structure of Mathematics.*

3.3.1 On the Object of Mathematics

Cassirer's investigations tackle three different, connected problems: (i) the problem of mathematical existence, (ii) the problem of nominal and real definitions and (iii) the problem of the extension to the infinite of methods that are reliable in the finite.

The problem of mathematical existence is the one that, as we said above, Hilbert was concerned with around 1900 and Husserl tries to answer in his *Double Lecture* of 1901. Hilbert had posed the question under which conditions the mathematical existence of an object or of a set of objects is assured. One basic condition had to be met, according to him: the theory of the introduced objects ought to be *consistent*.

Husserl had answered the question, in his *Double Lecture*, as follows: each *conservative* expansion of a *definite* structure is a new object that is mathematically existent or, *mutatis mutandis*, each *conservative* expansion of a structure, whose theory is *consistent* and *syntactically complete*, is a mathematically existent object.

²¹ Cp. Cassirer (1929), v ff.

Cassirer traces the problem of the mathematical existence back to Aristotle and sees it as strictly related to the traditional problems of the *real definitions*.²²

As early as Aristotle we find a remark suggesting that he found the essence of geometrical definition not in a mere conceptual explanation but only in an explanation embracing a theorem and proof of existence.

The definition of a mathematical object, say, for instance, of the set of natural numbers, does not necessarily consist, so Cassirer, of a list of conceptual marks that are constitutive of the *content* of the concept, but it may be given by exhibiting a “*modus generandi*”, a principle of construction of the set as well as a proof that the properties ascribed to that object by the definition of its concept are “necessarily contained and postulated in the [...] mode of production” of the object.²³ Such requirement is built on the idea, that Cassirer rightly attributes to Leibniz, that the definition of an object must be bound up with the condition of the *possibility* of that object. For not every combination of conceptual marks and correlated signs is actually possible. “Among the contents of thought there are some which, when one seeks to unite them synthetically, do not determine one another more closely in this synthesis, but instead negate one another.”²⁴ Indeed, even in the natural language it is easy to find combinations that are correct from the point of view of the grammaticality of the expression, but yield a countersense, when considered from the point of view of the meaning of the expressions. The “round square”, the “smallest fraction” or the “smallest velocity” are examples thereof.²⁵ The very same happens with combinations of mathematical signs.²⁶

According to Leibniz, we can avoid contradictions only if we correlate to each simple concept a simple sign and then set up certain universal rules of sign-combinations, which, taken together, make up a symbolic language with its own laws. A contradiction would then inevitably be manifested in the form of the signs themselves.²⁷

In view of Cassirer’s criticism of Brouwer, which we are going to consider below, let us add that as far as mathematical existence is concerned the transition from Leibniz’s rationalism to Kant’s transcendentalism is characterized by the role that

²² Cassirer (1929), 415f.

²³ Cassirer (1929), 418.

²⁴ Cassirer (1929), 417.

²⁵ The examples above are all Leibniz’s examples. Cp. e.g. his *Meditationes de Cognitione, Veritate et Ideis* (Leibniz 1999b), 585–592. Notice that both Husserl and Cassirer are followers of Leibniz. Indeed, both Husserl’s investigations on the symbolic nature of algorithms in the *Philosophy of Arithmetic* (Husserl 1970) and later on in the *Logical Investigations* (Husserl 1901) as well as Cassirer’s philosophy of symbolic forms develop out of Leibniz’s characterization of the *function of signs* in the *Meditations* and other works.

²⁶ Cassirer (1929), 415 f.

²⁷ Cassirer (1929), 415.

intuition plays in mathematical definitions: to define a mathematical concept means, according to Kant, to put it in relation with the pure intuition of time.²⁸

Headless of historical memory, the *Grundlagenstreit* contains all these elements. Formalism takes formal systems to be hypothetical deductive systems, justified only by their inner logical consistency and not by intuition and goes so far as to assume that if an axiom system is (syntactically) consistent then *there is* a “reality” that satisfies these axioms—notice that here a property of *semantic completeness* is, so to say, presupposed.²⁹

Intuitionism, at a variance, is characterized by the primary role attributed to the intuition in the *constitution* of the mathematical objects. “For Brouwer, who represents this view in its sharpest form, numerical thinking stands at the beginning of all thought: the elementary rules of logic have been abstracted from the theory of numbers, from arithmetic.”³⁰ Brouwer, so Cassirer, explicitly claims that both mathematics and logic refer originally to nothing other than finite quantities. They set up rules for such quantities and admit of no processes other than finite. For, as soon as the concept of the infinite comes into play, we enter in a completely new domain, for which the old instruments are inadequate and we are immediately involved in paradoxes and contradictions. “A remedy for these contradictions cannot be expected from the development of new instruments of thought, but only from a critical limitation of the possible objects of thought.”³¹ Cassirer writes:

Here [...] modern mathematics faces a true methodological dilemma. However it decides, something must be sacrificed. If mathematics intends to uphold its old claim to evidence, it would seem to be compelled to return to the primal source of this evidence, to the fundamental intuition of the whole number. But such a return, on the other hand, would seem to call for a heavy intellectual sacrifice which would close up wide and fruitful fields that classical analysis has conquered step by step. The final solution of this conflict among mathematicians is not in sight. But however it should fall out, the mere fact of such a conflict represents an important and fertile problem from the standpoint of pure epistemology; for in the shifting balance between these two views the epistemologist can clearly discern the diverse intellectual forces which have contributed to the building up of modern mathematics and which have determined its present form.

Cassirer is very well aware that the paradoxes at the core of *naïve* set theory all originate by the unconditioned acceptance of an unrestricted *comprehension principle*, which says that for each “property” *there is* the set whose elements are *all and only* the objects that have that property, and that such a set is an object in its full right. He observes: “[...] mathematical thinking placed unsuspecting trust in this mode of object formation: a set seemed to be determined as a unitary and inherently clear object if any criterion was given on the basis of which one could decide whether any thing whatsoever was an element belonging to this set. This

²⁸ Cassirer (1929), 422.

²⁹ As far as first-order formal systems are concerned, the semantic completeness theorem was proved by K. Gödel in 1929 (Gödel 1986).

³⁰ Cassirer (1929), 424.

³¹ Cassirer (1929), 424.

one requirement seemed to define the set and secure its existence as a legitimate mathematical object.”³²

Cassirer is also well aware of the strategies that had been employed until then to avoid the antinomies. One could either restrict the naive comprehension principle and claim that only the definable *subsets of a set* are sets, as axiomatic set-theory does in order to avoid the formation of too big sets (as, for instance, the set of all sets), or structure the universe of sets in a hierarchy of types and allow the *membership* relation to hold only between a set of a given type and a set of the next type, as (simple) type-theory does.

“But even if contradictions can be avoided through such prohibitions, a fundamental question still remains to be dealt with.”³³ For these are “pragmatic prohibitions”. They tell us nothing about the “methodological ground” of the antinomies.³⁴ Such solutions prove their “validity by [their] favorable consequences,” by the avoidance of the antinomies. They tell us *that* the antinomies are avoidable, but not *why* they are avoidable. They just help us “to avoid the manifestation of a certain symptom, but a doubt remains as to whether the disease behind the symptom has been truly diagnosed and cured. And as long as we have no certainty of this, we may expect it to break out in another place”.³⁵

Thus we are led back, once again, to the problem of mathematical existence. In particular, as to infinite totalities, how and by what extent infinite totalities may be considered to be “existent”.

Cassirer believes that is necessary to know the *law through which the totality is determined* or constructed. Such a law “not only embraces an infinite number of possible applications but produces them out of itself”.³⁶ One may think, for instance, of the operation of adding of a unity as the law through which the infinite totality of the positive integers is constructed.

The point is that Cassirer takes what is generated this way to be *not* concrete objects, but *abstract* logical objects. The field generated through the operation of adding a unity is, just as by Husserl, a *field of species*—we can think of a species as something that has instances; so for example the concept *three* is a species because, to say it roughly, all groups of three things fall under it.

It is a pure “process of iteration”³⁷ that is at the origin the construction of the numerical field and makes this possible. Thus positive integers are, according to Cassirer and in complete agreement with Husserl, *particularizations* of the law through which the field or totality of positive integers is constructed. Cassirer writes:³⁸

³² Cassirer (1929), 426.

³³ Cassirer (1929), 427–428.

³⁴ Cassirer (1929), 428.

³⁵ Cassirer (1929), 428.

³⁶ Cassirer (1929), 429.

³⁷ Cassirer (1929), 430.

³⁸ Cassirer (1929), 430.

What is essential [...] in this foundation from an *epistemological* point of view is that with it the *primacy of the concept of function over the concept of thing* is for the first time recognized in its full scope. If mathematics is led back to the “original intuition” of number, this intuition no longer means an intuition of concrete things but is taken as an intuition of a pure *process*. The starting point is a determinate sphere of *operations*; and it is these operations which first lead to the individuals that we designate as “numbers.” The “existence” of these individuals is demonstrated and demonstrable only through the disclosure of a principle, in accordance with which they can be *posited ad infinitum* in accordance with a predetermined rule. It is this mode of positing alone that makes it possible to master them fully in thought, for here our knowledge of the “law” is in the strict sense prior to our knowledge of “what is posited according to it”. It is from the operational sphere of number that the thing-sphere of the countable and the counted first unfolds. Only if it is permeated with this idealistic idea and is understood as an expression of it can modern intuitionism fully develop and maintain its usefulness for the critique of mathematical foundations. The idealism itself, to be sure, must here be understood as a strictly objective idealism: the sphere of mathematical objects must be grounded not in the psychological act of counting but in the pure idea of number.

It is this “*idealistic idea*” that distinguishes, so Cassirer, Weyl’s position in *The Continuum: A Critical Examination of the Foundation of Analysis* (Weyl 1918) from that of Brouwer. The “mathematical activity” Brouwer talks about should be, so Cassirer, “a purely intellectual activity; it does not occur in time but first makes possible a fundamental factor underlying time itself, the factor of ordering.”³⁹

Thus, according to Cassirer, Brouwer’s version of intuitionism does not sharply differentiate between “the purely ideal given”, that is the *principle of the operation*, and “the empirical given”, that is *the individual acts in which natural numbers are given*, which latter do have spatial and temporal determinations and are, therefore, empirical. Brouwer, so Cassirer, “seems to demand that every mathematical statement in which something is “given” be grounded in an individual act of “giving.” But this “threatens to blur the boundaries between the purely ideal given and the empirical given.”⁴⁰ Is Cassirer right? How are things really?

3.3.2 Brouwer’s Intuitionism

Since Oskar Becker’s critique focuses on Cassirer’s *idealistic* interpretation of intuitionism, let us previously ask the question whether Cassirer’s critique hits the nail on his head.

We restrict our analysis to the works of Brouwer explicitly mentioned by Cassirer in *The Object of Mathematics*. We will thus refer in particular to Brouwer’s inaugural lecture at the University of Amsterdam, *Intuitionism and Formalism*, held

³⁹ Cassirer (1929), 431.

⁴⁰ Cassirer (1929), 431.

on October 14, 1912⁴¹ and, to a lesser extent, also to the lecture *On the Significance of the Principle of the Excluded Middle in Mathematics, especially in Function Theory*⁴² given on 21 September 1923 in Marburg an der Lahn, as well as to *On the Foundation of Intuitionistic Mathematics II* appeared in 1926.⁴³ *Intuitionism and Formalism* takes up again many issues that Brouwer had already presented in his doctoral dissertation of 1907, *On the Foundations of Mathematics*,⁴⁴ though *Intuitionism and Formalism* is more focused on *set theory*, as we shall see.

Brouwer characterizes mathematics as the exact part of the mental activity of free creation of an (idealized) human subject, an activity which is founded on the pure intuition of time. This conception originates from a deep reflection on the very nature of reasoning as well as of time: constructive reasoning is an activity that unfolds in time, and the pure intuition of time is an act of the mind. It consist in the becoming aware of time and is the immediate source of continuity and discreteness as well as of the whole of mathematics.⁴⁵

Mathematical objects do not exist independently of the subject, they are *constituted* by the subject, they are mental constructions of the subject. There is no realm of abstract logical objects that are bearers of properties, that wait to be discovered by us and whose system of relations obtains independently of the knowing-subject.⁴⁶

The properties of mathematical objects are determined by the knowing-subject by means of sequences of mental constructions. In Brouwer's view there is no mathematical truth that subsists independently of such mental activity; saying that a proposition is true means that the subject has reached (immediate or mediate) *evidence* for it by means of appropriate mental constructions. This is an inherently *epistemic* conception of truth: the "truth" of a proposition depends essentially on the knowledge of the subject up to a given moment of his epistemic process. Accordingly, saying that a given proposition is *false* means something much stronger than saying that it is *not true*; namely, that at a time *t* of his epistemic process the subject has reached certainty that, independently of the way in which his knowledge will develop subsequently, he will never find evidence for that proposition.

Just like mathematics is not a description of a realm of abstract logical objects, mathematics is also not a language. Language plays in the intuitionistic conception

⁴¹ Cassirer—cp. Cassirer (1929), 425—refers to the version published in the *Bulletin of the American Mathematical Society. A Historical and Critical Review of Mathematical Science* in 1914. This work is contained in vol. 1 of Brouwer's *Collected Works* (Brouwer 1975a). We quote from the latter version, in the original pagination.

⁴² Brouwer (1975c). We quote from this lecture in the original pagination.

⁴³ Brouwer (1975b), 321–340.

⁴⁴ Brouwer (1975d). We quote from the original pagination.

⁴⁵ Brouwer (1975a), 4. Hereto also cp. the parallel passage in Brouwer (1975d), 17, as to the basic intuition of mathematics as the source of continuity.

⁴⁶ Hereto cp. Brouwer (1975a), 5, where Brouwer questions the assumption "of the existence of a world of mathematical objects, a world independent of the thinking individual, obeying the laws of classical logic".

of mathematics a secondary role: it serves to overcome the contingent restrictions and shortcomings of the human memory and to exchange information with other subjects. Also the role of logic is secondary: logic is a part of mathematics and is concerned with the most general schemas of mental constructions.⁴⁷

Very peculiar is the way in which Brouwer characterizes the difference between intuitionism and formalism with respect to the question *where mathematical exactness does exist*: “the intuitionist says: in the human intellect, the formalist says: on paper.”⁴⁸ “On paper” alludes to the fact that the consequences of a finite and *ad hoc* selected number of primitive propositions (axioms), which are chosen as reciprocally consistent and independent from one another, “have mathematical existence only when they have been represented in spoken or written language”.⁴⁹ At a variance, “in the human intellect” means that mathematical existence is founded in the human mind and ultimately, according to Brouwer, in the pure intuition of time.

In *Intuitionism and Formalism* he writes:⁵⁰

[The intuitionism] considers the falling apart of moments of life into qualitatively different parts, to be reunited only while remaining separated by time as the fundamental phenomenon of the human intellect, passing by abstracting from its emotional content into the fundamental phenomenon of mathematical thinking, the intuition of the bare two-oneness. This intuition of two-oneness, the basal intuition of mathematics, creates not only the numbers one and two, but also all finite ordinal numbers, inasmuch as one of the elements of the two-oneness may be thought of as a new two-oneness, which process may be repeated indefinitely; this gives rise still further to the smallest infinite ordinal number ω . Finally this basal intuition of mathematics, in which the connected and the separate, the continuous and the discrete are united, gives rise immediately to the intuition of the linear continuum, i.e., of the “between,” which is not exhaustible by the interposition of new units and which therefore can never be thought of as a mere collection of units.

Let us first notice that the above passage is rather obscure. Brouwer says that “the falling apart of moments of life [...] is [!] the fundamental phenomenon of the human intellect”. Does he rather mean that the *perception* of “the falling apart of moments of life [...] is the fundamental phenomenon of the human intellect”? That this interpretation is in line with Brouwer’s intentions is eventually confirmed

⁴⁷ Brouwer (1975d), 72 ff.

⁴⁸ Brouwer (1975a), 2.

⁴⁹ Brouwer (1975a), 1. Cp. the parallel passage in Brouwer (1975c), 1, where the Dutch mathematician criticizes formalism as follows: “[t]he *formalistic critique* [...] comes in essence to this: the *language accompanying the mathematical mental activity* is subjected to a mathematical examination. To such an examination the laws of theoretical logic present themselves as operators acting on primitive formulas or axioms, and one sets himself the goal of transforming these axioms in such a way that the linguistic effect of the operators mentioned [...] can no longer be disturbed by the appearance of the linguistic figure of a contradiction. We need by no means despair reaching this goal, but nothing of mathematical value will be thus gained: an incorrect theory, even if it cannot be inhibited by any contradiction that would refute it, is nonetheless incorrect, just as a criminal policy is nonetheless criminal even if it cannot be inhibited by any court that would curb it”.

⁵⁰ Brouwer (1975a), 2–3.

by the following excerpt from his doctoral Dissertation of 1907 where “the basic intuition of mathematics (and of every intellectual activity)” is described as the “the substratum [...] of any perception [!] of change”. Is also supported by the interpretation given by Dirk van Dalen in his essay *The Development of Brouwer’s Intuitionism*: “[...] the subject experiences [!] the passing of a present sensation, while the preceding one is retained as a past sensation”.⁵¹

But even if the question concerning the falling-apart-grasp were clarified, one could still wonder whether the human intellect is really capable of intuition. Haven’t we learned from Kant that human understanding is rather the source of discursive, conceptual knowledge, and not of intuition? If we put aside our hesitations and apply a *principle of charity*,⁵² then what Brouwer says here can be rendered as follows: only those objects that can be effectively constructed and ordered via the iteration of a simple rule of construction do have mathematical existence. He explicitly claims that in the construction of these objects “neither the ordinary language nor any symbolic language can have any other role than [...] to enable different individuals to build up the same [!] [object]”.⁵³

But, again, how should it be possible for different individuals to build up *the same* object? Does Brouwer mean that the *sameness* of the object is given by the uniformity of the modes of formation of the concept of that object in the human mind? Unfortunately, he doesn’t say anything to this regard in *Intuitionism and Formalism*.

Brouwer is critical towards the ways traditionally acknowledged for the formation of sets as, for instance, the act of “thinking together objects whatever”⁵⁴ as well as the correlated “relation of a set to its elements”.⁵⁵ He takes this way of sets-constitution to be at the origin of various axioms of set theory that are suggested by the practice with finite sets but are then systematically extended by the formalist to

⁵¹ van Dalen (2000), 120. The above interpretation is even confirmed by the late Brouwer himself in his *Cambridge Lectures on Intuitionism* of 1951 (Brouwer 1981). It is worth quoting the following passage in full: “[I]ntuitionistic mathematics is an essentially languageless activity of the mind which springs from the perception [!] of a motion of time. This perception of time can be described as the splitting of a life moment into two distinct things, one of which gives place to the other, but is preserved in memory. If the dyad thus born is stripped of every quality, what remains is the blank form of the common substratum of all dyads. And it is this common substratum, this common form, which is the basic intuition of mathematics.”

⁵² On the principle of charity cp., among others, Davidson (2005).

⁵³ Brouwer (1975a), 2–3.

⁵⁴ Cp. e.g. Dedekind (1888), 1 f.: “It very frequently happens that different things $a, b, c, \dots$ for some reason can be considered from a common point of view, can be associated in the mind, and we say that they form a system S ”. Also Cantor’s definition of a set goes in this direction: “By a “set” we are to understand any collection into a whole (*Zusammenfassung zu einem Ganzen*) M of definite and separate objects m of our intuition or our thought. These objects are called the “elements” of M ” (Cantor 1895, 481; Engl. transl. 85). One may also compare Husserl’s “origination of the concept of set through reflection on the collective mode of combination” in Husserl (1970), 17–21; Engl. transl. 18–22.

⁵⁵ Brouwer (1975a), 5.

infinite sets. This is, for instance, the case for all *closure axioms* of set theory that permit to generate new sets from old ones.⁵⁶

In this Brouwer is not alone. Also Cassirer sees a problem in the arbitrary combination of “everything with anything” which is at the basis of the intuitive notion of set. “The set”, so Cassirer, “is characterized from the outset by the form of mere aggregation [...]: fundamentally this means that regardless of any consideration [...] of meaning, anything can be united with anything into a conceptual totality.”⁵⁷ But, “if we assume that the realm of the conceivable is governed by any specific laws of meaning whatsoever, these laws must sooner or later set a limit to the arbitrary combination of everything with anything.”⁵⁸

As an example of the consequences generated by the introduction of “meaningless” set-theoretical concepts Brouwer mentions a well known paradox discovered by Cesare Burali-Forti and by him presented at the *Circolo Matematico di Palermo* on 28 March 1897. The paradox had been taken up again in a letter of Cantor to Dedekind dated Halle, 28 July 1899 and is quite simply derivable from the assumptions of naïve set theory: “the set of all ordinals is well-ordered, hence it has an ordinal. This ordinal is at once an element of the set of ordinals and greater than any ordinal in the set.”⁵⁹

Brouwer takes the *axiom of union*⁶⁰ to be responsible for the contradiction and stresses, once more completely in agreement with Cassirer, that “the only modifications that the discovery of paradoxes has brought about in the practice of formalism has been the abolition of those sets that had given rise to these paradoxes”.⁶¹ Quite at the opposite, the “modifications” introduced by the intuitionistic approach are not *ad hoc* but are based on conceptual insight—this is Brouwer’s and Cassirer’s point here. This attitude brings, for instance, to the intuitionistic acceptance of $\aleph_0$ (the cardinality of the set of natural numbers) as well as of the meaningfulness of the *concept* of denumerably infinite ordinal number $(\omega, \omega + 1, \dots, \omega + \omega, \dots)$, but to the rejection of the existence of the set of all denumerably infinite ordinal numbers and more generally of all transfinite cardinal numbers greater than $\aleph_0$.

⁵⁶ Among the principles and axioms extrapolated from the practice with finite sets, Brouwer explicitly mentions the *axiom of extensionality*, according to which a set is completely determined by its elements (so, two sets having the same elements are equal); the *principle* stating that for any two mathematical objects it is decided whether or not one of them belongs to the other; the *power-set axiom* (for every set X there is a set whose elements are all and only the subsets of X), the *axiom of choice* (given a family of non-empty and pairwise disjoint sets, there exists a set which contains exactly one element from each set belonging to the family, and nothing else); the *comprehension axiom*, according to which for any property there is a set containing exactly the objects having that property; the *union axiom* that says that for any family of sets, there is a set that contains exactly the elements of each of the sets belonging to the family. Cp. Brouwer (1975a), 4.

⁵⁷ Cassirer (1929), 426.

⁵⁸ Cassirer (1929), 427.

⁵⁹ van Heijenoort (1967), 104.

⁶⁰ See fn. 56.

⁶¹ Brouwer (1975a), 7.

Of interest to Cassirer is now the possibility of interpreting Brouwer's "original intuition" as a non-empirical intuition, as "an original, fundamental schema"⁶² under which it is possible to subsume the process of iteration that generates the number series. According to Cassirer it is important to establish the universality of the number-concept, even if one claims, as Brouwer does, that the number-concept has an intuition as its source, i.e., something that philosophy generally takes to be as *simple* and *singular*. Cassirer would like to claim, and indeed he does claim at a certain point, that the "original intuition which [the intuitionism] sets up as the source of number is anything but an intuition of empirical objects".⁶³

Within this context and set of problems Cassirer explicitly refers to the second part of Brouwer's work *On the Foundation of Intuitionistic Mathematics*.⁶⁴ Indeed, in both ways that Brouwer allows for creating new sets and that give rise, respectively, to *sets as properties* or *species* and to *sets as laws* or *spreads*, individual objects *are given by a law*, even if we are dealing with two different ways to carry out the genesis of a set. Roughly, *species* correspond to sets in the classical sense or, in Brouwer's words, to "properties supposable for mathematical entities previously acquired" satisfying the principle of extensionality with respect to a given equivalence relation between the mathematical objects in question.

At variance, in a *set as law* or *spread* each element "is constructed in stages: at each stage we choose, in accordance with certain rules, one more piece of the element, thus obtaining a better approximation of it. Specifically: a spread is created (i) by successive choices of natural numbers; each choice depends on the preceding ones, and is made either freely or in accordance with certain restrictions, and (ii) by assigning after each choice an object from a particular preexisting denumerable set. The restrictions in (i) and the correlations in (ii) are given by the *spreadlaw*, in reality a pair of laws, the *choice law* (which also determines whether the process will terminate or be continued, and in the second case there must be at least one permitted choice for the next step) and the *correlation law*. Attempting to describe the nature of the 'laws', Brouwer notes: "... we can say that a spread law yields an *instruction* according to which ...".⁶⁵ According to Cassirer, it is Hermann Weyl who rightly recognizes that such a law actually determines an ideal sphere of operations. In this he takes to consist the so-called "semi-intuitionistic" or "logical-idealistic" or "objective-idealistic" position that he ascribes to Weyl in *The Continuum. A Critical Examination of the Foundation of Analysis* (Weyl 1918).

⁶² Cassirer (1929), 402.

⁶³ Cassirer (1929), 401.

⁶⁴ Cassirer (1929), 401.

⁶⁵ Moschovakis and Vafeiadou (2020), 14–15.

3.4 On Weyl

“It is not the purpose of this work”, so Weyl in *The Continuum*, “to cover the “firm rock” on which the house of analysis is founded with a fake wooden structure of formalism—a structure which can fool the reader and, ultimately, the author into believing that it is the true foundation. Rather, I shall show that this house is to a large degree built on sand. I believe that I can replace this shifting foundation with pillars of enduring strength. They will not, however, support everything which today is generally considered to be securely grounded. I give up the rest, since I see no other possibility.”⁶⁶

Although *The Continuum* is primarily a mathematical treatise, Weyl does not avoid to look into philosophical issues and, as far as the epistemological side of logic is concerned, he explicitly refers to Husserl’s conception in the *Logical Investigations* (Husserl 1901) and in the *Ideas* (Husserl 1913).

Recall that Weyl had moved to Göttingen in 1904, three years after Husserl’s appointment as extraordinary professor at the Faculty of philosophy, and had got his PhD at the Faculty of mathematics under the supervision of David Hilbert. Weyl’s interest in the philosophy of Husserl was partly due to his wife, Helene Joseph (1893–1948), who was a student of Husserl. However, Weyl himself had attended a course on the *phenomenology of time* given by Husserl in Göttingen in the Winter term 1904/1905. Weyl had even presented a summary of Husserl’s *Philosophy of Arithmetic* in his *Habilitations-Vortrag* in Göttingen in 1910, something that shows that Weyl’s interest in Husserl’s philosophy dates much earlier than it is usually supposed and that he was acquainted not only with Husserl’s phenomenology, but also with Husserl’s work before the objectivist turn.⁶⁷

In Sect. 3.2.2 we have tried to provide evidence for our claim that Edmund Husserl is the source of the idea, resumed by Cassirer, that the numerical field is a field of species, that is, of ideal objects, which are the particularizations of an a priori law. On this basis we can also explain the sense and meaning of Weyl’s considerations about existential judgments in the first chapter of *The Continuum*. What there really is, with reference to the field of natural numbers, are “particularizations of the genus-concept (*Besonderungen des kategorialen Wesens*)” *natural number*, which make up a “closed system” of well-determined objects, namely, the *natural numbers*. The latter have an abstract logical existence and constitute “the extension” of the genus-concept in question.⁶⁸

In agreement with Cassirer, we take Weyl to defend, in *The Continuum*, the thesis that the arithmetic of natural numbers supplies us with a domain of species or of last specific differences of the genus *cardinal number*. Weyl actually says that natural numbers are individuals; however, “individuals” has to be interpreted here in the

⁶⁶ Weyl (1918), v.

⁶⁷ Weyl (1910), 111; hereto cp. Majer (1997), 40 ff.

⁶⁸ Weyl (1918), 4.

sense of “last specific differences”. Each individual is exclusively determined by its position in the natural number-series, i.e. it is exclusively determined by the order-relation.⁶⁹ Thus, when Cassirer argues, referring to Weyl, that “the category of natural number together with the original relation it embodies [...] determines the absolute operational field of mathematics” and that “from the process of iteration, of possible serial progression into the infinite, we can derive the fundamental insights regarding natural numbers on which all pure mathematics is logically built”⁷⁰ he is correctly interpreting Weyl, whose parallel passage reads: “[...] we will attain this insight only through our intuition of iteration, i.e., of the infinite repetition of a procedure. But from this intuition we also directly obtain the fundamental arithmetical insights into the natural numbers, on the basis of which the whole *mathesis pura* is logically constructed”.⁷¹

Weyl takes the concept of *natural number* to be a categorial concept, the natural numbers to be ideal objects⁷² and the natural number series with the natural order relation to be an “absolute sphere of operations.”⁷³ The system of natural numbers is the *only totality* that Weyl accepts in *The Continuum*. All other mathematical objects have to be *defined* in terms of natural numbers. Thus, his approach in *The Continuum* can be best characterized as a form of *definitionism*: natural numbers are granted and all other mathematical objects are accepted only if they can be obtained by successive definitions from the natural numbers.⁷⁴ Weyl takes classical logic to hold for the natural number system and also considers sets of natural numbers as mathematical objects but not, of course, the totality of sets of natural numbers.

If one wants to give an idea of the more properly formal aspects of Weyl’s system in *The Continuum*, the reconstruction given by Solomon Feferman in his paper *The Significance of Weyl’s Das Kontinuum*⁷⁵ turns out to be very helpful. Let us start by reviewing the choice of primitive logical operators and symbolic notation. The formal language $\mathcal{L}$ employed is a second-order language with identity, based on an alphabet which contains, besides the usual auxiliary symbols:

1. the usual logical constants (connectives and quantifiers), and the binary predicate of identity ($=$);
2. countably many individual variables: $x, y, z, \dots$ which are intended to vary on natural numbers;
3. countably many unary predicate variables: $X_1, X_2, X_3, \dots$ which are intended to vary on sets of natural numbers;
4. one 0-ary function letter: 0 (zero);

⁶⁹ Weyl (1918), 9.

⁷⁰ Cassirer (1929), 430.

⁷¹ Weyl (1918), 12.

⁷² Weyl (1918), 17.

⁷³ Weyl (1918), 19.

⁷⁴ Cp. Feferman (2000), 181.

⁷⁵ Feferman (2000). Quite recently, a substantially different reconstruction of Weyl’s system has been proposed by Arnon Avron in Avron (2020).

5. one 1-ary function letter: s^1 (successor);
6. other functions letters of arity greater than 0: $f^n, g^m, \dots$, intended to denote specific computable numerical functions;
7. one binary predicate constant: $\in$ (membership).

Individual terms $s, t, \dots$ of $\mathcal{L}$ are generated as usual from individual variables and constants by application of function letters. The atomic formulas are of the form: $t = s$ and $t \in X$. The remaining formulas of $\mathcal{L}$ are built up following the usual recursive procedure as in second-order languages, so in the construction of the formulas we allow for quantification not only of individual variables but also of predicate variables.

The axiom system adopted by Weyl corresponds to second order Peano Arithmetic PA_2 with a *restricted* comprehension schema, which is intended to fix which collections of natural numbers *do exist*, in the sense that they can be recognized as *sets*.

The unrestricted, or *full comprehension axiom schema* (CA) says that for *any* condition expressible in $\mathcal{L}$, that is for *any* formula $A(x)$ of $\mathcal{L}$, there exists the set $\{x \mid A(x)\}$ of all numbers x which satisfy the condition A . The acceptance of (CA) implies that *impredicative definitions* of sets (of numbers) are allowed: if the formula A contains quantifications on second-order variables, then the set $X := \{x \mid A(x)\}$ is impredicatively defined, since it is defined through a characteristic condition which refers to a totality to which X itself belongs. Impredicative definitions are not problematic if one pleads for the thesis that mathematical objects (here, sets of natural numbers) do exist independently of their definition: in this case, an impredicative definition of a set just *describes* through an appropriate condition a mathematical object which already exists.

Impredicative definitions are instead clearly incompatible with a thesis according to which mathematical objects (here, sets of natural numbers) do not exist independently from our definitions, or better are *constituted* by them. According to Weyl, while the set of natural numbers is already given to us, its subsets exist in so far as they can be constituted in the above sense. Accordingly, he accepts a *predicative* version of the comprehension schema CA intended to exclude impredicative definitions of sets of natural numbers.

On the basis of these considerations, let us call a $\mathcal{L}$ -formula A “*arithmetical*” (as it is nowadays usual) if it does not contain quantifications on second-order variables. Then the restricted, *arithmetical comprehension axiom schema* (ACA) postulated by Weyl is:

$$(ACA) \quad \exists X \forall x (x \in X \leftrightarrow A(x)) \quad \text{where } A \text{ is arithmetical.}$$

Thus the postulation of (ACA) in place of (CA) consists, intuitively, in the acceptance of the existence of all and only those sets of natural numbers which can be defined without quantifying over the collection of all sets of natural numbers. This is, according to Cassirer, the peculiar trait of Weyl’s position in *The Continuum*—the “*semi-intuitionistic (halbintuitionistische)* position”, as Becker puts

it.⁷⁶ In Cassirer's terminology (borrowed from previous works of Becker)⁷⁷ Weyl postulates the "*definiteness of extension (Umfangsdefinitheit)*"—which may be seen as corresponding, in the above formal reconstruction, to the acceptance of the axiom schema (ACA)—as the criterion for the existence of sets of natural numbers, whereas Brouwer requires the more restrictive criterion of the "*decidability of extension (Entscheidungsdefinitheit)*". Notice that accepting the "*definiteness of extension*", interpreted as (ACA), as a criterion for set-existence commits to the existence of *undecidable* sets of natural numbers, for an arithmetical formula may contain unbounded numerical quantifiers, and a set defined by such a formula is typically undecidable. Thus, the conceptual jump from Brouwer's "*decidability of extension*" to Weyl's "*definiteness of extension*" is definitely not trivial.

It is worth to quote in full the following passage⁷⁸

In the language of the theory of sets, the basic difference at which we are aiming here can most clearly be designated if we recall the diverse meanings that the concept "definite set" has assumed in the course of the theory's development. At first the concept was taken so broadly that a set was regarded as adequately determined if it could be established with certainty whether or not any random object of thought should be counted among the elements of the set or not. Every totality made up of definite elements in this sense is looked upon as a set. Soon paradoxes made it necessary to drop this unrestricted use of the concept of the set: the definiteness of the elements was now replaced by the definiteness of extension as the crucial requirement. Now not every aggregate defined by reference to an attribute or a law is held to constitute a valid mathematical object; in addition this aggregate must be ideally self-contained; all the elements of the set must be comprised in a circumscribed sphere of things, which may be delimited by a definite constructive principle. Then Brouwer went a step further by admitting only "decisively" definite totalities, so constituted that the question of whether they include elements with a prescribed attribute can always be decided by purely finite processes.

The postulate of "extensive definiteness," but not necessarily that of "decisive definiteness," would correspond to the concept of "constructibility" as we have attempted to formulate it. For "decisive definiteness" requires that the construction be really carried to its conclusion, while "extensive definiteness" is satisfied with the ideal possibility of construction.

The last paragraph of the above passage explains the reason why Cassirer is sympathetic with "Weyl's formulation of intuitionism" and will be discussed in the next section.

⁷⁶ Cassirer (1929), 433 ff. The term "*semi-intuitionistic*" is typically associated, after Heyting (1934), to the position of Borel, Baire and Lebesgue. However, the term was previously employed in Becker (1927), 461 and Becker (1930), 36 (97) to refer to Weyl's position in *The Continuum*. Cp. Hesselning (2003), 8.

⁷⁷ Cp. Becker (1923), 403 ff.; Becker (1927).

⁷⁸ Cassirer (1929), 432–433.

3.5 Oskar Becker on Cassirer

Cassirer's critique of Brouwer grounds on a distinction made by Leibniz in the *fifth* letter to Clarke between "*observabilité*" and "*actual observation*". It is worth depicting the general framework of this discussion.⁷⁹

The Leibniz–Clarke correspondence is a scientific and philosophical debate that had occurred during the years 1715 and 1716 between Leibniz and Samuel Clarke (1675–1729), an English follower of Isaac Newton. The controversial point in the passage to which Cassirer refers is Newton's absolute concept of *space*.⁸⁰ Clarke pleads for an absolute space (*espace réel absolu*) within which things are. Such absolute space is indivisible and has an invariable structure which is not altered by the things within it. We have no direct access to it, since the absolute space is not a substance but "a property of an *incorporeal* substance",⁸¹ that is of God.

According to Leibniz, space is the order of coexistence of bodies and, as such, it is a relation.⁸² Space is no real entity but a mere *ens rationis*.

In the passage Cassirer has in mind, Clarke is arguing that one should distinguish between absolute and relative motion, which latter is merely a change of the order or situation of bodies with respect to each other. "There may be *real Motion*, where is *none relative*; and *relative Motion*, where there is *none real*"—so Clarke.⁸³ According to Leibniz there is no motion other than relative; motion is a change of situation of the bodies among themselves, that is, there is motion when the order of relations among the bodies is changed. In particular, even if motion can be said to be independent from *actual observation* (*being observed*) it cannot be said to be independent from the *observabilité* (*being able to be observed*), i.e., from the possibility of being observed.

Cassirer extrapolates this distinction from its context and applies it to mathematics. He distinguishes between "constructability" or "*possible construction*" and "*actual construction*." An object, or a set of objects, may be said to be "existent" in mathematics, when its existence can be taken to be "provable" or "constructible" even if not "actually proved" or "construed". Cassirer indicates this position as Weyl's position not only in the *The Continuum* but also in later works. He also claims that "constructability" or "provability" in this sense of "possible proof" is tantamount to requiring the "*definiteness of the extension*" of the concept whose existence is in question, but not its "*decisive definiteness*".

Becker replies to these considerations by Cassirer with two different kinds of criticism. The first is a philological clarification contained in a footnote and runs as follows: the acceptance of the arithmetical comprehension schema (ACA) as

⁷⁹ Cp. Becker (1930), 36 (98) and fn. 133.

⁸⁰ Leibniz (1978), 383. 9–13; 403.52.

⁸¹ Leibniz (1978), 383. 8.

⁸² Cp. e.g. Leibniz (1962) II, 13; 17.

⁸³ Leibniz (1978), 384.

existence-criterion represents admittedly Weyl's position *only* in the *The Continuum*. In the paper *The new Foundational Crisis of Mathematics* of 1921⁸⁴ as well as in later works Weyl explicitly adopts Brouwer's version of intuitionism. Thus, Cassirer's claim that Weyl takes the "*definiteness* of the *extension*" as existence-criterion altogether is misleading since "it stems from a flashback in the "*semi-intuitionistic*" position of 1918 that Weyl, a little later in the same text [Weyl 1921], *explicitly* rejects in favor of Brouwer's intuitionism".⁸⁵ According to Becker there are no essential differences between Weyl's position from 1921 onwards and Brouwer's position: "both authors require the *effective* presentation of a construction by the proof of existence, that is, as far as sets are concerned, the "*decidability* [*Entscheidungsdefinitheit*]"'.⁸⁶

The second criticism is more serious and amounts to the objection that the distinction between "provability" and "proof", or between "constructability" and "construction", introduces a modal connotation that makes no sense as far as mathematical proofs are concerned. It is worth quoting Becker's argument at length:⁸⁷

Does the distinction between [...] possibility and reality make sense at all in the field of mathematics? Can the positive *possibility* of a construction or of a proof be shown, in mathematics, in some other way—than exhibiting *de facto* the construction or the proof?

In the realm of empirical observation and historical knowledge there are quite many degrees [*Stufenfolgen*] of more or less probable "*real*" possibilities, up to the absolute *real* impossibility. [...] In the realm of pure mathematics there is nothing similar. If one wishes, one can call a construction or a proof "possible", if one can show that the *assumption* of its provability, or of its existence, are consistent with the other accepted premises. But, with this, it is not at all said, how this "possible" construction or this "possible" proof should look like. While, in the "empirical" case, a fictional observer could perform fictional observations, in the "purely" mathematical case an imagined construction and a "real" construction would be immediately equivalent, they would not be distinguishable.

The concept of *constructability* in the sense of *possibility* of being constructed can be *prima facie* characterized, so Becker, only in a negative way. For how should it be possible to show how a proof is made that does not exist yet? "It seems not to be possible", so Becker here, to give to the concept of *constructability* "a *positive* content besides the mere *privative* one of *non*-contradictoriness."⁸⁸

Oskar Becker is very subtle in using precisely this reflection to elaborate his critique of Cassirer's interpretation of Hilbert's ideal elements.

Indeed, the *provability* of a proposition or, respectively the *constructability* of a theory, the latter conceived as a set of true propositions, comes to coincide with the *certainty* that it is not possible to derive a contradiction from that proposition or, respectively, from that set of propositions. In this way the *positivity* that seems to be

⁸⁴ Weyl (1921), 53.

⁸⁵ Cp. Becker (1930), 36 (98–99) and fn. 135.

⁸⁶ *Ibid.*

⁸⁷ Cp. Becker (1930), 37–38 (99–100).

⁸⁸ Cp. Becker (1930), 38 (100).

inherent in the concept of *possibility* is recast as *actual construction of a proof of non-contradiction*. Becker writes: “It is *essential* for the recent consistency proofs of the Hilbert school that they *merely* prove the *impossibility* of any contradiction, the *impossibility* of the occurrence of the formula $0 \neq 0$, or of any other equivalent inconsistent statement, in a valid formal-mathematical deduction. Hilbert’s method makes nothing more.”⁸⁹ He continues as follows: “Neither is Hilbert’s method able, nor it aims at all to provide “ideal” structures that would complete, in a somehow *formal-ontological suitable* way (say, as set-theoretical constructions) the already introduced “intuitionistic” structures. Therefore, [Hilbert’s method] is fundamentally different from *all* previous methods for introducing “ideal” mathematical objects.”⁹⁰

Let us try to understand Becker’s point here, setting aside, for the moment, from the word “intuitionistic” in the latter quotation. Becker is saying that Hilbert’s method, when it is successful, demonstrates that there is no finite syntactical mathematical object, a “formal proof”, which terminates with an occurrence of the formula $0 \neq 0$, or another contradiction equivalent to it. Thus, he seems to be implying that a *syntactical proof of the consistency of a theory* is *not* tantamount with *the existence of a model for that theory*, as we would say nowadays, or with *the existence of some structure* that satisfies the theory’s axioms. In other words, he seems to be aware of the fact that the “completeness theorem” cannot be taken for granted but has to be proven. This is not that surprising because Gödel’s *completeness theorem* for first-order logic was firstly proved in 1929 (Gödel 1986), even if it was not immediately clear to everyone that such a theorem, together with Gödel’s compactness theorem could be presented as the *model existence theorem* that was later made fully explicit by Leon Henkin in his PhD’s thesis (Henkin 1949).

Now, how does Becker come from this point to his critique of Cassirer’s interpretation of Hilbert’s ideal elements? He writes: “[...] Cassirer tries to single out a *more positive* characterization of the sense and the justification of the addition of ideal elements, beyond the mere non-contradictoriness and thus, he also is eager to connote the concept of “possibility” of a “formalistic” construction in a *positive* way. Cassirer assumes that, for Hilbert, to add ideal (transfinite) propositions to the finitary propositions is justified by the consistency of the whole theory once the addition is made, and by a sort of permanence of the formal laws.”⁹¹

To understand what is at issue, it is important first of all to notice that Cassirer seems to assume that Hilbert’s finitary mathematics can be identified *tout court* with intuitionistic mathematics. This untenable interpretation of the informal concept of *finitary methods*, which is central to Hilbert’s program, was indeed shared by many scholars during the 1920s and up to the beginnings of the 1930s (see e.g. von Neumann 1931); it is only around 1933, after the results of Gödel and Gentzen showing the equiconsistency of intuitionistic and classical formal theories

⁸⁹ *Ibid.*

⁹⁰ *Ibid.*

⁹¹ Becker (1930), 39 (100).

of arithmetic, that it became clear to all that such an identification is impossible in the light of Gödel's second incompleteness theorem.

He seems to believe that the expansion of finitary logic with infinitary tools is, in a way, analogous to the stepwise *conservative* expansion of the field of the positive integers up to the reals. Since it is possible to define the integers as pairs of natural numbers and, consequently, it is possible to transform back each proposition on the integers in a proposition on natural numbers and since it is possible to represent rational numbers as pairs of integers as well as real numbers as suitable pair of sets of rational numbers, it should also be possible to transform back each proposition containing ideal elements in a finitary proposition. However, finitary logic is classical propositional logic and its extension with ideal elements is nothing more than predicate first order logic. Thus, the expansion consists, in this case, in the introduction of the quantifiers. But then, as Becker observes, it is *not* always possible to transform back propositions containing quantifiers into finitary propositions, since the latter may very well contain infinite chains of conjunctions and/or disjunctions or of both. "Cassirer", so Becker, "[...] is wrong [...] as far as the ideal *propositions* are concerned. Judgements upon Hilbert's "ideal propositions" can *in no way* be transformed back [...] into finitary propositions (which hold also for the *intuitionistic* logic). *There is no—even so complex—"collection" of "finitary" propositions that is able to represent a "transfinite" ("ideal") proposition.*"⁹²

Moreover, it is not possible to describe the relation between classical and intuitionistic logic as analogous to the relation that obtains between, say, complex and real numbers, as Cassirer claims. Namely, it is possible to consider real numbers as a special case of complex numbers, but it is not possible to consider intuitionistic logic as a special case of classical logic, for there are properties and relationships that classical logic is not able to account for. For instance, intuitionistic logic considers a certain proposition p , its negation $\neg p$, and its double negation $\neg\neg p$ as three different propositions, while classical logic is just able to account for the difference between p and $\neg p$. Thus, intuitionistic logic seems to be richer than classical logic from a conceptual point of view.

Now, a reader a little versed in logic might ask: why does Becker maintain this position? Is it not the case that intuitionistic logic is a proper subset of classical logic? If classical logic is presented, for instance, in form of an axiomatic synthetic calculus, say as Hilbert-Bernays-style calculus, then intuitionistic logic is obtained from classical logic by dropping one axiom schema, for example, the schema of the excluded middle ($A \vee \neg A$) or the double negation schema ($\neg\neg A \rightarrow A$). So, what is the point of criticizing Cassirer by saying that classical logical has rather to be considered a fragment of intuitionistic logic and not vice versa? To put it with Becker, "the facts (*sit venia verbo!*) are against Cassirer."⁹³ Indeed, even if classical logic, when considered from an *extensional* point of view, is stronger than intuitionistic logic (it singles out more tautologies) and even if the

⁹² *Ibid.*

⁹³ Becker (1930), 41 (102).

formal propositional language employed is apparently the same in both logics, the connectives in intuitionistic logic are *not* interdefinable. This means that classical logic evens out differences that intuitionistic logic, on the contrary, is capable to account for. Becker's view is moreover confirmed by the different translations that permit to see classical logic as a fragment of intuitionistic logic. A first translation was devised by Valery Glivenko in 1929 (Glivenko 1929) and was known to Becker when writing *On the Logic of Modalities*. The translation maps each formula A to its double negation $\neg\neg A$. Glivenko's theorem, which holds *only* for classical propositional logic but *not* for first order logic, states that for any propositional formula A , A is a *classical* tautology *if and only if* $\neg\neg A$ is an *intuitionistic* tautology. Herefrom it follows as a corollary the more general statement:

$$T \vdash_{\text{CPC}} A \iff T^{\neg\neg} \vdash_{\text{IPC}} \neg\neg A$$

In words: a propositional formula A is derivable in the classical propositional calculus **CPC** from a set of propositional formulas T if and only if its double negation $\neg\neg A$ is derivable in the intuitionistic calculus **IPC** from the set of propositional formulas $T^{\neg\neg}$, where $T^{\neg\neg}$ is obtained from T by replacing each formula B in T with its double negation $\neg\neg B$.

A still more important translation is the so called “*negative translation*” that permits to see not only classical propositional logic but also first order logic as a fragment of intuitionistic logic. This translation was introduced in 1933 by Kurt Gödel in a short paper with the title *On Intuitionistic Arithmetic and Number Theory* (Gödel 1933b).⁹⁴ Gödel observes that under a suitable translation “the classical propositional calculus turns out to be a subsystem of the intuitionistic one”.⁹⁵ The negative translation maps each atomic formula to its double negation, each disjunction $(\dots \vee \dots)$ —via De Morgan's law—to $\neg(\neg\dots \wedge \neg\dots)$ and each existential assertion of the form $\exists x A$ into $\neg\forall x\neg A$. Gödel goes on to show that “something similar holds for all of arithmetic and number theory [...]. Here, too, we can give an interpretation of the classical notions in terms of the intuitionistic ones *so that all propositions provable from the classical axioms hold for intuitionism as well*”.⁹⁶

For the purposes of our interpretation of Becker's critical remarks concerning Cassirer's analysis of Hilbert's *ideal elements* is important to take into account Gödel's own judgement on his result: “*the system of intuitionistic arithmetic and number theory is only apparently narrower than the classical one, and in truth contains it, albeit with a somewhat deviant interpretation.*”⁹⁷ This is exactly Becker's point against Cassirer in Becker (1930). He writes: “Hilbert's logic “extended” with

⁹⁴ The “negative translation” is often referred to in the literature as the “Gödel-Gentzen” translation, see Gentzen (1936).

⁹⁵ Gödel (1933b), 287.

⁹⁶ Gödel (1933b), 287–288.

⁹⁷ *Ibid.*

the ideal proportions is nothing else than the “ordinary” logical calculus of Schröder, Peano, Frege and Russell-Whitehead. It is *much simpler* than the “intuitionistic” logic, as recently set up in detail by Heyting. By no way, Hilbert’s system is to Brouwer’s and Heyting’s system as complex algebra is to real algebra! [...] One cannot find the new and much more complex logical relations of intuitionistic logic enclosed as “special cases” in those of classical logic. On the contrary, it is classical logic that can be seen, in a sense, as a “part” of the new logic [...].⁹⁸

⁹⁸ Becker (1930), 41 (102 f.).

Chapter 4

Oskar Becker: *On the Logic of Modalities*



On the Logic of Modalities

[1]

by

Oskar Becker (Freiburg im Breisgau)

Preliminary Remark

The following short essay contains a first attempt to confront with the currently emerging problem of the logic of modalities on the basis of a peculiar method. Such method is determined, on the one side, by the *calculus of logic*, as the now indispensable tool of formal-logical research and, on the other side, by the standpoint of *phenomenology*. The concurrent usage of these methods of research, which are so different as to their essence and to their methodological technique, could appear to be questionable and is, actually, not bare of difficulties. Nevertheless, this seems to be unavoidable, if one does not want to end up in two “polar” unilateralities, namely, the mathematical combination of mostly empty concept-constructs, on the one hand, and the quite “shortsighted” description of obvious, more or less arbitrarily assembled concrete cases, on the other hand, which latter has been called, jokingly, “empiricism of the apriori”.

[Thus] it is perhaps forgivable, if the application of our method is still far from being perfect. One will criticize especially the preponderance of the formal over the phenomenological. Nevertheless, there is reason to hope that the clarification of the formal-mathematical structures will serve to pave the way for phenomenological research.

Unfortunately, even the formal problems of such a quite restricted field as the rank order and the reduction of nested modalities are by no way completely solved. Special care has been taken in singling out most clearly the open problems and in not hiding the still existing gaps in the hope that future research will be able to close them up. |

While the first Part, which comes after a general introduction, shortly dwelling [2] also on historical issues, treats *the problems of the formal rank order and reduction of modalities* by way of a *calculus* and—although just sketchy—interprets it phenomenologically, the second Part is concerned, completely independently from the first one, with the task of throwing new light on the philosophical, i.e. logical-ontological problem of the mathematical “intuitionism” from the standpoint of a non-formal logic of modality. These investigations were inspired by Ernst Cassirer’s recently published discussion of Brouwer’s thoughts to which we will refer continuously.¹

Both these quite independent parts should be seen as first experiments on this still sparsely investigated field. The author will hopefully add others afterwards.

I. On the Rank Order and Reduction of Logical Modalities

Introduction

Although already Aristotle has addressed extensively the logic of modalities (in the last chapters of his *On Interpretation* and in his *Prior Analytics*),² still the attempt of its presentation *in form of a calculus* was made on the turn of the twentieth century. And it was in this context that the idea of *arbitrarily composing and iterating modalities* (reiterated self-application of modalities), which is the object of our investigations in the first part of this essay, made its appearance for the first time.

¹ [Eds.] Becker refers here to the third volume of Cassirer’s *Philosophie der symbolischen Formen*, entitled *Phänomenologie der Erkenntnis*, appeared in Berlin in 1929 (Cassirer 1929); English translation: E. Cassirer, *The Philosophy of Symbolic Forms*, vol. 3, *The Phenomenology of Knowledge* (Cassirer 1957). Responsibility for translations is ours, even when we refer to, benefit from, or simply echo published translations.

All the works explicitly quoted in *On the Logic of Modalities* have been included in the Bibliography of the present volume.

² [Eds.] Cp. *On Interpretation* (Aristotle 2014a), Chs. 12–14 and *Prior Analytics* (Aristotle 2014f) A8–22.

Already in 1906, in his work *Symbolic Logic and its Applications* (London, Longmans),³ as well in a series of journal papers,⁴ Hugh MacColl has developed a logical calculus that takes into account five “truth values”, instead of the usual two: “true” and “false”. These five truth values originate from the combination of the logical “qualities” *affirmation*⁵ and *negation*⁶ with the so-called “modalities” of judgements: “necessity”, “actuality” (= truth) and “possibility” (while the so-called “limitation” that occurs in Kant as third judgment-category⁷ stays unconsidered). The fundamental judgement-predicates, according to MacColl, are:⁸ “(...) is certain” or “(...) is necessary”; “(...) is impossible”; “(...) is true”; “(...) is false”; “(...) is variable” = “(...) is not certain and not impossible”, | the latter being [3] the logical product⁹ of two “possibility”-predicates, which can be rendered, even if not that precisely, by “(...) is possibly true” = “(...) is not impossible” and “(...) is possibly false” = “(...) is not necessary” [“(...) is unnecessary”] or “(...) is uncertain”.

C. I. Lewis, who in his famous work *A Survey on Symbolic Logic* (Berkeley 1918) refers, though very briefly, to H. MacColl,¹⁰ has developed, on his turn, a system of elementary propositional logic¹¹ that introduces, besides the usual negation, the *impossibility* as a “primitive idea”, and then defines possibility and necessity (as well as their opposites) in terms of it.¹²

³ [Eds.] MacColl (1906).

⁴ [Eds.] Becker refers here to a series of eight papers by MacColl on *Symbolical Reasoning*. Cp. in particular: MacColl (1880), MacColl (1897), and MacColl (1900).

⁵ [Eds.] Our emphasis.

⁶ [Eds.] Our emphasis.

⁷ [Eds.] Cp. Kant (1998), *Critique of Pure Reason* A80/B106.

⁸ [Eds.] For the sake of clarity we adopt henceforth Frege’s way of writing predicates, that has the advantage of making clear the unsaturatedness of the predicate.

⁹ As usually we denote by “logical product” the *conjunctive* combination (“and”) of two (or more) judgements. The fact that [the “logical product”] is recently also be taken to stand for the *disjunctive* combination (“or”) seems to us, in spite of certain, for our part, neglectable formal advantages, to be a not really useful alteration of a linguistic practice that relies, after all, on a relevant tradition and diffusion.

¹⁰ Cp. Lewis (1918), 108, 292. [Eds.] Lewis refers at this passage to MacColl (1906). Starting from the first chapter H. MacColl introduces here his logical operators step-by-step on the basis of examples.

¹¹ Cp. Lewis (1918), Ch. V, 291–339. An amendment hereto [is to be found] in the *Journal of Philosophy, Psychology and Scientific Methods*, now enclosed in Lewis and Langford (1932). [Eds.] O. Becker refers here to the fact that the system for modal propositional logic presented in Lewis (1918) collapses into classical logic, as Emil L. Post had pointed out and Lewis had acknowledged in his paper *Strict Implication: an Emendation* published in *The Journal of Philosophy, Psychology and Scientific Methods* in 1920 (Lewis 1920). The amended system will appear in 1932 in Lewis and Langford (1932), where Lewis displays his modal systems **S1** to **S5**.

¹² [Eds.] Indeed, Lewis (1918) takes the operator “it is impossible that” as primitive, instead of the negation and one of the modal operators $\Box$ or $\Diamond$ (“it is necessary that” or, “it is possible that”), as is nowadays standard.

While MacColl, in harmony with the natural language and at a variance with the usual systems for propositional logic, introduced, for reasons of principles, five modalities, C. I. Lewis was guided by a specific purpose when setting up his “*system of strict implication*”; namely, the avoidance of certain “*peculiar* (i.e. paradoxical) *propositions*”, which inevitably occur in classical propositional logic. As typical examples can be taken the paradox-seeming sentences: “A true proposition is implied by any proposition”, “A false proposition implies any proposition”. This and similar sentences are unavoidable, if one works with the usual [Philonian, Fregean] concept of implication. That they appear to be paradoxical depends on the fact that the idea expressed in the natural language by “if ... then ...” does *not* coincide with the [Philonian, Fregean] concept of implication. Lewis strives to find a new concept of implication, even though related to the old one. He calls it “*strict implication*” and [he takes it to be] much closer [than the usual one] to the thought expressed by “if ... then ...”. To define it, he basically needs just to add the new *negative* concept of *impossibility*.¹³ While, namely, the “classical implication” between p and q —in symbols: “ $p \subset q$ ” if one follows Lewis (“ $p \supset q$ ” if one follows Russell | and “ $p \rightarrow q$ ” if one follows Hilbert)—can be defined as the *negation* of the logical product of p and the negation of q , in symbols: [4]

$$p \subset q = -(p \times -q)$$

“ p strictly implies q ”, in symbols “ $p < q$ ”,¹⁴ means that the logical product of p and the negation of q is *impossible*:¹⁵

$$p < q = \sim (p \times -q)$$

(thus, “ $\sim$ ” is the symbol for the *impossibility*, while “ $-$ ” is the symbol for the simple negation). Shortly: “ p implies q ” means in the *classical* sense:

“‘ p and not- q ’ is *false*”, while, in the “*strict*” sense, it means: “‘ p and not- q ’ is *impossible*, that is, [‘ p and not- q ’] is *necessarily false*”. This is connected with the fact that the definition of the strict implication emphasizes—in a way that we cannot expose here more in detail—the *necessity* of the [implicative] combination, which emerges also in the popular idea of “necessary consequence”.

The aim of the present essay is now closely related to both MacColl’s and Lewis’s investigations. The ultimate goal of our investigations is to develop an elementary logical calculus that takes adequately into account the *modalities* of the statement, namely in such a way *that the so-called elementary decision problem is solvable*, as

¹³ [Eds.] Our emphasis. We borrow here from Frege the usage of indicating concepts in italics.

¹⁴ [Eds.] Lewis’s symbol “ $\supset$ ” is typeset as “ $<$ ” in Becker’s *On the Logic of Modalities*.

¹⁵ [Eds.] Notice Becker’s constant oscillation between the predicate “(...) is *impossible*”, which denotes the property of being impossible and the operator “*it is impossible that* (...)”, which takes a sentence as input and gives a modalized sentence as output.

in the ordinary propositional calculus.¹⁶ The particular interest of such a calculus—regardless of the fact that the problem of modalities has become, as many claim, one of the next big “logical” problems at all—lies in the following: Brouwer’s approach to a pure finite logic, based solely on the *clarity of evidence* (*Klarheitsevidenz*, Husserl),¹⁷ can probably be developed, from a formal calculatory point of view, as a closed system in the form of a logic of modalities.

As H. MacColl has already noticed in one of his first works on the subject,¹⁸ the following predicates | of a statement can be treated as modalities: “(...) is known to be true”, “(...) is known to be false” and “(...) is neither known to be true nor known to be false”.¹⁹ These can be rendered, with respect to Brouwer’s primitive logical concepts, by: “(...) is provably true” (i.e., “(...) is provable” or, “(...) is constructively producible”); “(...) is provably false” (i.e., “(...) is confutable” or, “(...) is impossible to prove”); and “(...) is undecided” (in the sense that neither a proof of existence nor a proof of impossibility can be given or, which is the same, that [the statement] is neither a “tautology” nor a contradiction.)²⁰

It is obvious that the “*tertium non datur*” does not hold here, in the sense that besides the provably true and the provably false, or, in other words, besides the

[5]

¹⁶ To gain insight into the elementary *decision problem* cp. e.g. H. Behmann, *Mathematics and Logic* (*Mathematik und Logik*), (Mathemat.-physik. Bibl. Bd. 71), Leipzig-Berlin 1927, 14 f. (Behmann 1927); H. Weyl 1928 [*Philosophie der Mathematik und der Naturwissenschaften*] in the *Handbook of Philosophy* (*Handbuch der Philosophie*), Section II A, 13 ff. (Weyl 1927); Hilbert-Ackermann, *Outlines of Mathematical Logic* (*Grundzüge der mathematischen Logik*), (Berlin 1928), §§4, 7–9 (Hilbert and Ackermann 1928).

¹⁷ Cp. *Formal and Transcendental Logic*, *Yearbook for Philosophy and Phenomenological Research*, vol. X, 1 ff.; in particular 49 ff. [o.p.n.] (Husserl 1929).

¹⁸ H. MacColl, *Symbolic Reasoning*, II, *Mind*, New Series 6 (1897), 493–510; in particular: 509 (MacColl 1897).

¹⁹ [Eds.] What Becker ascribes here to MacColl is not at all a peculiarity of MacColl alone, but, rather, a *leitmotiv* in the history of modal logic. The Schoolmen recognized many kinds of modalities. Ockham, for instance, considered in his *Summa logicae* (c. 1323, in Ockham 1967) a plurality of modalities: “There are [...] more modes, which make propositions modal than [the alethic ones]. For just as one proposition is necessary, another impossible, another possible, another contingent, so one proposition is true, another false, another known, another unknown; another spoken, another written, another conceived, another believed, another opined, another doubted, and so on. [...] And therefore, just as a proposition is called “modal” in which the mode “possible” or “necessary” or “contingent” or “impossible” [...] occurs, so a proposition in which one of the previous modes occurs can equally reasonably be called “modal”. Ockham, *Summa logicae*, Book II, *On Propositions*, Ch. 1, *On the Classification of Propositions in General*. Also cp. the following quotation from the essay “*De propositionibus modalibus* (*On Modal Propositions*)” that is often attributed to Thomas Aquinas, even if it has been actually written by a later author: “The modes which determine the proposition are six: “true”, “false”, “necessary” “possible”, “impossible” and “contingent” (Modi autem qui compositionem determinant, sunt sex, scilicet: verum, falsum, necessarium, impossibile, possibile, contingens).” Cp. Thomas of Aquin, *Opera Omnia* (Thomas of Aquin 1980), Suppl. VI: *Reportationes*, *Opuscula dubiae authenticitatis*, 579–580.

²⁰ For a logical-phenomenological interpretation of this kind of “indeterminacy” see Oskar Becker’s *Mathematical Existence* (Becker 1927) (*Yearbook for Philosophy and Phenomenological Research* vol. VIII, 441 ff., in particular: 496 ff. and 775 ff. [o.p.n.]).

certain (the necessarily true) and the impossible (hereto, recall MacColl's predicates "(...) is certain" and "(...) is impossible"), no third is given. Moreover, this "third", which is at once neither necessary nor impossible, can first be analyzed into its two components (the "uncertain" = the "possibly false" and the "not impossible" = the "possibly true"),²¹ and thus can be characterized in a mere *privative* way—solely by the lack of necessity or, respectively, of impossibility (which latter can also be seen as "contradictoriness (*Widersprüchlichkeit*)", i.e. as *positive* predicate). In this privative characterization lies the possibility to scale the indeterminacy present here in different levels that correspond to the phenomenological *differences in evidence* (*Zugangsmöglichkeiten*). [For instance] there can be immediate evidence solely for the intuitive fulfillment of the meaning-intention,²² or for its intuitive frustration, or else there can be lack of immediate evidence or, more exactly, immediate evidence of that lack. But even the ultimate evidence of the lack itself can lack primary evidence, which is acknowledged through a third level evidence, and so forth *in indefinitum*. All these infinitely many cases are contained in the above "indeterminacy".²³

Now, also in the calculus of modalities, there can be found some parallels with the formal intuitionistic logic, as is outlined by Brouwer himself and by R. Wavre,²⁴ and, quite recently, systematically exposed by A. Heyting.²⁵ Especially | Heyting's presentation would require a more precise comparison with the foundations, the method and the results of Lewis's system. However, some peculiar difficulties arise

[6]

²¹ [Eds.] Becker writes, at this passage, "impossible" instead of "not impossible", but it is clearly a lapse.

²² [Eds.] Cp. Husserl (Husserl 1901), *LU* VI, 28: "[I]n fulfillment, the object is given intuitively in the same way in which the mere meaning means it".

²³ Cp. Oskar Becker, *Mathematische Existenz* 775 ff. [o.p.n] (Becker 1927). [Eds.] For ease of comparison, we quote the parallel text in *Mathematische Existenz*: "It should first be observed that intuitionistic logic does not distinguish between "true" and "false", but rather between "true" and "absurdity". "Being true" means "being actually (or, constructively) provable", while "to be an absurdity" means "to be demonstrably contradictory". In basic phenomenological terminology one can say: "true" = "there is a "synthesis" of fulfillment of the judgement-intention" or, "the coincidence between the intended and the intuited [meaning is realized]; "absurdity" = "there is a "synthesis" of frustration of the judgement intention" or, [there is] a "synthesis" of conflict (*Widerstreit*), which always presupposes a partial synthesis. It is evident that there is here no complete disjunction, in the sense of the *tertium non datur*, between fulfillment and frustration, or between coincidence and conflict. If we define "false" as "not true", this possibility, which represents the third, corresponds to the "lack of coincidence", which is to be distinguished both from the coincidence and from (*positively* evident) conflict, as well as the "lack of fulfillment" is to be distinguished both from the fulfillment and from the frustration."

²⁴ [Eds.] Rolin-Louis Wavre (1896–1949) was a Swiss mathematician, logician and philosopher of mathematics. He was a follower of Brouwer's intuitionism.

²⁵ Cp. Brouwer, *Jahresbericht d. D. Math. Vereinigung* 33, 251 ff. (Brouwer 1925a); *Journal für reine u. angew. Math.* Bd. 154, 1 ff. (Brouwer 1925b)—R. Wavre, *Revue de Métaphysique et de Morale* 33, 65 ff. (Wavre 1926); A. Heyting, *Berliner Berichte, math.-phys. Kl.*, 1930 II: "*Die formalen Regeln der intuitionistischen Logik*" (Heyting 1930).

in this case. Thereon it will be briefly reported in the appendix.

§1.

The “System of strict implication” of C. I. Lewis.

I. Undefined Concepts:²⁶

1. *Propositions*: p, q, r , etc.
2. *Negation*: $\neg p$; i.e. “ p is false”.
3. *Impossibility*: $\sim p$; i.e. “ p is impossible”.
4. *Logical Product*: pq or $p \times q$; i.e. “ p and q are both true” or “ p is true and q is true”.
5. *Equivalence*: $p = q$, at the same time the defining relation.²⁷

II. Simplest Definitions:

A. Modalities of a Proposition:

Lewis enumerates five of them:

- (1) p : “ p is true”
- (2) $\neg p$: “ p is false”
- (3) $\sim p$: “ p is impossible”
- (4) $\neg \sim p$: “It is false that p is impossible”, i.e. “ p is possible”
- (5) $\sim \neg p$: “It is impossible that p is false”, i.e. “ p is necessarily true”

Lewis claims that also the next more complicated modalities in his symbolical system are irreducible:

- (6) $\neg \sim \neg p$: “ p is not necessary”, i.e. “ p is possibly false” (this is the negative counterpart to (5)).
- (7) $\sim \neg \sim p$: “ p is impossibly possible” (*not* [to be read as] “ p is impossibly true”).
- (8) $\neg \sim \neg \sim p$: “ p is not impossibly possible”, i.e. “ p is possibly possible”.

...“and so forth” (where, however, the law underlying the progression is not clear.)²⁸

The more complicated modalities are not treated further by Lewis. Notice that the *iterations of the impossibility*: |

[7]

$$\sim \sim p, \sim \sim \sim p, \dots, (\sim)^n p$$

²⁶ [Eds.] The undefined concepts correspond to the primitive symbols of the formal language under consideration.

²⁷ [Eds.] Here to see the Appendix, Sect. 5.1.

²⁸ [Eds.] Becker alludes here to the question, that it is not clear to him, and also to Lewis, whether the number of irreducible modalities in **S3** is finite or not. He seems to lean more towards the second option. Parry (1939) proved instead that **S3** has exactly 42 irreducible modalities.

are not mentioned.²⁹ There is a gap here in the “*system of strict implication*”. We will see that this gap can be filled by adding a new axiom (or, respectively, several new axioms) and that, this being done, only the six modalities (1) to (6) turn out to be irreducible.³⁰

B. Simple Relations between Propositions:

Definitions:

1.01 *Compatibility (Consistency)*

$$p \circ q = \sim (pq) \text{ Def.}$$

1.02 *Strict Implication*

$$p < q = \sim (p - q) \text{ Def.}$$

1.03 *Material Implication*

$$p \subset q = \sim (p - q) \text{ Def.}^{31}$$

1.04 *Strict Logical Sum*

$$p \wedge q = \sim (-p - q) \text{ Def.}^{32}$$

1.05 *Material Logical Sum*

$$p + q = \sim (-p - q) \text{ Def.}$$

1.06 *Strict Equivalence*

$$(p = q) = (p < q)(q < p) \text{ Def.}$$

1.07 *Material Equivalence*

$$(p \equiv q) = (p \subset q)(q \subset p)$$

There hold the following simple relationships between the “strict” and the “material” relations:

$p < q = \sim (p \circ \sim q)$ $p \wedge q = \sim (\sim p \circ \sim q)$ $(p = q) = \sim (p \circ \sim q) \times \sim (q \circ \sim p)$	$p \subset q = \sim (p - q)$ $p + q = \sim (\sim p - q)$ $(p \equiv q) = \sim (p - q) \times \sim (q - p)$
--	--

To the “compatibility”, “ $\circ$ ”, on the side of the “strict” relations, there corresponds the logical product on the side of the “usual” relations. Where classical logic has a conjunction between two propositions, the logic of “strict” implication has just their “compatibility”, that is, the not-impossibility (i.e., the possibility) of their conjunction. While the usual relations express only the factual implication, disjunction, equivalence . . . , the “strict” relations provide the *necessary* [counterparts thereof, i.e., the *necessary*] implicative, disjunctive, equivalent . . . logical connection. |

III. Axioms (Propositions that do not need Proof):

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²⁹ [Eds.] See the previous footnote. Lewis does not raise at all the issue of *how many non-equivalent nestings of modal operators* there are. It is Becker who poses and investigates this problem.

³⁰ [Eds.] Becker refers here to the extended system designated in this volume as **S3'**.

³¹ This relation is denoted by Russell with “ $\supset$ ”, by Hilbert with “ $\rightarrow$ ”.

³² [Eds.] Caution! The symbol “ $\wedge$ ” used by Lewis and Becker to denote the strict disjunction (“ $p \wedge q$ ” reads “necessarily, p or q ”) is nowadays currently used to denote the Boolean conjunction.

1.1	$pq < qp$	1.5	$p < -(-p)$
1.2	$qp < p$	1.6	$(p < q)(q < r) < (p < r)$
1.3	$p < pp$	1.7	$\sim p < -p$
1.4	$p(qr) < q(pr)$	1.8 *	$(p < q) < (\sim q < \sim p)^{33}$

In place of the axiom (1.8*) Lewis originally postulated a similar axiom 1.8 containing the sign of strict equivalence ($=$). 1.8 was equivalent to $(1.8^*) \times (1.8^{**})$, where (1.8^{**}) is

$$1.8^{**} \quad (\sim q < \sim p) < (p < q)$$

and, thus:

$$(1.8) \quad p < q = \sim q < \sim p.$$

In the above mentioned “emendation” axiom (1.8^{**}) was then dropped, i.e. only (1.8^*) was required, which appeared, in the original version, as Theorem 2 and was easily derivable from 1.8. E. L. Post showed that 1.8^{**} (2.21 in Lewis) leads to prove

$$\sim p = -p,$$

whereby Lewis’s system comes to coincide with the usual (Russellian) logical calculus. (Cp. the mentioned “emendation”).³⁴

However, that 1.8^{**} ($= 2.21$) as well as its consequences are de facto unsound can also be shown by means of suitable counterexamples, namely as soon as one sharply distinguishes between *necessity* and merely *contingent truth*, or between *impossibility* and *contingent falsity*.

Example One may think of a sequence of numbers generated by drawing arbitrarily many times, one after another, numbered balls from an urn, whereby, each time, the

³³ [Eds.] After the careful presentation of the Primitive notions (I), of the Definitions (II) and of the Axioms (III) of Lewis’s system, one would expect the specification of the (primitive) Inference rules of the system. Oddly enough, Becker forgets to provide this essential information to the reader. Concerning this point, see the Appendix, Sect. 5.1.

³⁴ [Eds.] Lewis’s two-pages *Emendation* (already mentioned in fn. 11) appears in the *The Journal of Philosophy, Psychology and Scientific Methods* on May 20, 1920 (Lewis 1920). It begins with the following words: “Mr. E. L. Post of the department of mathematics in Columbia University, calls my attention to an error in the development of the system of “Strict Implication,” as presented in Chapter V. of *A Survey of Symbolic Logic*. The postulate 1.8. $(p < q) = (\sim q < \sim p)$, is equivalent to the pair, 2.2. $(p < q) < (\sim q < \sim p)$ [and] 2.21. $(\sim q < \sim p) < (p < q)$. Of these, 2.21, “If ‘q is impossible’ implies ‘p is impossible,’ then p implies q” is false. It is consistent with the other principles assumed, but is incompatible with the intended meaning of the primitive idea of the “impossibility,” and with the distinction of this from the idea of simple falsity.” (Lewis 1920, 300). For further details see the Appendix, Sect. 5.1.

drawn ball is put back in the urn, before drawing the next ball. *It is unknown* how many balls are in the urn and how they are numbered.

Let “ p ” stand for: “The number 17 appears among the first hundred places of the sequence.” Let “ q ” stand for: “The number 17 appears among the first 200 places of the sequence.”

Clearly

$$p < q$$

does hold.

Furthermore, it is also the case that:

$$(p < q) < (\sim q < \sim p).$$

This means (rendered in a free manner): If $p < q$ holds, as shown, and if $\sim q$ holds, then $\sim p$ holds. That is: “If it is *impossible* that 17 | appears among the first 200 places of the sequence, then it is also *impossible* that 17 appears among the first 100 places.” [9]

However, the converse

$$(\sim q < \sim p) < (p < q)$$

does not hold, as our example—suitably modified—does show. Namely, let now “ q ” and “ p ” stand for:

q : “19 appears among the first 100 places.”

p : “19 appears among the first 200 places.”

Then

$$\sim q < \sim p$$

does hold.

Actually, the *impossibility* that 19 occurs in the sequence (at a variance with its contingent, i.e., *incidental* not-occurring) can be due exclusively to the fact that no balls with the number 19 are contained in the urn. This impossibility holds for all places of the sequence, if it holds for some.

However, from $\sim q < \sim p$ it does *in no way* follow $p < q$, i.e. the implication: “If 19 appears among the first 200 places, then it necessarily appears within the first 100 places”, since this sentence is trivially false.³⁵

³⁵ [Eds.] Indeed, the number 19 might appear after the first 100 places.

Similarly, in Lewis's system, it is possible to prove the following implications, but *not their converses*:³⁶

$$(p < q) < (-\sim p < -\sim q)$$

$$(p < q) < (\sim -p < \sim -q)$$

That is:³⁷ If $p < q$ does hold, then it also holds: "The necessity of p implies the necessity of q " and "the possibility of p implies the possibility of q "—but not conversely.

In the first case, an example for the invalidity of the converse would be the following: *all* balls in our urn are numbered with 19, " p " stands for "19 occurs at the 75th place", and " q " stands for "19 occurs at the 77th place". It then surely holds that if 19 *necessarily* occurs at some place, it—necessarily—occurs at all places. But from this, it does not follow in any way that if—by a random numbering of the balls—the number 19 occurs *incidentally* at the 75th place, it *must* occur *incidentally* at the 77th place, too.³⁸

As to the invalidity of the converse, in the second case, it is easy to find examples in everyday life. Consider the following reasoning: "If I catch the train in Freiburg, then it is *possible* (but not certain!) that I catch the connecting train in Basel." |

Then it holds: "If I can *possibly* catch the train in Freiburg, then I *might* catch the train in Basel" (the first possibility opens the second one). But this does not imply: "If I *really* get the first train, then I *really* get the second."³⁹ Also in the case of an *ontologically* intended possibility ([“ontologically” to be understood] in the Aristotelian sense), [i.e.,] in the case of a “potentiality”, it does *not* hold: "If the potentiality of Q lies in the potentiality of P , then the actuality of Q lies in the actuality of P ." For instance, the next generation lies potentially in the actuality of the present generation, and in the former lies already, although only potentially, the

[10]

³⁶ [Eds.] Becker writes $(p < q) < (-\sim p < -\sim p)$ and $(p < q) < (\sim -p < \sim -p)$, in this order. But it is clearly a lapse. In the notation adopted in this volume, the two intended implications amount to $(p \rightarrow q) \rightarrow (\Diamond p \rightarrow \Diamond q)$ and, respectively, $(p \rightarrow q) \rightarrow (\Box p \rightarrow \Box q)$.

³⁷ [Eds.] Notice that here Becker puts into words the two formulas in inverse order.

³⁸ [Eds.] The counterexample is correct, although the presentation, by starting with "all balls in our urn are numbered with 19", may be misleading. The point is the following: $\sim -p < \sim -q$ or, in our notation: $(\Box p \rightarrow \Box q)$, is true. Indeed, suppose that $\sim -p$ (i.e., $\Box p$) is true, that is: it is *necessary* that 19 occurs at the 75th place. This *necessity* can be founded exclusively in that *all* balls in the urn are numbered with 19; so it is also *necessary* that 19 occurs at the 77th place, and thus $\sim -q$ (i.e., $\Box q$) is true. On the other side, $(p < q)$ or, in our notation $(p \rightarrow q)$, is trivially false.

³⁹ [Eds.] A counterexample built on Becker's "balls model" might be the following: Let " p " stand for "7 is picked in the first draw" and " q " for "7 is picked the second draw." Then $\sim -p < \sim -q$, in our notation: $(\Diamond p \rightarrow \Diamond q)$, is true: if it is possible to pick a 7 in the first draw, then at least one ball in the urn must be numbered with 7, and so, given that drawn balls are replaced in the urn before the next draw, it must be also possible to pick a 7 in the second draw. On the other side, $(p < q)$ or, in our notation $\Box(p \rightarrow q)$, is trivially false.

after next generation. But this does not imply that if the second generation is actual, also the third one is already actual.⁴⁰

It is furthermore worth to be noticed that also from the [strict] implication of two possibilities neither the implication of the corresponding necessities follows, nor vice versa.⁴¹ By contrast, [from the strict implication of two possibilities] it does follow the inverse implication of the corresponding impossibilities and, respectively, in the case of the necessities, [it follows the inverse implication] of the [corresponding] “unnecessities”.⁴²

As to the independence of the first two implications above, one can create his own examples according to the following method—that would have been applicable also in the previous cases.

Let an urn contain two balls, each of which is either white (*w*) or black (*b*). First, *one* ball is drawn and *not* replaced in the urn, next, the *other* ball [is drawn and *not* replaced in the urn]. There are three possible ball-distributions in the urn (*ww*, *wb* = *bw*, *bb*) and four draw results (*ww*, *wb*, *bw*, *bb*) are possible.

Let us first show the following:⁴³

From: *p* is necessary < *q* is necessary

it does *not* follow: *p* is possible < *q* is possible.

Let “*p*” and “*q*” stand for:

p : *w* is picked in the first draw.

q : *w* is picked in the second draw.

Then it holds: if *p* is necessary, then also *q* is necessary. For, *p* is *necessary* if and only if the ball distribution is (*ww*); but then also *q* is necessary. However, it does *not* hold that if *p* is *possible*, | then also *q* is *possible*. For, if the ball distribution is (*wb*) = (*bw*), then it may well be the case, under certain circumstances, that *p* [11]

⁴⁰ This is interesting from a historical point of view since Aristotle himself (Metaph. Θ 4; 1047 b, 26–30) derives from the implication of [two] potentialities the implication of [two] actualities. Unfortunately, his explanation is not clear to me (b 28–30: τὸ γὰρ δυνατόν εἶναι ἐξ ἀνάγκης τὸ B εἶναι, εἰ τὸ A δυνατόν, τοῦτο σημαίνει, ἔάν ᾗ τὸ A καὶ ὅτε καὶ ὡς ᾗ δυνατόν εἶναι, καὶ ἐκείνο τότε καὶ οὕτως εἶναι ἀναγκαῖον). It may be that the text is not correct.

[Eds.] Aristotle’s text reads as follows: “For, to say that B must be possible, if A is possible, means that if A is real at the time when, and in the way in which, it was assumed that it was possible for it to be real, then B must be real at that time and in that way.”

⁴¹ [Eds.] Put into our symbolism, Becker is saying here that the implications ($\Diamond p \rightarrow \Diamond q$) and $\Box p \rightarrow \Box q$ are *independent*, in the sense that (i) $(\Diamond p \rightarrow \Diamond q) \rightarrow (\Box p \rightarrow \Box q)$ and (ii) $(\Box p \rightarrow \Box q) \rightarrow (\Diamond p \rightarrow \Diamond q)$ are not valid.

⁴² [Eds.] Put into our symbolism, Becker is saying here that the implications: (i) $(\Diamond p \rightarrow \Diamond q) \rightarrow (\neg \Diamond q \rightarrow \neg \Diamond p)$ and (ii) $(\Box p \rightarrow \Box q) \rightarrow (\neg \Box q \rightarrow \neg \Box p)$ are valid.

⁴³ [Eds.] Put into our symbolism, what Becker is going to show in the following reads as follows: $(\Box p \rightarrow \Box q) \not\rightarrow (\Diamond p \rightarrow \Diamond q)$.

is possible and q is impossible; namely, if w is picked in the first draw. For, in this case, it is clearly *possible* that w is picked in the first draw, (since this is *de facto true*),⁴⁴ but it is impossible that w is picked in the second draw, since there is no white ball left in the urn.

Let us now show that from “ p is possible $<$ q is possible” it does not follow “ p is necessary $<$ q is necessary”.

Let here “ p ” and “ q ”—under the same background assumptions—stand for:

p : w is picked in the second draw.

q : w is picked in the first draw.

Then it holds—with *any* ball distribution!—: “If p is possible, then q is possible”, i.e. “if w *can* be picked up in the second draw, i.e., if there is a w in the urn at all, it *could* have picked up also in the first draw.” But, it is by no means always true: “If p is necessary, then q is necessary.” Indeed, if the ball distribution is (wb) and b is picked up in the first draw, then p is necessary (w occurs necessarily in the second draw),⁴⁵ but q is false.

§2.

Formal Completion

of Lewis’s System into a Closed

System with Six Irreducible Modalities

Before going into the actual meaning of the reduction problem of the infinitely many nested modalities, which arise through the iteration and composition of the symbols “ $\sim$ ” and “ $-$ ”, we present a purely formal investigation, by which Lewis’s system becomes a closed system, thanks to the addition of a further axiom. This can be done in several ways. Here we will consider two of them.

The assumptions introduced by Lewis are (apparently) not sufficient to obtain a closed system of irreducible modalities. |

Therefore we add to Lewis’s axioms the new axiom 1.9:⁴⁶

[12]

⁴⁴ [Eds.] Intuitively, it is clear that p implies $\Diamond p$. Actually, $p \rightarrow \Diamond p$ is a theorem of **S3**.

⁴⁵ The difference between “ p is contingently true” and “ p is necessary” *cannot* be shown in the example used. This would require *three* balls, (wwb) or (wbb) , and a black one had to be picked up in the first draw. Then, *after* the first draw, there would still be (ww) or (wb) in the urn and, if w would be picked up in the second draw, p would be either necessarily true or contingently true.

⁴⁶ [Eds.] In the notation adopted in this volume: $\Box(\Diamond A \rightarrow \Box\Diamond A)$. This is the schema $(\Box E)$, which is a $\Box$ -version of the schema 5 or E , one of the characteristic axiom schemas of Lewis’s system **S5** (see Sect. 2.3).

$$1.9 \quad -(\sim p) < \sim(\sim p)$$

This is, to a certain extent, an *arbitrary* choice of the additional axiom. Later on, we will make different choices and draw the corresponding consequences as well.

It should be first pointed out that for a number of Lewis axioms 1.1–1.8*, the corresponding equivalences can be proven on the basis of the axiom system itself; namely: $pq = qp$, $p = pp$, $p(qr) = q(pr)$, $p = -(-p)$.⁴⁷ By contrast, the equivalences corresponding to 1.2, 1.6 do *not* hold, as well as

$$p < q = \sim q < \sim p,$$

does *not* hold, as it has been shown in detail above,⁴⁸ while

$$p < q = -q < -p$$

does hold. Thus, Lewis's Theorem 2.2⁴⁹ does hold, but neither 1.8 nor 2.21⁵⁰ do hold. As a consequence, all theorems, whose proofs contain the theorems 1.8 and 2.21 in question, must be discarded.

Luckily, 2.51

$$-(-p) = p$$

does *not* belong hereto, and is indeed to be regarded as proven in Lewis's system.

For our part, we have to prove the equivalence

$$-(\sim p) = \sim(\sim p)$$

that corresponds to our new axiom 1.9 or, which is sufficient in view of the postulation of 1.9, [we have to prove] $\sim(\sim p) < -(\sim p)$.⁵¹ This can be easily done with the aid of 1.7.⁵² If we replace in 1.7 p by $\sim p$, we immediately obtain

⁴⁷ [Eds.] Becker is saying here that the formulas obtained from Lewis axioms 1.1, 1.3, 1.4 and 1.5 by replacing “<” with “=” (strict equivalence) are provable in Lewis calculus. In the notation adopted in this volume: (i) $\Box(p \wedge q \leftrightarrow q \wedge p)$ (commutativity of the conjunction); (ii) $\Box(p \leftrightarrow p \wedge p)$ (idempotence of the conjunction); (iii) $\Box(p \wedge (q \wedge r)) \leftrightarrow (q \wedge (p \wedge r))$; (iv) $\Box(p \leftrightarrow \neg(\neg p))$ (law of double negation).

⁴⁸ [Eds.] See Becker's counterexample on p. 18. Axiom 1.2 is $qp < p$, axiom 1.6 is $(p < q)(q < r) < (p < r)$, while $p < q = \sim q < \sim p$ is axiom 1.8, that is the one that was responsible for the collapse of modalities in Lewis's original system.

⁴⁹ [Eds.] Theorem 2.2 is $(p < q) < (\sim q < \sim p)$.

⁵⁰ [Eds.] Theorem 2.21 is $(\sim q < \sim p) < (p < q)$.

⁵¹ [Eds.] As we said above, 1.9 is $(\Box E)$: $\Box(\Diamond A \rightarrow \Box \Diamond A)$. Thus, Oskar Becker just has to prove the other direction, namely: $\Box(\Box \Diamond A \rightarrow \Diamond A)$.

⁵² [Eds.] Axiom 1.7 is $\sim p \rightarrow -p$, that is $\Box(\neg \Diamond p \rightarrow \neg p)$.

$$\sim (\sim p) < -(\sim p)$$

quod erat demonstrandum.

With this, the basis of a closed calculus of modalities is set up, since the above mentioned problem of reducing all possible nested modalities to the six basic modalities (1) to (6), can now be solved.

The theorems

$$(\alpha) \quad p = -(-p) \qquad (\beta) \quad -(\sim p) = \sim(\sim p)$$

are in fact sufficient for this purpose. |

Indeed, given an arbitrary sequence of “–” and “~” symbols, one can at first [13] drop all repetitions (“powers”) of “–” according to the rules:⁵³

$$(\alpha\alpha) \quad (-)^{2n} p = p \qquad \text{and} \qquad (\beta\beta) \quad (-)^{2n+1} p = -p$$

The reason thereof becomes immediately apparent from (α) .

After that, one only has single occurrences of the symbol “–” and powers of the symbol “~”.

One can transform such a power $(\sim)^m$ by sufficiently many applications of (β) as follows:

For $m = 2n$ it holds:⁵⁴

$$(\gamma\gamma) \qquad (\sim)^{2n} p = (-)^{2n-1} \sim p = - \sim p$$

For $m = 2n + 1$ holds:⁵⁵

⁵³ [Eds.] Given a certain symbol S , Becker denotes by “ S^n ” the string $SS \dots S$ (n -times) and calls it the n -th power of “ S ”. Here he is saying that an even number n of consecutive occurrences of “–” is reducible to 0 through $(\alpha\alpha)$, while an odd number n of consecutive occurrences of “–” is reducible to 1 through $(\beta\beta)$.

⁵⁴ [Eds.] Becker says here that an even number of occurrences of “~” is reducible to “– ~” through $(\gamma\gamma)$.

⁵⁵ The formulas $(\gamma\gamma)$ and $(\delta\delta)$ contain Brouwer’s rule for the *iteration of the absurdity*. Indeed, taking $(\beta) : - \sim p = \sim \sim p$ into account, $(\gamma\gamma)$ says: $(\sim)^{2n} p = \sim \sim p$, which, together with $(\delta\delta) : (\sim)^{2n+1} p = \sim p$, yields the following: an odd “power” of the absurdity is equivalent to the single absurdity, an even power of the absurdity is equivalent to the second power, i.e. to the “absurdity of absurdity”. But this is just Brouwer’s rule.

By contrast, the axiom 1.9, i.e. (β) , is *not necessary* to justify Brouwer’s rule. One can in fact replace it by the weaker “Brouwerian axiom” $p < \sim \sim p$. For the derivation of the “absurdity rule” cp. e.g. Becker *Mathematische Existenz* 339 [o.p.n.] (Becker 1927) as well as below, p. 90.

[Eds.] For the ease of the reader, we quote here Becker’s derivation of Brouwer’s rule in the *Addendum* to §4 (*Problems of Brouwer’s logic. The iteration of absurdity*) of *Mathematische Existenz*, 339 (Becker 1927). “Brouwer succeeds in proving the theorem that the third power of the absurdity is equivalent to the first. (Hence: $abs^{2n} p = abs^2 p$ and $abs^{2n+1} p = abs p$). The proof runs as follows: let p and q be two propositions, and assume the principle

$$(\delta\delta) \quad (\sim)^{2n+1} p = (-)^{2n} \sim p = \sim p$$

The new symbols “ $-$ ”, which eventually originate by the application of $(\gamma\gamma)$, can be eliminated, in case they generate powers of “ $-$ ”, by means of $(\alpha\alpha)$, while new resulting powers of “ $\sim$ ” can be eliminated using (β) , and so on. The result is in any case either a single “ $-$ ” or a single “ $\sim$ ” or a shorter or longer sequence of alternating “ $-$ ” and “ $\sim$ ”, as for instance:

$$- \sim - \sim - \sim - \sim - p$$

One can transform the groups of “ $(- \sim)$ ” into groups of “ $(\sim\sim)$ ” via application of (β) . This yields:

$$\begin{aligned} \sim\sim\sim\sim\sim\sim\sim - p &= (\sim)^8 - p \\ &= - \sim - p \end{aligned}$$

This is modality (6) “ p is possibly false”.

In general one has

1. $(- \sim)^n p = (\sim\sim)^n p = (\sim)^{2n} p = - \sim p$ (modality 4)
2. $- (- \sim)^n p = -- \sim p = \sim p$ (modality 3)
3. $(- \sim)^n - p = - \sim - p$ (modality 6)

$$(p \rightarrow q) \rightarrow (abs\ q \rightarrow abs\ p)$$

By taking here p and $abs^2 p$ respectively in place of p and q , one has:

$$(p \rightarrow abs^2 p) \rightarrow (abs^3 p \rightarrow abs\ p)$$

Hence, from $p \rightarrow abs^2 p$ [...], it follows

$$abs^3 p \rightarrow abs\ p$$

On the other hand, it follows by substitution of $abs\ p$ for p in the valid formula $p \rightarrow abs^2 p$: $abs\ p \rightarrow abs^3 p$. Hence

$$abs^3 p \leftrightarrow abs\ p$$

Q.e.d.”

Here Becker shows (writing “ abs ” for the intuitionistic negation “ $\neg$ ”) how, using the two intuitionistically valid laws $(A \rightarrow B) \rightarrow (\neg B \rightarrow \neg A)$ (*weak contraposition*) and $A \rightarrow \neg\neg A$ (*weak double negation*), one derives $\neg\neg\neg A \leftrightarrow \neg A$, known as Brouwer’s law. If one writes the weak double negation law with “ $\sim$ ” in place of “ $\neg$ ”, one gets the modal schema $A \rightarrow \sim\sim A$, that is $A \rightarrow \Box\Diamond A$ in our notation, that is the schema B known as Brouwer’s schema. The key point of Becker’s observation at the above place of *On the Logic of Modalities* is that the characteristic axiom of his calculus $S3'$, which we have called $\Box(E)$ in this volume, is not necessary to justify $(\gamma\gamma)$ and $(\delta\delta)$, since the weaker schema $\Box(B)$ is already sufficient for this purpose.

$$4. \quad -(-\sim)^n - p = --- \sim -p = \sim -p \quad (\text{modality 5}) \quad |$$

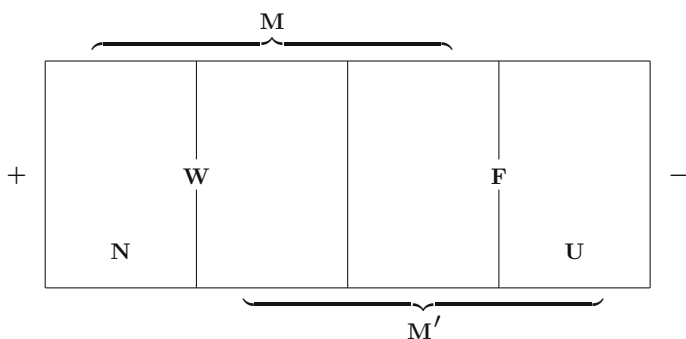
It can be seen that by this procedure, which is fairly simple in practice, every combination of modalities can be lead back to one of the six elementary modalities. [14]

The whole issue becomes clearer, if one replaces “ $\sim p$ ” i.e., “it is impossible, that p ”, with “ $- p$ ” i.e., “it is necessary, that non- p ”. This is *prima facie* just a different symbolism; but it introduces the *positive* modalities as primitive, in closer connection to natural language.

In the following, the six basic modalities are presented in both notations:

- | | | | |
|-----|-------------------------------------|---|--|
| (1) | p (true) | : | p (true) |
| (2) | $- p$ (false) | : | $- p$ (false) |
| (3) | $\sim p$ (impossible) | : | $\cdot - p$ (necessarily false) |
| (4) | $- \sim p$ (not impossible) | : | $- \cdot - p$ (not necessarily false) |
| (5) | $\sim - p$ (impossibly false) | : | $\cdot p$ (necessarily true) |
| (6) | $- \sim - p$ (not impossibly false) | : | $- \cdot p$ (not necessarily true) ⁵⁶ |

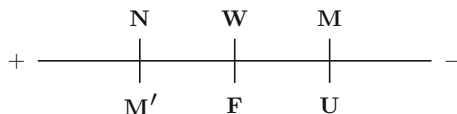
The relations between the six basic modalities can be illustrated by the following figure:



or, in abbreviated form:⁵⁷

⁵⁶ By (5) one has, more precisely: $\sim - p = \cdot - - p = \cdot p$; by (6): $- \sim - p = - \cdot - - p = - \cdot p$.

⁵⁷ [Eds.] The implication relations between the positive modalities are to be read in the diagram from the left to the right side (from + to -). In the symbolism adopted in this volume: $\Box p \rightarrow p \rightarrow \Diamond p$. The implication relations between the negative modalities are to be read in the inverse direction (from - to +): $\neg \Diamond p \rightarrow \neg \Box p \rightarrow \Diamond \neg p$.



Here “*N*” stands for “necessary”, “*W* [(*wahr*)]” for “true”, “*F*” for “false”, “*U* [*unmöglich*]” for “impossible”, “*M* [*möglich*]” for “possible” (or “not impossible”), “*M'*” for “possibly false” (or “not necessary”).

The big rectangle represents the *universe of discourse*. | [The dichotomies]:⁵⁸ [15]

$$W \text{ and } \overline{W} = F$$

$$N \text{ and } \overline{N} = M'$$

$$U \text{ and } \overline{U} = M$$

partition it completely. Hence, there is a threefold “*law of excluded middle*”:

1. *p* is either true or false (not true),
2. *p* is either necessary or possibly false (not necessary),
3. *p* is either possible (not impossible) or impossible.

But, obviously, it does *not* hold:

“*p* is either necessary (tautological) or impossible (contradictory)”.

In this case *datur tertium*, namely the area common to the rectangles *M* and *M'* (in the picture), which lies between *N* and *U* and, together with them, fills out the “*universe*.”⁵⁹

Thus, one has: *N* or *M* × *M'* or *U*: “*p* is either necessary, or impossible, or both possibly true and possibly false” (i.e. “undetermined”, or “neither necessary nor impossible”).

Furthermore it holds:

$$N < W < M \quad \text{and} \quad U < F < M'$$

This implies the following order among modalities from positive to negative:

$$+ \quad N \quad W \quad M \qquad M' \quad F \quad U \quad -$$

⁵⁸ [Eds.] The dichotomies are: true/false, $\Box/\neg\Box$; $\neg\Diamond/\Diamond$.

⁵⁹ In the ‘*logic of necessities*’ the “*law of excluded middle*” does *not* hold, but the principle of contradiction does. (“No proposition can be necessary and impossible at once”).—By contrast, in the case of the ‘*logic of possibilities*’ the law of excluded middle does hold (“A proposition is always possibly true or (*vel*) possibly false”) but the principle of contradiction (“no proposition can be possibly true and possibly false at once”) does not hold.—Such relations between propositions are called, in traditional logic, “*contrariety*” in the first case, and “*subcontrariety*” in the second case.

The compatibility relations can be easily deduced, as well.

Remark to §2.

By replacing axiom 1.9 by axiom 1.9*.⁶⁰

$$(1.9^*) \quad \sim (\sim p) < \sim (-p)$$

one essentially obtains the same system [sic!] as far as the rank order of modalities is concerned.

In this case, since⁶¹

[16]

$$\sim (-p) < \sim (\sim p)$$

does hold by 1.7 and 1.8*, one gets the equation:⁶²

$$\begin{aligned} & \sim -p = \sim \sim p \\ & \text{(instead of } - \sim p = \sim \sim p \text{).} \end{aligned}$$

Thus, $\sim \sim p$ is *stronger* than p (in contrast to Brouwer's conception).⁶³ From a *purely formal point of view*, this approach also seems to be possible without contradiction; even if it has perhaps no concrete meaning, it shows, however, the *independency of 1.9 from the remaining axioms*.

§3.

On Another Possible Formal Completion

of Lewis's Axiomatic System:

The Ten Modalities Calculus

As we already said, in place of the axiom

⁶⁰ [Eds.] As we said above, 1.9 is the schema $\Box(E)$, while 1.9* is the schema that we have called in this volume (see p. 22) " $\Box(E^*)$ ", i.e., $\Box(\Box \Diamond A \rightarrow \Box A)$, which characterizes the system $\mathbf{S3}^*$, a variant of $\mathbf{S3}'$ with $\Box(E^*)$ instead of $\Box(E)$. Becker's claim is that $\mathbf{S3}^*$ has six irreducible modalities, like $\mathbf{S3}'$, and that these are ordered with respect to logical strength in the same way as their are ordered in $\mathbf{S3}'$. Unfortunately, $\mathbf{S3}^*$ collapses—see Sect. 5.9.1 of the Appendix for a detailed proof.

⁶¹ [Eds.] $\sim (-p) < \sim (\sim p)$ is $\Box(\Box p \rightarrow \Box \Diamond p)$.

⁶² [Eds.] $\sim (-p) = \sim (\sim p)$ is $\Box(\Box p \leftrightarrow \Box \Diamond p)$.

⁶³ [Eds.] Becker is saying here that, if we consider an instance of Brouwer's schema B, say: $p \rightarrow \Box \Diamond p$, then p is stronger than $\Box \Diamond p$, while in the new calculus $\mathbf{S3}^*$ the other way round is the case.

$$(1.8^*) (p < q) < (\sim q < \sim p)$$

Lewis had originally postulated the stronger axiom

$$(1.8) (p < q) = (\sim q < \sim p)$$

which is equivalent to the logical product $(1.8^*) \times (1.8^{**})$, where

$$(1.8^{**}) (\sim q < \sim p) < (p < q).$$

Post showed that 1.8 implies the equivalence

$$\sim p = -p$$

(i.e. not only (1.7): $\sim p < -p$ but also its converse $-p < \sim p$). But, with this, the introduction of $\sim p$ becomes obviously superfluous and the whole calculus turns out to be identical with the classical one.⁶⁴

However, dropping 1.8** gives rise to a significant lacuna in Lewis's system. For, if no new axiom is added in its place, one can, indeed, still prove that the classical calculus (in the usual Russellian definition) is contained as a part in the new modalities calculus, yet it is not possible to set up the latter as a closed calculus of modalities, because

1. there arise infinitely many modalities that are not reducible one to another; |
2. there is not even a linear rank order among the positive or negative modalities [17]
according to their implicative strength;
3. not even the iterations or "powers" of the simplest modalities "impossible", "necessary" [and] "possible" are reducible to a finite number of basic modalities. Only the iterations of the negation, depending on whether they are even or odd, can be reduced to true or false.

The simplest modalities feature the following relations of implication:⁶⁵

$$\sim -p < - - p < - \sim p \quad (\text{by 1.7})$$

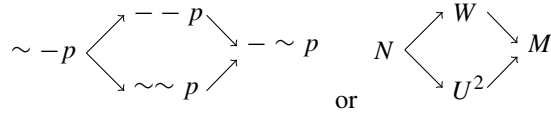
$$\sim -p < \sim \sim p < - \sim p \quad (\text{by 1.8* and 1.7})$$

Thus, in the form of an "arrow diagram" (where " $\longrightarrow$ " replaces " $<$ ");⁶⁶

⁶⁴ [Eds.] The collapse of modalities, as already said.

⁶⁵ [Eds.] The first formula displays the same order among basic modalities we have seen above: $\square \rightarrow - \rightarrow \diamond$, i.e.: necessity $\rightarrow$ truth $\rightarrow$ possibility. The second formula, says, in symbols: $\square \rightarrow \square \diamond \rightarrow \diamond$, in words: "what is necessary, is also necessarily possible and hence is possible."

⁶⁶ [Eds.] In words, the bifurcation in the diagrams can be rendered as follows: the *necessary* strictly implies both the *true* and the *necessarily possible*, each of which, taken separately, strictly implies the *possible*.



The relation between $p = \text{---} p$ and $\sim\sim p$, that is between W and U^2 , remains undetermined.

One can now add “Brouwer’s *axiom*” to this setting (this is the weakest addition we propose):

$$\begin{array}{ccc}
 p & = & \text{---} p < \sim\sim p \\
 W & \rightarrow & U^2
 \end{array} \quad (1.91)$$

Truth implies the absurdity of the absurdity (but not conversely!).⁶⁷

In this way one gets the linear rank order:

$$\begin{array}{ccccccc}
 \sim -p & < & \text{---} p & < & \sim\sim p & < & - \sim p \\
 N & \rightarrow & W & \rightarrow & U^2 & \rightarrow & M
 \end{array}$$

Brouwer has shown that using (1.91) and (1.8*) it is possible to evaluate the powers of the absurdity:⁶⁸

$$U^{2n+1} = U, \quad U^{2n} = U^2$$

In addition, one has [in classical logic]:

$$F^{2n+1} = F, \quad F^{2n} = W$$

But this is not sufficient to reduce all compositions of modalities to a finite number.

For instance, alternating compositions like

$$\begin{array}{ccc}
 \sim - \sim - \sim - p & \text{or} & - \sim\sim - \sim - \sim - p \\
 N^3 p & & MN^3 p
 \end{array}$$

can not be reduced in this way. |

However, if we add an axiom (1.92) that allows to operate with the iterations of [the] necessity [operator], it is possible to reduce all modalities, if not to the 6 basic modalities N, W, M, M', F, U considered above, to 10 basic modalities. [18]

Already in Lewis we have:⁶⁹

$$\sim - \sim - p < \sim - p \quad \text{or} \quad \cdot \cdot p < \cdot p \quad (N^2 \rightarrow N)$$

Indeed, by 1.7 it holds:

$$\begin{aligned} \sim - \sim - p &= \sim (- \sim - p) < -(- \sim - p) \\ &= - - (\sim - p) \\ &= \sim - p \end{aligned}$$

As an axiom we choose the converse of the above implication, namely:

$$(1.92) \quad \sim - p < \sim - \sim - p \quad \text{or} \quad \cdot p < \cdot \cdot p \quad (N \rightarrow N^2)$$

While already according to Lewis it holds that “If it is necessary that p is necessary, then it is true that p is necessary”, now it also holds that “If it is true that p is necessary, then it is necessary that p is necessary”, i.e. the necessity should entail its own necessity, like the truth entails its own truth. From this it follows immediately $N^n = N$.

Since

$$\begin{aligned} - \sim p &= - \sim - - p = -(\sim -) - p \\ - \sim - \sim p &= - \sim - \sim - - p = -(\sim - \sim -) - p, \end{aligned}$$

and in general

$$(- \sim)^n p = -(- \sim)^n - p, \quad \text{i.e. } M^n = FN^n F$$

hold, then it also holds:

⁶⁷ [Eds.] Cp. fn. 55. Becker writes “Absurdität”; he usually employs this word to refer to the intuitionistic negation, but in the present context “Absurdität” means “impossibility”.

⁶⁸ [Eds.] Becker is saying here that a number k of consecutive occurrences of “ U^2 ”, i.e. “ $\Box\Diamond$ ” is reducible to one single occurrence of “ $\Box\Diamond$ ”, if k is odd, and to two consecutive occurrences of “ $\Box\Diamond$ ”, if k is even.

⁶⁹ [Eds.] Becker is going to claim the following. The schema $\Box(\Box\Box A \rightarrow \Box A)$, being an instance of the boxed-version of the schema **T**, already holds in Lewis’s system **S3**. If we now add, as an additional axiom, (1.92), i.e. $\Box(\Box A \rightarrow \Box\Box A)$, the “boxed-version” of the schema 4, then we get the equivalence $\Box(\Box\Box A \leftrightarrow \Box A)$ that permits to reduce all powers of $\Box$ to one single $\Box$ and so also, because of the interdefinability of $\Box$ and $\Diamond$, also all powers of $\Diamond$ to one single $\Diamond$.

$$M^n = FN^nF = FNF = M$$

i.e. also all powers of the possibility [operator] are reducible to one.

If one postulates (1.91) $\times$ (1.92) one can thus set up a ten modalities calculus.

Systematic Reduction to the Ten Basic Modalities

Besides the truth W , there arise the irreducible modalities: $F, U, U^2; N, M, M'; U^2F, FU^2, FU^2F$.

One can namely reduce the following “powers”:

$$\begin{array}{lll} (-)^{2n}p = p; & (-)^{2n+1}p = -p & (F^{2n} = W, F^{2n+1} = F) \\ (\sim)^{2n}p = \sim^2 p; & (\sim)^{2n+1}p = \sim p & (U^{2n} = U^2, U^{2n+1} = U) \\ (\sim -)^m p = \sim - p & \text{and also} & (- \sim)^m p = - \sim p \\ (N^m = N) & & (M^m = M) \quad | \end{array}$$

To get an overview it is more convenient to use the dot-dash notation, where [19]
 $\sim - = \cdot (N)$ and $\sim = \cdot - (NF)$.⁷⁰ The reduction proceeds in this case as follows:

- Any dot-dash sequence can be first lead back, by reduction of the N and F powers, to a list of alternating N and F signs (dots and dashes).
- There are four types of such alternating sequences, depending on their initial and final symbols: NN, NF, FN, FF , i.e. $\cdot \cdot, \cdot -, - \cdot, --$.

So one has⁷¹ : i.e. in Lewis's notation:

$$\begin{array}{lll} \text{Type I:} & (\cdot -)^l & : \quad \sim^l \sim = \sim^{l+1} - \\ \text{Type II:} & (\cdot -)^l & : \quad \sim^l = \sim^l \\ \text{Type III:} & -(\cdot -)^l & : \quad - \sim^l \sim = - \sim^{l+1} - \\ \text{Type IV:} & -(\cdot -)^l & : \quad - \sim^l = - \sim^l \end{array}$$

⁷⁰ [Eds.] “ $\cdot$ ” corresponds, in the notation adopted in this volume, to “ $\Box$ ”.

⁷¹ [Eds.] Type I begins with a “ $\Box$ ” and ends with a “ $\Box$ ”, thus, Type I looks as follows: “ $\Box \dots \Box$ ”. In between there must be an alternation of “ $\Box$ ” and negations (starting with “ $\neg$ ” and ending with “ $\neg$ ”), as for instance in: “ $\Box \neg \Box \neg \Box \neg \Box$ ”. Becker abbreviates this as follows: “ $(\Box \neg)^l \Box$ ”, where $l \geq 1$. (Notice that Becker groups the modal operators from the left). Type II begins with a “ $\Box$ ” and ends with a negation “ $\neg$ ”. Thus, Type II looks as follows: “ $\Box \dots \neg$ ”. In between there is nothing, or there can be an alternation of “ $\neg$ ” and “ $\Box$ ” as in “ $\Box \neg \Box \neg \Box \neg$ ”. Becker abbreviates this as follows: “ $(\Box \neg)^l$ ”, where $l \geq 1$. Type III begins with a negation “ $\neg$ ” and ends with a “ $\Box$ ”, as in: “ $\neg \Box \neg \Box \neg \Box$ ”. Becker abbreviates this as follows: “ $\neg(\Box \neg)^l \Box$ ”, where $l \geq 0$. Type IV begins with a negation “ $\neg$ ” and ends with a negation “ $\neg$ ”, as in: “ $\neg \Box \neg \Box \neg \Box \neg$ ”. Becker abbreviates this as follows: “ $\neg(\Box \neg)^l$ ”, where $l \geq 1$.

From now on Lewis's notation is more convenient. Two subcases are possible here, namely:⁷²

$$\begin{array}{lll} \text{(A)} & l = 2n; & l + 1 = 2n + 1 \\ \text{(B)} & l = 2n - 1; & l + 1 = 2n \end{array}$$

There result eight cases:⁷³

IA: $\sim^{2n+1} - = \sim -$	N	IB: $\sim^{2n} - = \sim^2 -$	$U^2 F$
IIA: $\sim^{2n} = \sim^2$	U^2	IIB: $\sim^{2n-1} = \sim$	U
IIIA: $- \sim^{2n+1} = - \sim -$	$FUF = M'$	IIIB: $- \sim^{2n} - = - \sim^2 -$	$FU^2 F$
IVA: $- \sim^{2n} = - \sim^2$	FU^2	IVB: $- \sim^{2n-1} = - \sim$	M

To these must be added “W”, “F” and the simple “-” that is not an alternating sequence in the dot-line notation.

Thus, one has the following irreducible ten basic modalities:⁷⁴

$$N, W, M, M', F, U; U^2, FU^2F, U^2F, FU^2$$

Among the pure modi

$$\begin{array}{l} U^2, FU^2F \text{ (even number of } U \text{ and } F \text{ “factors” are positive,} \\ U^2F, FU^2 \text{ (odd number of } U \text{ and } F \text{ “factors” are negative.} \end{array}$$

As to U^2 , we already know that, according to Brouwer's axiom, it lies between W and M .⁷⁵ ($W \rightarrow U^2 \rightarrow M$) |

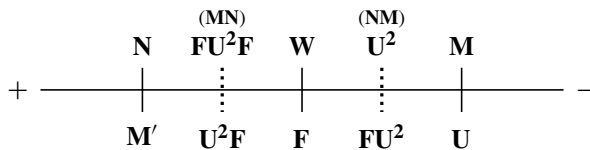
We can depict the situation clearly by means of the picture we introduced above [20]

⁷² [Eds.] Becker is distinguishing here the two cases: (A) l is an even number (and so $l + 1$ is odd), and (B) l is an odd number (and so $l + 1$ is even).

⁷³ [Eds.] Becker considers in the following each of the four Types he has considered immediately above, with l even and, respectively, with l odd. Since we have four types and two possibilities for each type (l even, l odd), we have exactly eight cases. In the notation adopted in this volume the eight cases are: $\Box$ (IA); $\Box\Diamond\rightarrow$ (IB), $\Box\Diamond$ (IIA), $\rightarrow\Diamond$ (IIB), $\Diamond\rightarrow$ (IIIA), $\Diamond\Box$ (IIIB), $\Diamond\Box\rightarrow$ (IVA), $\Diamond$ (IVB).

⁷⁴ [Eds.] The previous eight modalities with the addition of true and false.

⁷⁵ We do not know whether Brouwer's axiom (1.91) and the axiom (1.92) are independent, although this is probable. At any rate, (1.92) does not rule directly the rank order of the modalities. [Eds.] (1.91) is the schema $\Box(B)$ and (1.92) is the schema $\Box(4)$. As we will see in the Appendix, Sect. 5.9.2, the answer to Becker's question is instead *negative*: while (1.91) is not a theorem of $S3 + (1.92)$, it turns out that (1.92) is a theorem of $S3 + (1.91)$. This is indeed the reason why the “Ten modalities system” boils down to the “Six modalities system”.



Explanation

- (1) U^2 is represented by an area that, on the one side, contains W and, on the other side, is contained in M , since it holds: $W < U^2 < M$.⁷⁶
- (2) FU^2 is the area that added to U^2 gives the “universe”.⁷⁷ (Similarly, $FW = F$; $FN = M'$; $FU = M$)
- (3) U^2F is the area symmetrical to U^2 with respect to the central axis (the line connecting W and F), provided that one changes the sign. (Similarly, N is symmetrical to U ; M is symmetrical to M' ; W is symmetrical to F .)
- (4) FU^2F is the area that added to U^2F gives the “universe”.

Validation of the claims (2), (3), (4):

- (a) (2) says:

$$U < FU^2 < F$$

$$\sim p < -\sim\sim p < -p$$

De facto:⁷⁸

$$\sim p = --(\sim p) < -\sim(\sim p) \quad (\text{since } --q < -\sim q)$$

and

$$-(\sim\sim p) < -p \quad (\text{since } p < \sim\sim p \text{ by 1.91})$$

- (b) (3) says:

$$F < U^2F < M'$$

⁷⁶ [Eds.] Becker says here that $\Box\Diamond$ stands between the true (W) and the possible ($\Diamond$).

⁷⁷ [Eds.] That is, $\neg\Box\Diamond$ is the “complement” of $\Box\Diamond$.

⁷⁸ [Eds.] Becker’s proof of (i) $\sim p < --\sim p$ and (ii) $-\sim\sim p < -p$ runs as follows. (i) By double negation, $\sim p < --\sim p$; on the other side, $\sim\sim p < -\sim p$ is a substitution instance of axiom (1.7), yielding $--\sim p < -\sim\sim p$ by contraposition; thus, by transitivity of $<$ (axiom (1.6)), $\sim p < -\sim\sim p$ follows. (ii) $-\sim\sim p < -p$ follows by contraposition from $p < \sim\sim p$, which is axiom (1.91).

$$-p < \sim\sim -p < -\sim -p$$

De facto is:⁷⁹

$$--(-p) < \sim\sim (-p) \quad (\text{by 1.91})$$

and

$$\sim(\sim -p) < -(\sim -p) \quad (\text{by 1.7})$$

(c) (4) says:

$$N < FU^2F < W$$

$$\sim -p < -\sim^2 -p < p$$

Indeed:⁸⁰

$$--(\sim -p) < -\sim(\sim -p) \quad (\text{since } --q < -\sim q)$$

Furthermore:

$$p < \sim\sim p \quad (1.91)$$

$$(-p) < \sim\sim (-p)$$

$$-(\sim\sim (-p)) < -(-p)$$

i.e.

$$-\sim\sim -p < --p = p \mid$$

Result:

[21]

We obtain the following chains of implications:

$$\begin{aligned} N &< FU^2F < W < U^2 < M \\ U &< FU^2 < F < U^2F < M' \end{aligned}$$

⁷⁹ [Eds.] Becker's proof of (i) $-p < \sim\sim -p$ and (ii) $\sim\sim -p < -\sim -p$ runs as follows. (i) is a substitution instance of axiom (1.91). Notice that the "--" that Becker puts in front of $(-p)$ is a lapse. (ii) is a substitution instance of axiom (1.7).

⁸⁰ [Eds.] Becker's proof of (i) $\sim -p < -\sim\sim -p$ and (ii) $-\sim\sim -p < p$ runs as follows. (i) follows from $\sim\sim -p < -\sim -p$, which is a substitution instance of axiom (1.7), by contraposition and double negation. (ii) follows from $-p < \sim\sim -p$, which is a substitution instance of axiom (1.91), by contraposition and double negation.

as well as the following polar rank order:⁸¹

$$(+)N, FU^2F, W, U^2, M \mid M', U^2F, F, FU^2, U(-)$$

§4.

The Material Meaning of the Six- and Ten-

Modalities-Calculi.

The reduction rule of the six modalities system implies that in an expression of the form:

$$(F)N^{x_1}FN^{x_2}F \dots N^{x_p}(F)$$

(if one thinks of all “powers” as written in a resolved form, as, for instance, NN instead of N^2 , NNN instead of N^3 etc.) only the rightmost N remains, while *all* other N are dropped. But then, of course also the remaining F -powers are reducible ($F^{2q} = F^2 = W$; $F^{2q+1} = F$).⁸²

What is now the *material* meaning of such reduction rule for the N s?

(a) Firstly, it means a reduction of the N -powers *alone*: $N^x = N$. This can be derived by induction from $N^2 = N$.

Thus, the necessity of the necessity is the same as the simple necessity. Similarly, the possibility of the possibility is the same as the simple possibility.⁸³

Now, $N^2 = N$ means the logical product of two implications, namely:

$$(N^2 \rightarrow N)(N \rightarrow N^2)$$

The first implication⁸⁴ $N^2 \rightarrow N$ is self evident, since it is just a special case of the general principle $N \rightarrow W$. Indeed, it can also be written as $N(N) \rightarrow N(W)$.⁸⁵

⁸¹ [Eds.] That is, a (linear) ordering of the five positive modalities N, FU^2F, W, U^2, M , together with the *specular* (linear) ordering of the negative modalities M', U^2F, F, FU^2, U .

⁸² [Eds.] Becker is (correctly) saying here that in the “six modalities calculus” any combination of N ’s and F ’s containing at least one occurrence of N reduces to one of the following four combinations: N, FN, NF, FNF .

⁸³ This results formally in the following way: $M = FNF$; $M^2 = FNFNF = FNNF = FNF = M$. Therefrom it follows by complete induction: $M^x = M$, since $M^{x+1} = M^{x-1+2} = M^{x-1}M^2 = M^{x-1}M = M^x$.

⁸⁴ [Eds.] Recall that, by saying that a certain modality Λ “implies” another modality Σ , and by writing “ $\Lambda \rightarrow \Sigma$ ”, Becker means that the formula $\Box(\Lambda p \rightarrow \Sigma p)$ (or, equivalently, $\Box(\Lambda A \rightarrow \Sigma A)$ for all A) is a theorem of the calculus under consideration, see p. 74. This “arrow notation” is henceforth systematically adopted by Becker.

⁸⁵ In Lewis’s notation: $\sim -(\sim -p) < \sim -p$. [Eds.] This follows indeed by the substitution instance $\sim - \sim -p < -- \sim -p$ of axiom (1.7), corresponding to the boxed version of T , by double negation.

Things are different with the second implication $N \rightarrow N^2$: the simple necessity implies its own necessity. Or: the *iteration of the modality* “necessity” (and [of the modality] “possibility”; and hence the iteration of every *positive*, direct or “oblique” modality)⁸⁶ *has no effect. It is nothing.* |

What is the concrete (phenomenological) meaning of such implication?

[22]

[Such an implication] says that necessity is to be seen as an *absolute* modality. [It says that] necessity is necessity *par excellence*, it has no degrees. If such an “absolute” necessity does obtain, then it is itself necessary, it has, in a sense, an *ideal* character.

That this is not, from every point of view, true, is shown by Husserl’s distinction between contingent and formal apriori⁸⁷—or, better: “necessary apriori”. If one considers the corresponding essential laws [*Wesensgesetze*] and [the corresponding] essential necessities [*Wesensnotwendigkeiten*], one finds immediately the conceptual pair: “contingent necessity vs. ideal necessity”. In this case, N^2 , the “necessary necessity” is definitely different from the “contingent” one, that is, from the “simple” necessity N . The apriori structure of the color-manifold, for instance, is indeed necessary, but it is difficult to gain insight in the ground of this eidetic structure, for, [the necessity] seems to pertain, so to say, “by chance” to this “hyletic region”. From the standpoint of formal constitutive phenomenology the existence of hyletic regions governed by determined essential laws [*Wesensgesetze*] is necessary, however, such laws are not determined individually, but just on the basis of a certain formal scaffold. Thus, the “material” necessity is subordinated to the formal one and the fact that some material-apriori structures, among those that are *formally* possible, are realized, while other ones are not, is accidental.⁸⁸ The formal necessities [*formale Notwendigkeiten*] are those that obtain necessarily—just like formal phenomenology, as a whole, refers back to itself and finds in itself its own justification⁸⁹—, the material necessities (in particular, the “hyletic” necessities

⁸⁶ [Eds.] The “direct modi” are, according to Oskar Becker, *true* and *false*, while the “oblique modi” are *necessary* and *possible*. For Becker’s use of the pair “direct”/“oblique”, also at a variance with Brentano’s use, see below fn. 147, as well as Becker’s *Additional Remark* on p. 111 ff.

⁸⁷ Husserl, *FTL* 1929, §6 in: Jahrbuch X, 25 (Husserl 1929).

⁸⁸ [Eds.] For the ease of the reader we quote the main points of the parallel place in Husserl, *FTL* 1929, §6, 25-26 [o.p.n.] (Husserl 1929): “The universality of logic [...] is not simply an a priori or eidetic universality; rather it is [...] a *formal* universality. [...] We could say [...]: *Reason itself*, including theoretical reason in particular, *is a form concept*. [...] In a certain sense every eidetic cognition is a product of “pure” reason—*pure form of all empiria* [...]; but not every eidetic cognition is pure in a *second sense*; the one pertaining to *form as a principle*. An a priori proposition about all *sounds* as such, about sound meant with “pure” universality, is pure only in the first sense; it is [...] a “contingent” a priori. It has in the *eidos sound* a materially determined core, which goes beyond the realm of the universality of “principles” [...] and restricts it to the “contingent” province of ideally possible sounds. “Pure” reason is not only above everything empirically factual, but also above every sphere of hyletic, materially determined, essences. It is the title for the self-contained system of pure principles that precede every hyletic, materially determinate, a priori [...]”

⁸⁹ Husserl, *FTL*, §§102, 104 in: Jahrbuch X, 236 ff., 241 ff. (Husserl 1929).

in the narrow sense)⁹⁰ are those, whose validity must be accepted as a “fact” in a higher sense, to some extent as an “essential fact [*Wesenstasache*]”—where this expression is not more paradoxical than the expression “contingent a priori”.

Now, [the point is that] through the thesis $N \rightarrow N^2$ such “contingent” essential necessities [*Wesensnotwendigkeiten*], such “essential facts [*Wesenstatsache*]” are neglected. |

Similarly, as far as possibilities are concerned. Though, here, the problematic implication between the two implications arising from the equation $M = M^2$ is not $M \rightarrow M^2$ (for, this means $W(M) \rightarrow M(M)$: “The *truth* of the possibility implies the possibility of the possibility”, which is obvious!), but $M^2 \rightarrow M$.⁹¹

[23]

To put it in other words, the question is whether the possibility of the possibility implies the simple possibility. This is indeed the case, if one considers the possibility as an “ideal” or “formal” possibility, i.e., in a very broad and, so to say, absolute sense [of possibility] that does not admit “degrees”.⁹² This case coincides with that of the formal necessity, as it [actually] has to be, given the logical equivalence of the relations $N \rightarrow N^2$ and $M^2 \rightarrow M$.

(b) The reduction of the *N*-powers is clearly not sufficient to provide the reduction rule of the *six-modalities-calculus* (§2). Indeed, with it, we just obtained a *ten-modalities-calculus* (§3). For a further reduction the implication $M \rightarrow NM$ is sufficient here (§2, namely: $NM = NFNF = U^2$) or also: $M' = FN \rightarrow NFN = NM'$. This means: *If a proposition is possibly true (or false), then it is necessarily possibly true (or false)*. Or, briefly: *Every possibility implies its own necessity*.⁹³ This does *not* follow from the assumptions made in (a). However, these assumptions can be completed with an observation that makes the proposition of the necessity of any possibility evident.

⁹⁰ Husserl, *Ideen zu einer reinen Phänomenologie*, §§65, 85 in: Jahrbuch I, 1, 122 ff., 171ff. (Husserl 1913).

[Eds.] For convenience of the reader we quote a few lines from §85 of the *Ideas*, tellingly entitled “*Sensuous* ὅλη, *intensive* μορφή”: “[...] this remarkable duality and unity of *sensuous* ὅλη and *intensive* μορφή plays a dominant role in the whole phenomenological sphere. [...] Sensuous data present themselves as stuffs for intensive formings, or sense-bestowings [...]. We need a [...] term which expresses the whole group of [sensuous data] [...] and we choose for it the expression “hyletic-data” or “stuff-data”, likewise simply “sensuous stuffs.”

⁹¹ Formally, this can actually be derived from $N \rightarrow N^2$, because the latter, applied to a false proposition, yields: $N(F) \rightarrow N^2(F)$, hence by contraposition $FN^2F \rightarrow FNF$ or $M^2 \rightarrow M$. Conversely, from $M^2 \rightarrow M$ we can obtain $N \rightarrow N^2$. Namely: $(FN^2F \rightarrow FNF) \rightarrow (FFNF \rightarrow FFN^2F) \rightarrow (NFF \rightarrow N^2FF) = (N \rightarrow N^2)$.

[Eds.] The way the latter derivation is presented is misleading. Actually, from $M^2 \rightarrow M$ applied to a false proposition one gets $M^2F \rightarrow MF$, whence by contraposition it follows $FMF \rightarrow FM^2F$, i.e. $N \rightarrow NFFN$, that is $N \rightarrow N^2$.

⁹² It is not yet necessary to leave the sphere of the *positive* possibility and go into that of the *lack* [*privatio*] of impossibility (or, “non-contradiction [*Widerspruchslosigkeit*]”) (cp. below p. 100). But, rather, the possibility in question is an ideal “factual” possibility (which is not the same as *material*!), and thus, relevant from a formal-ontological standpoint.

⁹³ [Eds.] This is Becker’s axiom (1.9), cp. p. 68.

The reduction of the *N*- and *M*-powers was grounded on the concept of an absolute, “ideal” (from a logical point of view strongest) necessity and, respectively, on the concept of a (logically weakest) possibility that cannot be surpassed or, respectively, be further weakened. These two modalities determine, in a sense, an “ideal” or “absolute” sphere, against which the “direct” modi are, or can be, “contingent”. | One can now say that the necessity lies already in the essence of this ideal sphere, so that applying it explicitly to a modality of the sphere has an obvious legitimation and achieves nothing new. [24]

Thus considered, the six-modalities-calculus seems to have, with respect to the ten-modalities-calculus, the advantage that the logical setting is stronger: indeed, the capability of reducing the *M*- and *N*-powers alone seems to be an unforgivable half measure.

True, there are paradoxical theorems also in the six-modalities-calculus, as for instance:

$$MN = FNFN = FFN = N.$$

The possibility of a necessity is equivalent to the necessity itself,⁹⁴ that is, for a necessity, its own possibility and its own “reality” are one and the same. However, also this paradox can be justified by the concept of a purely *ideal* sphere. This sphere of ideal possibilities is *such that* any structure in it is necessary in an essential way [*wesensnotwendig*]. For, necessity inheres to the “ideal” as such (one may think of the Aristotelian ἀίδιον) and any possibility is immediately realized in it.

Nevertheless, one might find domains, in which the *M*- and *N*-powers have no meaning, but nevertheless *NM* and *M*, as well as *MN* and *N*, are distinct.

(c) There is still the question of how the Brouwer axiom, $W \rightarrow NM$,⁹⁵ can be reasonably justified.

For, even if one distinguishes between the necessary possibility and the simple possibility, still the first cannot have the same logical strength as the truth. A necessary possibility is not yet a truth, even if the other way round does hold: the truth is always necessarily possible.

The consequence $MN \rightarrow W$ of the Brouwer axiom (i.e., $FNMF \rightarrow FWF$, where $FNMF = FNFNFF = MN$)⁹⁶ is, at first sight, less plausible. However, the truth is *de facto* a necessary condition of the necessity, i.e. the necessity of *p* is not even possible, if *p* is not true; but this means nothing else than *MN* implies the true.

* * *

In the following another calculus should be developed, where *all* restrictions to the possibility of building nested modalities that are not necessary from a formal

⁹⁴ [Eds.] $\Diamond\Box p \rightarrow \Box p$ follows from axiom (1.9), $\Diamond p \rightarrow \Box\Diamond p$, by dualization; $\Box p \rightarrow \Diamond\Box p$ is a substitution instance of axiom (1.7).

⁹⁵ [Eds.] $\Box(B) : A \rightarrow \Diamond\Box A$.

⁹⁶ [Eds.] Becker is considering here the dual form of the schema $\Box(B) : \Box(\Diamond\Box A \rightarrow A)$.

calculatory point of view are as far as possible dropped. |

§5.

[25]

On a Calculus of Modalities

with Minimal Requirements

that Still Allows for a Linear Rank Order.

(a) One can try to free oneself from all requirements that are imposed by special *material* [*sachliche*] assumptions, and introduce solely the most general *formal* conditions of a calculus of modalities. Firstly, one can drop the requirement that the reduction to a finite number of primitive modalities be possible. Such requirement is indeed immediately dropped as soon as one allows all *N*-powers to be different modalities.

One will, however, keep the general requirement of a linear rank order of modalities, whereby the implicational relation of any two distinct modalities is univocally determined. Otherwise a modality could no longer be univocally determined by its “rank of logical strength”. Such requirement should be in any case the upper bound of our formal freedom.

Now, the question is, on the one hand, whether Lewis’s calculus satisfies this requirement and, on the other hand, whether suitable axioms for a theory of the rank order of modalities can be established independently from Lewis’s calculus. The answer to the first question is negative, while the answer to the second question is positive.

Indeed, the question whether p is stronger than $\sim\sim p$, or conversely, cannot⁹⁷ be decided within the *not-completed* axiomatic system of Lewis, as we explained above (see p. 74).⁹⁸ For this reason, a linear rank order of *all* modalities is, [in that calculus], a priori impossible.

By contrast, the second question can be answered in the sense that one can specify a number of simple *operational rules* on modalities concerning one and the same proposition p , which help to set up a theory of rank order. These rules should not prove anything that is not already provable in the *not-completed* calculus of Lewis, from which they can be easily deduced. They will be successively completed by suitable *additional axioms*.

One can conceive of the modalities as a kind of *operations* that can be applied to the same proposition p . They can be composed and, hence, also iterated. (E.g.: $\sim p$, $\neg p$, $\sim\neg p$, $\sim\sim p$ or, in our notation, Up , Fp , Np , U^2p or, even more shortly: U ,

⁹⁷ However, a *proof* of independence is still missing.

⁹⁸ [Eds.] Oskar Becker is claiming here that in Lewis’s system **S3** neither $\Box(A \rightarrow \Box\Diamond A)$ nor $\Box(\Box\Diamond A \rightarrow A)$ are provable. Indeed, he is right: see the Appendix, Sects. 5.9.2, resp. 5.9.1.

$F, N = UF, U^2$; undetermined operations will be denoted by capital Greek letters like Λ, Ω and so on.) |

Rule I: Each non-elementary modality results from composition of the two elementary modalities N and F (or $U = NF$ and F). The composition (“multiplication”) is associative, but *not* commutative. (So, right and left multiplication are to be distinguished.) [26]

Rule II: An implication (and hence also an equation) between two (simple or composed) modalities can be *right* multiplied without loosing its validity.

Definition: A composed modality is *positive* (*negative*), if it contains an even (odd) number of simple negative elementary modalities (U, F).⁹⁹

(Any composed modality can be represented via N and F , or U and F ; any *positive* modality via M and N .)

Rule III: An implication between two modalities may be, preserving validity, either left multiplied with a *positive* modality or, under *reversion of the implicational relation*, left multiplied with a *negative* modality.

In symbols, Rule II and III can be rendered as follows:¹⁰⁰

$$\begin{aligned} (\Lambda \rightarrow \Lambda') &\rightarrow (\Lambda\Theta \rightarrow \Lambda'\Theta) \\ (\Lambda \rightarrow \Lambda') &\rightarrow (\Pi\Lambda \rightarrow \Pi\Lambda'), & \text{where } \Pi \text{ is positive} \\ (\Lambda \rightarrow \Lambda') &\rightarrow (\Sigma\Lambda' \rightarrow \Sigma\Lambda), & \text{where } \Sigma \text{ is negative} \end{aligned}$$

Remark: The “multiplications” mentioned in Rule II and III cannot be substituted by “divisions”, even if these divisions “make sense”; that is, performed multiplications cannot be withdrawn. $\Lambda \rightarrow \Lambda'$ does *not* follow from $\Pi\Lambda \rightarrow \Pi\Lambda'$.¹⁰¹

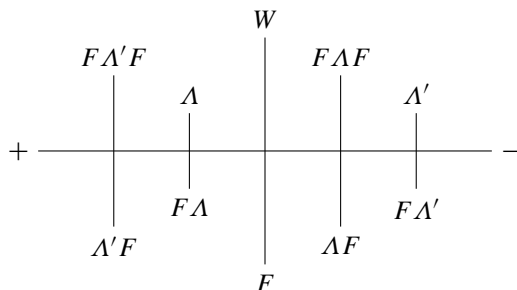
In particular, if one applies to an implication $\Lambda \rightarrow \Lambda'$ the right, respectively the left multiplication with F , then $\Lambda F \rightarrow \Lambda' F$, respectively $F\Lambda' \rightarrow F\Lambda$, follow. This can be illustrated by a graphical presentation, which we used for the finite calculi in §§2–3: the left multiplication with F means “completion to the universe”, the right multiplication with F means transition to the symmetric modality under change of sign.¹⁰²

⁹⁹ [Eds.] Thus, for instance, $\Box$ and $\neg\Box\neg$ (that is: $\Diamond$) are positive, while $\Box\neg$ (equivalent to $\sim$, *impossible*) is negative.

¹⁰⁰ [Eds.] Let Λ, Λ' and Σ be basic or nested modalities, and let Λ be at least as strong as Λ' (that is, it is provable that $\Lambda\Lambda \rightarrow \Lambda'\Lambda$ for any Λ). Rule II then says that $\Lambda\Sigma$ shall be at least as strong as $\Lambda'\Sigma$. And Rule then III says that $\Sigma\Lambda$ shall be at least as strong as $\Sigma\Lambda'$, provided Σ is *positive*, while $\Sigma\Lambda'$ shall be at least as strong as $\Sigma\Lambda$, provided Σ is *negative*. One can prove (see the Appendix, Sect. 5.9.3.) that Lewis’s system **S3**, and more generally any axiomatic extension of **S2**, is closed under these rules.

¹⁰¹ From the viewpoint of Lewis’s calculus this is a consequence of the facts discussed above on p. 63 ff. Notice, however, that here we always deal with the same (unmodalized) proposition p , and not with two different propositions p, q like there.

¹⁰² [Eds.] If Λ is a positive (negative) modality in the diagram, then ΛF is the negative (positive) modality that is located specularly to Λ with respect to the vertical axis.



I

One can easily see that the following implications hold simultaneously:

[27]

$$\Lambda \rightarrow \Lambda', \Lambda F \rightarrow \Lambda' F, F\Lambda' \rightarrow F\Lambda, F\Lambda' F \rightarrow F\Lambda F$$

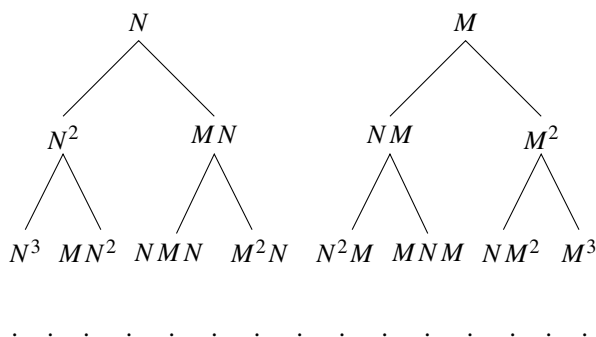
Rule IV: $FF = W$ holds (falsity of the falsity = truth). W can be arbitrarily added or omitted.

* * *

(b) For what concerns the problem of obtaining a ranking of all modalities under the weakest possible requirements, the postulations in the 4 “rules” are obviously not sufficient. Indeed, not even the relation $W \rightarrow U^2$ can be deduced from them. Which additional requirements must be introduced?

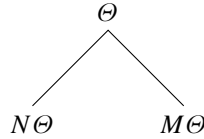
At first, it is convenient to restrict our considerations to the *positive* modalities. They contain arbitrarily many N ’s and an even number of F ’s, so can also be seen as composed of N ’s and $FN F$ ’s = M ’s.

In a purely combinatorial way, the following options arise:



according to the general schema¹⁰³

¹⁰³ [Eds.] The inductive procedure by which the tree-like diagram is generated is the following: *Stage 1*: draw the “root” nodes N (to the left) and M (to the right); *Stage $k + 1$* : for each node A



or equivalently—as one can easily check—following the rule that the left half of the i -th series is obtained by right multiplying the $(i - 1)$ -th series with N , and the right half by right multiplying it with M .¹⁰⁴

The implicational relations are at first not determined by this purely combinatorial generation schema, but the *second* generation procedure allows important preliminary conclusions to this aim.

Namely, since by Rule II the validity of an implication is preserved by right multiplication, it is deduced immediately | that *if* the $(i - 1)$ -th row of the schema is ordered from left to right by decreasing logical strength (so that every member implies the subsequent one), then the same will hold for each half of the i -th row. On the other hand, nothing is said about the implicational relations between members from different half rows.¹⁰⁵ [28]

So, initially $N \rightarrow M$ holds, as is well known; *therefore* also $N^2 \rightarrow MN$ and $NM \rightarrow M^2$, whereas nothing is as yet said about the relation between MN and NM .

One can get one step further if one takes into account that $N \rightarrow W \rightarrow M$. Hence it follows by Rule II: $N^2 \rightarrow N \rightarrow MN$, and furthermore $N^3 \rightarrow N^2 \rightarrow MN^2$, $NMN \rightarrow MN \rightarrow M^2N$ and so on.

But we see that in this way a linear rank order is not obtained.

To this aim, we need the relation $MN \rightarrow NM$ ¹⁰⁶ or, to include also W in the schema: $MN \rightarrow W \rightarrow NM$. It is sufficient for this purpose to require $W \rightarrow NM$

drawn at stage k , add two child nodes, NA (to the left) and MA (to the right). For $k \geq 1$, the k -th row is the list (in the left-to-right order) of the nodes added at stage k . Clearly, the k -th row has 2^k nodes.

¹⁰⁴ [Eds.] Becker is pointing out a second, provably equivalent inductive way for generating the tree-like diagram, according to which the 2^{k+1} nodes of the $k + 1$ -th row are obtained from the 2^k nodes $A_1, \dots, A_{2^k}$ of the k -th row as follows (in the left-to-right order): $A_1N, \dots, A_{2^k}N$ (left half), $A_1M, \dots, A_{2^k}M$ (right half).

¹⁰⁵ [Eds.] On the basis of the second generation procedure (see the previous fn.), one has that *if* the members of the k -th row are linearly ordered w.r. to logical strength, i.e. if $A_1 \rightarrow A_2 \rightarrow \dots \rightarrow A_{(2^k)-1} \rightarrow A_{2^k}$, then, by Rule II, both the first and the second half of the $k + 1$ -th row are linearly ordered: $A_1N \rightarrow A_2N \rightarrow \dots \rightarrow A_{(2^k)-1}N \rightarrow A_{2^k}N$ as well as $A_1M \rightarrow A_2M \rightarrow \dots \rightarrow A_{(2^k)-1}M \rightarrow A_{2^k}M$. On the other side, no information can be derived as to the implicational relation between Σ and Π , where Σ (Π) is a node/modality in the first (second) half of the row.

¹⁰⁶ [Eds.] Notice, *en passant*, that the implication $MN \rightarrow NM$ considered here by Becker corresponds to the schema $\Box(\Diamond\Box A \rightarrow \Box\Diamond A)$ (or $\Diamond\Box A \rightarrow \Box\Diamond A$ in the unboxed version) that, added to the minimal normal modal system **K**, yields the system which is characterized, in Kripke's semantics, by the class of all frames in which the accessibility relation R is *confluent*, that is, satisfies the condition $\forall x \forall y \forall z \in W (xRy \wedge xRz \rightarrow \exists v \in W (yRv \wedge zRv))$.

or $W \rightarrow U^2$, i.e. “Brouwer’s axiom” (see above p. 74). Indeed, this axiom yields, by Rules II and III via left and right multiplication with F : $FNMF \rightarrow FWF$ or $MN \rightarrow W$ (since $FWF = FF = W$ (Rule IV) and, by $M = FNF$, $FNMF = FNFNFF = MN$).

Hence Brouwer’s axiom is just sufficient to order the first two rows (with W as middle element) in this way: $N^2 \rightarrow N \rightarrow MN \rightarrow W \rightarrow NM \rightarrow M \rightarrow M^2$, where W can be also omitted due to the transitivity of the “ $\rightarrow$ ”-relation.

From this we now immediately get, as above [i.e. by right multiplication with N], a linear rank order separately for each half of the 3rd row (where W does of course *not* occur), and indeed not only that but also, as one might fail to notice, the linear order of both half schemas up to the third row included, so e.g. the left one, as follows:¹⁰⁷

$$N^3 \rightarrow N^2 \rightarrow MN^2 \rightarrow N \rightarrow NMN \rightarrow MN \rightarrow M^2N$$

(Only $MN^2 \rightarrow N \rightarrow NMN$ could be not immediate, but this follows from $MN \rightarrow W \rightarrow NM$ by right multiplication with N , according to Rule II.)

Now, however, it remains *not* determined the relation between M^2N , N^2M and W , and this must be settled by a generalization of Brouwer’s axiom: $W \rightarrow N^2M$, from which $M^2N \rightarrow W$ follows immediately, as above.

Thereby the whole schema is ordered up to the third row.

One sees that we can continue in this way, provided we generalize Brouwer’s axiom stepwise, i.e. by postulating $W \rightarrow N^3M$, $W \rightarrow N^4M$, ..., $W \rightarrow N^iM$. It easily follows by complete induction *that the “generalized Brouwer’s | axiom”, i.e. $W \rightarrow N^iM$, is sufficient to linearly order (with the order type η of the rational numbers) the whole unbounded schema of MN -combinations, i.e. all positive modalities including W .*¹⁰⁸

[29]

The generalized Brouwerian axiom is plausible for the same reason as the simple one is: the necessity of the necessity etc. of a *possibility* is logically never as strong as the truth. And further, the possibility of the possibility etc. of the *necessity* of something that is not true, is also not given (cp. §4c above, p. 84).¹⁰⁹

¹⁰⁷ [Eds.] In the previous paragraph Becker has shown how, by using Brouwer’s axiom, one gets the linear ordering $N^2 \rightarrow N \rightarrow MN \rightarrow W \rightarrow NM \rightarrow M \rightarrow M^2$ of all the modalities contained in the first and the second row, plus W . Now, taking also into account the second procedure of generation of the tree, by right-multiplication with N (resp. M), one gets the linear ordering $N^3 \rightarrow N^2 \rightarrow MN^2 \rightarrow N \rightarrow NMN \rightarrow MN \rightarrow M^2N$ (resp. $N^2M \rightarrow NM \rightarrow MNM \rightarrow M \rightarrow NM^2 \rightarrow M^2 \rightarrow M^3$) of all the modalities occurring in the left (resp. right) subtree up to the third row.

¹⁰⁸ [Eds.] As soon as all the iterations $p \rightarrow \Box^n \Diamond p$ ($n \geq 1$) of Brouwer’s axiom are added to the calculus, the infinitely many modalities occurring in the whole tree-like diagram turn out to form, with respect to the relation “ $\rightarrow$ ” of implication, a *dense, countable linearly ordered set without first and last element*. Any such ordered set is order-isomorphic to the set $\mathbb{Q}$ of the rational numbers, whose order type is usually denoted by η .

¹⁰⁹ The choice of the implicative relation $MN \rightarrow W \rightarrow NM$ in place of $NM \rightarrow W \rightarrow MN$, which at first sight appears to be formally equivalent to it, is not devoid of a purely formal relevance.

(c) As a result, the *ranking problem* for our calculus is solved. The remaining second question is now the *reduction problem*. Our calculus has infinitely many irreducible modalities, as it follows already from the distinctness of $N, N^2, N^3, \dots$ ¹¹⁰ However, *this does not yet mean that all distinct MN-combinations represent different positive modalities*. In fact some types may be reduced to others. To which extent is this the case?

To start with, the simple Brouwer's axiom yields a proof of the *periodicity of the U-powers*. ($U = U^3, U^4 = U^2$; in general: $U = U^{2n+1}, U^2 = U^{2n}$.) The proof is the following, as we know: from $W \rightarrow U^2$ (Brouwer's axiom) it follows, by *right* multiplication with U according to Rule II, $U \rightarrow U^3$; on the other hand, by *left* multiplication with U we get, according to Rule III, $U^3 \rightarrow U$; thus $(U \rightarrow U^3) \times (U^3 \rightarrow U)$ or $U = U^3$.

This proof can be considerably generalized. Namely, let Λ be an *arbitrary negative* modality with the property $W \rightarrow \Lambda^2$; then right multiplication with Λ yields $\Lambda \rightarrow \Lambda^3$, while on the other hand left multiplication yields $\Lambda^3 \rightarrow \Lambda$, so $\Lambda = \Lambda^3$ as above. Analogously, $\Lambda^3 \rightarrow \Lambda$ and $\Lambda \rightarrow \Lambda^3$ follow from $\Lambda^2 \rightarrow W$, thus again $\Lambda = \Lambda^3$.

What is the most general form of Λ ? One can see from the *MN* schema that all the positive modalities which are implied by W (and *only* these) end with M . So Λ^2 , represented as MN -“product”, must end with M (in the “analogous” case, with N).¹¹¹

So, once represented in *FN*-form, Λ^2 has to contain an even number of F and end with F (since $M = FNF$). Thus Λ , in turn, contains an odd number of F and *is also ending with F*. Therefore Λ has the forms¹¹² [30]

$$\Lambda = N^{x_1} F N^{x_2} F \dots F N^{x_{2s+1}} F$$

Indeed, the essential principle which is used in the text to produce the linear ordering depends on this choice. A change would bring along drastic modifications of the ranking relations in the whole schema. Altogether, it can be questioned whether a linear rank order of *all* modalities is still possible. But the issue needs to be further investigated.

[Eds.] On can prove that with the variant $NM \rightarrow W \rightarrow MN$ suggested by Becker the system collapses, see Sect. 5.9.3 of the Appendix.

¹¹⁰ [Eds.] Unfortunately, Becker's claim appears to be not correct. See the discussion in Sect. 5.9.3 of the Appendix.

¹¹¹ [Eds.] In the two previous paragraphs Becker has proved that, in the extended system under consideration, (i) the negative modality U , that satisfies $W \rightarrow U^2$ by the simple Brouwer's axiom, obeys the reduction rules $U = U^{2n+1}$ and $U^2 = U^{2n}$, and (ii) more generally, *any* modality Λ that (*) is negative and satisfies $W \rightarrow \Lambda^2$, obeys analogous reduction rules. Now Becker is trying to identify the general form of such a Λ . The first step is to look at the “*MN* schema”, that is the tree-like diagram on p. 87, and observe that all the positive modalities which are implied by (resp., which imply) W are contained in the tree on the right (on the left) and end with M (with N). Since Λ^2 is positive it Λ is negative, it follows that, for any Λ satisfying (*), Λ^2 ends with M . Analogously, any negative modality Σ satisfying $\Sigma^2 \rightarrow W$ is such that Σ^2 ends with N .

¹¹² [Eds.] Λ has an odd number of F separated by powers of N . Since Λ ends with F , either the number of N -powers is odd and Λ begins with a N -power, or the number of N -powers is even and Λ begins with a F .

$$\text{or } \Lambda = FN^{y_1} FN^{y_2} F \dots FN^{y_{2s}} F$$

Hence:

$$\begin{aligned} \Lambda^2 &= N^{x_1} FN^{x_2} \dots N^{x_{2s+1}} FN^{x_1} \dots N^{x_{2s+1}} F \\ \text{or } \Lambda^2 &= FN^{y_1} FN^{y_2} F \dots N^{y_{2s}+y_1} FN^{y_2} \dots N^{y_{2s}} F \end{aligned}$$

Or, written in MN form:

$$\begin{aligned} \Lambda^2 &= N^{x_1} M^{x_2} \dots M^{x_{2s}} N^{x_{2s+1}} M^{x_1} \dots N^{x_{2s}} M^{x_{2s+1}} \\ \text{respectively } \Lambda^2 &= M^{y_1} N^{y_2} \dots M^{y_{2s-1}} N^{y_{2s}+y_1} M^{y_2} \dots N^{y_{2s-1}} M^{y_{2s}} \end{aligned}$$

For these and the “analogous” forms, i.e. for *all negative* forms Λ it holds: $\Lambda^2 = \Lambda^{2m}$; $\Lambda = \Lambda^{2m+1}$.¹¹³ Obviously, we get thereby far-reaching reductions. Some simple examples are:¹¹⁴

$$NMN = N; NMN^2 = N^2; N^2MN = N^3, MNMN = MN$$

The detailed consequences of this reduction are not easily assessable and therefore cannot be further considered here. There appears to be a difficult combinatorial problem here, which must be left over to the mathematicians.

Appendix to Part I

The Logic of Modalities and the Brouwer–Heyting

“Intuitionistic” Logical Calculus

A. Heyting has developed a calculus of high mathematical perfection in his essay “*The formal Rules of Intuitionistic Logic (Die formalen Regeln der intuitionistischen Logik)*”.¹¹⁵ As set out in the introduction, this intuitionistic formal logic asks for a comparison with the calculus of modalities (see above, p. 59 ff).

¹¹³ [Eds.] As proved three paragraphs above.

¹¹⁴ [Eds.] It is not clear how the (correct) reductions in the examples follow from the above general result. Actually, the first reduction $NMN = N$ is obtained as follows: from $W \rightarrow NM$ (Brouwer’s axiom) and Rule II (right-multiplication by N) one gets $N \rightarrow NMN$, and from $MN \rightarrow W$ (dual form of Brouwer’s axiom) and Rule III (left-multiplication by N) one gets $NMN \rightarrow N$. The second reduction follows from the first one by means of Rule II. The third and the fourth reductions follow from the first one by means of Rule III. Notice that there is a lapse in Becker’s text: “ $N^2MN = N^3$ ” should read “ $N^2MN = N^2$ ”.

¹¹⁵ Sitzungsberichte der Preußischen Akademie der Wissenschaften, phys.-math. Klasse, 1930 II, 3 ff. (Heyting 1930).

Heyting's calculus is obtained from Russell's classical elementary logical calculus by dropping all axioms, definitions and theorems which either depend on the principle of *tertium non datur* or have such principle as a consequence.

The four *fundamental notions* $\supset$ (implication), $\wedge$ (conjunction, "and"), $\vee$ (disjunction, "or"), $\neg$ (negation of a special kind) are introduced. *None of these fundamental notions can be reduced to the remaining others!* |

Further, the following 11 axioms hold for these notions:

[31]

- 2.1 $a \supset a \wedge a$
- 2.11 $a \wedge b \supset b \wedge a$
- 2.12 $a \supset b \cdot \supset \cdot a \wedge c \supset b \wedge c$
- 2.13 $a \supset b \cdot \wedge \cdot b \supset c \cdot \supset \cdot a \supset c$
- 2.14 $b \supset \cdot a \supset b$
- 2.15 $a \wedge \cdot a \supset b \cdot \supset b$
- 3.1 $a \supset a \vee b$
- 3.11 $a \vee b \supset b \vee a$
- 3.12 $a \supset c \cdot \wedge \cdot b \supset c \cdot \supset \cdot a \vee b \supset c$
- 4.1 $\neg a \supset \cdot a \supset b$
- 4.11 $a \supset b \cdot \wedge \cdot a \supset \neg b \cdot \supset \neg a$

All these propositions are valid in Russell's calculus, if one "translates" the symbols as follows:

Heyting : $\supset, \wedge, \vee, \neg$

Russell : $\supset, \cdot, \vee, \sim$

By contrast, many theorems and even definitions of Russell's calculus do not hold in Heyting's calculus.¹¹⁶ How is now Heyting's calculus related to the uncompleted and the completed Lewis's calculus?

First of all, the question concerning an appropriate "translation" of the symbols arises.

Since the H.-calculus is contained in the R.-calculus and, in turn, the R.-calculus is contained in the L.-calculus (if one translates: R.: $\supset, \cdot, \vee, \sim$; L.: $\subset, \times, +, -$), the answer seems easy: the H.-calculus (under the translation: H.: $\supset, \wedge, \vee, \neg$; L.: $\subset, \times, +, -$) is of course also contained in the L.-calculus. However, this is a worthless triviality. Indeed, the purpose of a comparison between intuitionistic and modal logic can only be to make the deficits of the former with respect to the latter comprehensible by interpreting the intuitionistic notions by the *specific* modal-logical notions, that is by Lewis's "strict" notions (*strict implication, strict logical sum, impossibility*).

This viewpoint suggests the following translation: H.: $\supset, \wedge, \vee, \neg$; L.: $<, \times, \wedge, \sim$.

¹¹⁶ [Eds.] There is clearly a lapse in Becker's text here: he writes "many theorems and even definitions of Heyting's calculus do not hold in Russell's calculus".

Once the comparison is carried out in these terms, it turns out, as expected, that in Heyting's axioms *more* is essentially assumed than in the Lewis's system. Unfortunately, *so* much is assumed under this interpretation, that the whole Lewis's system collapses. |

Namely, in Heyting's system it holds (Axiom 3.1 and Theorem 3.22, cf. p. 8 of [32] Heyting's essay):

$$a \vee a \supset \subset a$$

The corresponding L.-translation is:

$$p \wedge p = p$$

Now in the L.-system it holds:¹¹⁷

$$p \wedge p = \sim (-p \times -p) = \sim -p$$

(by definition L. (1.04), see above p. 62).

Hence the theorem in question means nothing other than $\sim -p = p$ does hold. So $N = W$. If we substitute p by $-p$, we get $\sim - -p = -p$ or $\sim p = -p$. I.e.: *The peculiarity of the Lewis's system vanishes and the latter is reduced to the ordinary classical calculus without "oblique" modalities.*

This result may seem paradoxical, yet it can be explained by the fact that the system of *primitive* notions in Lewis is different from Heyting's one. Lewis's system analyses the notions more deeply, indeed it has two negations (further to the negation F also one symbol for the *obliquitas* of the simple positive modality N) but in return it is able to represent all other relations solely by means of the logical product. This provides a bigger flexibility of the deductive technique and therefore, the "same" axioms are "stronger" in the L.-system than in the H.-system.¹¹⁸

Since both the translations discussed so far turned out to be unusable, a third one will be considered, which perhaps may accomplish our goal, as far as our present investigations enable us to claim.

Let now the correspondence be fixed as follows: Heyting: $\supset, \wedge, \vee, \neg$; Lewis: $\subset, \times, +, \sim$ (i.e. *material implication, logical product, material logical sum, impossibility*). So this time the "ordinary" relations are chosen, except the negation, which is translated as *impossibility*.

¹¹⁷ [Eds.] Recall that in Lewis's and Becker's notation the symbol " $\wedge$ " denotes the strict disjunction. Thus, the translation of $a \vee a \supset \subset a$ is, in our notation, $\Box(\Box(a \vee a) \leftrightarrow a)$, from which $\Box(\Box a \leftrightarrow a)$, that is the collapse formula, might be easily derived in S3.

¹¹⁸ On the other hand, the L.-axiom (1.5) $p < -(-p)$ is presumably not expressible in the H.-system, thus the H.-system should not completely contain the L.-system in the interpretation under consideration. It is anyway worthless to investigate this fact in detail, since this "translation" is in any case unusable because of the mentioned reasons.

This choice immediately leads to the consequence that the Heyting's axioms (1)–(9), all of which do not contain “ $\neg$ ”, can be directly | transcribed in the Lewis's system. This is because they hold in Russell's “*System of material implication*”, which in turn is fully contained in the Lewis's system. [33]

On the contrary, once we translate Heyting's $\neg$ as $\sim$ (*impossible!*), (10) and (11) do not hold in Russell's system.¹¹⁹

The question is now whether the translations of (10) and (11) hold in the original Lewis's system, or in an appropriate extension of it.

Translated into Lewis's notation, the axioms in question read as follows:

$$\begin{aligned} (10)' & \quad \sim p \subset (p \subset q) \\ (11)' & \quad (p \subset q)(p \subset \sim q) \subset \sim p \end{aligned}$$

(10)' does hold also in Lewis's system. Namely, since $\sim p \subset \neg p$ [is an axiom of Lewis's system], also $\sim p \subset \neg p$ (α) does hold¹²⁰ and, furthermore:

$$(10)^* \quad \neg p \subset (p \subset q)$$

since this holds in Russell's system. (The corresponding theorem in the “*Principia Mathematica*” has the number *2.21.) But (α) and (10)* imply (10)', using to the syllogistic principle, that also holds for the material implication.

By contrast, (11)' is apparently not provable in the original, unextended Lewis's system.¹²¹

This arises also by the fact that the theorem $\neg\neg\neg p \equiv \neg p$ is a consequence of Heyting's axioms (1)–(11) (H. 4.31 and 4.32).¹²² This theorem is indeed weaker than our $\sim\sim\sim p = \sim p$ ($U^3 = U$, cp. above, p. 69, 75, 90), since the former contains the “material” and not the “strict” equivalence.

At this point a further investigation must begin, with the aim to assess *whether and which additions must be made to the extended Lewis's system (Calculus of 10 Modalities, Calculus of 6 Modalities) in order to validate the translation of Heyting's axiom (11)*. Further problems can nevertheless arise because of the difference of the undefined notions in Heyting's and Lewis's system. The solution of these tasks and the overcoming of these difficulties shall be left to future work.

¹¹⁹ One should not forget that Russell used the symbol “ $\sim$ ” in the sense of Lewis's symbol “ $\neg$ ”.

¹²⁰ [Eds.] $(A \rightarrow B) \rightarrow (A \rightarrow B)$ is a theorem of **S3**.

¹²¹ [Eds.] Indeed, it is easily seen that (11)' does not hold even in **S5**. Actually, this is an immediate consequence of the following general fact: let **S** be any modal system extending **S2** (so that $T : \Box A \rightarrow A$ is a theorem of **S**). Then **S** + (11)' collapses. Take indeed the following instance of (11)': $(\Box p \subset p)(\Box p \subset \sim p) \subset \sim p$. Then by *T* and MP we get $(\Box p \subset \sim p) \subset \sim p$ and then, by $\neg\Box p \subset (\Box p \subset \sim p)$ and transitivity, $\neg\Box p \subset \sim p$, that is by contraposition $\Diamond p \subset \Box p$. By using (the dual form of) *T* again we get $p \subset \Box p$ and so the collapse.

¹²² [Eds.] Becker writes “ $\sim\sim\sim p \equiv \sim p$ ”, which is clearly a lapse.

Before concluding, however, let us give some hints. Heyting proves—thankfully—the mutual independence of his axioms and eventually also the unprovability of the *tertium non datur* in his system. For this purpose, he specifies | a system of three elements, 0, 1 and 2, and an associated “group”,¹²³ in which *all* axioms are satisfied whereas the *Law of excluded middle* is *not* satisfied (Heyting, l.c. p. 17, group XII).¹²⁴ He adds the following interpretation of the elements: “0 stands for an arbitrary true proposition, 1 for an arbitrary false proposition, and 2 for a proposition, which cannot be false but is also not proven.” Apparently, here Heyting is assuming three irreducible modalities, which exclude a fourth one. (*Quantum non datur!*) This remark goes together with the observation concerning a work of Glivenko,¹²⁵ who had proved:¹²⁶

[34]

1. If the proposition *a* is provable in classical logic, then the proposition “*a* cannot be false” ($\neg\neg a$) is provable in Heyting’s system.
2. If the proposition “*a* is false” is provable in classical logic, then it is provable in Heyting’s system ($\neg a$).

But this means that in Heyting’s system it holds: “Either $\neg a$ or $\neg\neg a$ ”, where $\neg\neg a$ means that *a* is either provably true or unprovable but true.¹²⁷

¹²³ By the well-known [matrixes] method which was developed by Bernays. Cf. Bernays, *Axiomatische Untersuchung des Aussagenkalküls der “Principia Mathematica”*. Math. Zeitschr. 25 (1926), p. 305 (Bernays 1926).

¹²⁴ [Eds.] Heyting’s “group” (matrix) XII features the three values 0, 1, 2, where 0 is the designated value, and the following truth-tables for the intuitionistic connectives (for binary connectives, the first element is above and the second is on the left):

$\supset$	0	1	2	$\wedge$	0	1	2	$\vee$	0	1	2	$\neg$	0	1	2
0	0	0	0	0	0	1	2	0	0	0	0				
1	1	0	1	1	1	1	1	1	0	1	2		1	0	1
2	2	0	0	2	2	1	2	2	0	2	2				

¹²⁵ Glivenko, Acad. R. de Belgique, Bull. de la classe des Sciences, s. 5, t. 15, p. 183 (1929) (Glivenko 1929).

¹²⁶ [Eds.] Here Becker mentions Glivenko’s main result, according to which a formula *A* is provable in the classical propositional calculus if and only if its double negation $\neg\neg A$ is provable in the intuitionistic propositional calculus. A straightforward consequence thereof (in view of Brouwer’s law $\neg\neg A \leftrightarrow \neg A$) is that a negative propositional formula $\neg B$ is classically provable if and only if it is intuitionistically provable.

¹²⁷ [Eds.] Becker seems to be assuming here that the schema $\neg A \vee \neg\neg A$ (*WEM*: *weak excluded middle*) is intuitionistically valid and, more generally, that intuitionistic propositional logic can be somehow seen as a *three-valued logic*. Both assumptions are however false. The schema *WEM* is *not* provable in the intuitionistic propositional calculus: the intermediate logic obtained by adding to *IPC* the axiom schema *WEM* (known as Jankov’s logic (Jankov 1968)) is a *proper* extension of *IPC*. As to the second assumption (which was indeed shared by other scholars at the time Becker was writing), a fundamental result proved in 1932 by Gödel (Gödel 1932), states that *IPC* cannot be viewed as a system of many-valued logic with a finite number of truth values: it is not possible to find a group of *n*-valued truth tables which characterizes the intuitionistic tautologies, in the

This suggests a close connection to the Calculus of Six Modalities. One would interpret the “provable truth” as necessity, the falseness as “impossibility” and the unprovable truth *either* as “variability” in the sense of MacColl (see above p. 57 f.), i.e. as $M \times M'$, or as “contingent” truth, i.e. $W \times M'$.

However, that nothing conclusive is obtained in this way is illustrated by Heyting’s table

$\neg$	0	1	2
	1	0	1

which says the following:

$$\neg 0 = 1; \neg 1 = 0; \neg 2 = 1 \text{ or:}$$

1. If a is provably true (so necessary), then $\neg a$ is provably false (impossible).
2. If a is provably false (impossible), then $\neg a$ is provably true (necessary).
3. If a is not provably true, i.e. not (provably?) false and also not provably true (i.e. if a has $M \times M'$ or $W \times M'$ as modality), then $\neg a$ is provably false (impossible). [35]

One cannot see clearly how these hypotheses, in particular the third one, can be interpreted in the language of the Calculus of Six Modalities. Especially, if one takes into consideration that $\neg a$ should be rendered as $\sim p$. Also the proper definition of the meaning of “2” which is given by Heyting is not completely clear. “2” shall mean a proposition “which cannot be false but also whose truth is not proven.” Does this mean that the proposition can neither be provably true nor *provably* false (so neither tautological nor contradictory, i.e. neither necessary nor impossible), or else that it can neither be provable true nor false *in general* (so it also cannot be contingently false), and consequently is contingently true?

Alas, we have to leave also this problem open to the reader’s keen wit.

II. The Logical Interpretation of the Mathematical Intuitionism from the Standpoint of Modality

With Special Consideration of E. Cassirer’s “Philosophy of Symbolic Forms”

Besides the formal parallels mentioned in the *Introduction*, the problem of modalities is also very important for the non-formal “philosophical” understanding of the intuitionistic foundation of mathematics. This becomes apparent in the

same way as the usual 2-valued truth tables characterize the classical tautologies. These wrong assumptions undermine the significance of the paragraphs that follow.

most recent philosophical examination, originated by Brouwer's investigations, of the critical exposition of intuitionism given by Ernst Cassirer in Part III of his *Philosophy of Symbolic Forms: Phenomenology of Knowledge*.¹²⁸ Such criticism achieves its proper sharpness through the introduction of the difference between possibility and actuality at a crucial point (*Ibid.*, 431 ff.).

Cassirer starts from the premise that intuitionism assumes the "primacy of the concept of function [*Funktionsbegriff*] over the concept of thing [*Dingbegriff*]" and obtains the individuals [that we call] "numbers" from the idea of *operation*.¹²⁹ The knowledge of the "law" is here prior, in the strict | sense, to the knowledge of "what is posited by the law".¹³⁰ The intuitionism is permeated with this *idealistic idea*, however, in the sense of a strict objective idealism. The object domain of mathematics must be grounded not in the psychological *act of counting* but in the pure *idea of number*.¹³¹ Furthermore, [Cassirer embraces] Brouwer's idea that

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¹²⁸ Berlin 1929 (Cassirer 1929). In particular Part III (of the 3rd volume), Ch. IV: *The Object of Mathematics*, 415 ff. (Page numbers refer henceforth to the original pagination.)

¹²⁹ [Eds.] Cp. Cassirer (1929), Part. III, Ch. 4, 430. As we said in Ch. 2, 30 ff., Cassirer, whether he knows it or not, is echoing Husserl at this place. Hereto cp., for instance, Husserl's *double lecture* presented to the Mathematische Gesellschaft in Göttingen in Winter 1901, Engl. transl. in Husserl (1970): "We now take up operation systems [*Operationssysteme*] which do not exclude the introduction of individuals that give rise to operative results" (422). Also cp. Husserl's *Philosophy of Arithmetic* (1891), Engl. transl. in Husserl (1970) 404: "Numbers are the lowest species of operations in a system of operations."

¹³⁰ [Eds.] Cp. Cassirer, *cit.* 430: "What is essential and crucial in this principle from an *epistemological* point of view is that with it the primacy of the *functional concept* over the *thing concept* is for the first time recognized in its full scope. If mathematics is led back to the primal intuition (*Urintuition*) of number, this intuition no longer means an intuition of concrete things but is taken as an intuition of a pure *process*. The starting point is a determinate sphere of *operations*; and it is these operations which first lead to the individuals that we designate as "numbers." The "existence" of these individuals is demonstrated and demonstrable only through the disclosure of a principle, in accordance with which they can be *posited ad infinitum* in accordance with a predetermined rule. It is this mode of positing alone that makes it possible to master them fully in thought, for here our knowledge of the "law" is in the strict sense prior to our knowledge of what is posited according to it. It is from the operational sphere of number that the thing-sphere of the countable and the counted first unfolds. Only if it is permeated with this *idealistic idea* and is understood as an expression of it can modern "intuitionism" fully develop and maintain its usefulness for the critique of mathematical foundations. The idealism itself, to be sure, must here be understood as a strictly "objective" idealism: the sphere of mathematical objects must be grounded not in the psychological act of counting but in the pure idea of number."

¹³¹ [Eds.] As we said in Ch.2, p. 33 f., it is worth considering the early Husserl's reading of the expression "to be grounded in the essence of" or "to be grounded in the concept of this or that" to grasp their use in Becker and Cassirer at this place. In the *Philosophy of Arithmetic*, for instance, Husserl speaks of "operations which are grounded in the idea of the cardinal number" (Husserl 1970, 435). In the *Logik Vorlesung 1986*, (Husserl 2001, 241), Husserl writes: "the general logical laws divide in several groups: in laws which *have their roots* in the concept of proposition, in laws which *have their roots* in the concept of concept, in laws which *have their roots* in the concept of object." A law is *grounded in a concept* if and only if it is analytically included in that concept. So the laws of commutativity and associativity, for example, are grounded in the concept of addition by being analytically included in this concept. Hereto cp. Centrone (2010), 109ff.

all mathematics is “far more an activity than a theory”, under the assumption, however, that the mathematical activity is a purely intellectual one; it does not *occur* in time but first *makes possible* a fundamental factor underlying time itself, the factor of “ordering”. Mathematical activity may therefore not be dissolved into an “aggregate of individual actions which stand to one another in a relation of *empirical* ‘succession’,” so that they can only ‘*gradually*’ build up a totality”. But, rather, the whole is prior to its parts and stands at the beginning as “*principle* of operation”, as generative law. By contrast, Brouwer demands that every mathematical statement of the form “there is” be grounded in an *individual act* of “giving” and, in this way, he blurs the boundaries between the *purely ideal given* and the *empirical given*. One should here go back to a well-known distinction made by Leibniz: just as Leibniz distinguished between *observabilité* (possible observation, observability) and *observation* (actual observation),¹³² one should, nowadays, introduce, besides the concept of (*actual*) construction also the concept of “constructability” (*possible* construction).¹³³ To grasp the concept of mathematical existence, the *possibility* of the construction, or—as one also may say—the possibility of the proof (provability, “*démonstrabilité*”), should be sufficient. This amounts to the claim that for the existence of a set just “*the definiteness of extension* [*Umfangsdefinitheit*]” and

¹³² Leibniz’s correspondence with Clarke: Leibniz’s fifth letter, §52 [Leibniz 1875, VII, p. 403].

¹³³ [Eds.] Leibniz is answering at this place to the following remark by Clarke: “13. If the World be *Finite* in Dimensions, it is *moveable* by the Power of God; and therefore my Argument drawn from that *moveableness*, is conclusive. *Two Places*, though *exactly alike*, are not the *same Place*. Nor is the *Motion* or *Rest of the Universe*, the same *State*; because a man shut up in the Cabin cannot perceive whether the Ship sails or not, so long as it moves uniformly. The *Motion* of the Ship, though the man perceives it not, is a *real different State*, and has *real different Effects*, and upon a *sudden stop* would have *other real Effects*; and so likewise would an indiscernible Motion of the Universe. To this Argument, no Answer has been given. It is largely insisted on by Sir Isaac Newton in his *Mathematical Principles* (Definit. 8) where, from the Consideration of the *Properties, Causes, and Effects* of Motion, he shows the difference between *real Motion*, or a Body’s being carried from one part of Space to another; and *relative Motion*, which is merely a change of the *Order or Situation* of Bodies *with respect to each other*. This Argument is a Mathematical one; showing, from *real Effects*, that there may be *real Motion*, where is *none relative*; and *relative Motion*, where there is *none real*: And is not to be answered, by merely asserting the contrary.” (Leibniz 1875, VII, 384). Leibniz answers as follows: “52. Pour prouver que l’espace sans les corps est quelque réalité absolue, on m’avoit objecté que l’univers matériel fini se pourroit promener dans l’espace. J’ay répondu, qu’il ne paroist point raisonnable que l’univers matériel soit fini; et quand on le supposeroit, il est deraisonnable qu’il ait du mouvement, autrement qu’en tant que ses parties changent de situation entre elles: parce qu’un tel mouvement ne produiroit aucun changement observable, et seroit sans but. Autre chose est quand ses parties changent de situation entr’elles, car alors on y reconnoist un mouvement dans l’espace, mais qui consiste dans l’ordres des rapports, qui sont changés. On replique maintenant, que la verité du mouvement est independante de l’observation, et qu’un vaisseau peut avancer sans que celui qui est dedans s’en apperçoive. Je reponds que le mouvement est independent de l’observation, mais qu’il n’est point independent de l’observabilité. Il n’y a point de mouvement, quand il n’y a point de changement observable. Et même quand il n’y a point de changement observable, il n’y a point de changement du tout. Le contraire est fondé sur la supposition d’un espace reel absolu, que j’ay refuté demonstrativement par le principe du besoin d’une raison suffisante des choses.” (Leibniz 1875, VII, 403–404).

not the “*decidability* [*Entscheidungsdefintheit*]” thereof¹³⁴—as in Brouwer—is required, which leads to the “semi-intuitionistic” position.¹³⁵ |

If one wants to choose between the “semi-intuitionistic” (logical-idealistic, or objective-idealistic) and the proper “intuitionistic” (the *so called* “empirical”) view, one is immediately led, as one sees, to a problem of modalities. Indeed, the question arises: Does the distinction between (non-empty, “positive”) possibility and reality make sense at all in the field of mathematics? Can the positive *possibility* of a construction or of a proof be shown, in mathematics, in some other way—than exhibiting de facto the construction or the proof?

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In the realm of empirical observation and historical knowledge there are quite many degrees [*Stufenfolgen*] of more or less probable “*real*” possibilities, up to the absolute *real* impossibility. One can still believe as really possible that humans from the future visit other planets with space ships, or at least circumnavigate the moon and directly observe its far side, which, at present, is inaccessible to us. One would take it for very improbable, although not completely impossible, that a Petronius or a Livius would be born again. But one would no longer take it for really possible that humans reach the center of the earth or of the sun or of a far fixed star, perhaps even of a nebula. Just as little as it may ever happen to us again that we are given a poem written in a Proto-Indo-Germanic language. Nevertheless, it still belongs to the *sense of reality* of those celestial bodies and of that proto-language, which we all take to be [or to have been] existent, that it is

¹³⁴ [Eds.] Becker is distinguishing here two criteria for set-existence, using the terminology he adopted in Becker (1923), 403 ff., while analyzing and elaborating Husserl’s concept of *definite manifold*. The first criterion, the *definiteness of extension*, reflects Russell’s *vicious circle principle* and the ban of *impredicative* definitions: in order to be said to exist, a set X must be *predicatively* defined, that is, defined as the set of all objects x that satisfy a certain condition $A(x)$, $X = \{x \mid A(x)\}$, where the condition $A(x)$ does not contain quantifications ranging over domains to which X belongs. The second criterion, the *decidability of extension*, is much more restrictive: in order to be said to exist, a set X must be *decidable*, i.e. an effective procedure (an algorithm) must be given by means of which one can decide, for any object a , whether a belongs to X or not. As to Becker’s distinction between *Umfangsdefintheit* and *Entscheidungsdefintheit* see, among others, Hesseling (2003), 148 ff.

¹³⁵ This is Weyl’s position in 1918, in his *The Continuum* (Weyl 1918). In 1921 (in the paper “The new Foundational Crisis of Mathematics”, Math. Zeitschr. Vol. 10 (Weyl 1921), 53) he basically adopted Brouwer’s intuitionism, while, later on (in his “*Radbemerkungen* [zu Hauptproblemen der Mathematik]”, Math. Zeitschr. 20 (1924) (Weyl 1924), in the “Symposium” article [*Die heutige Erkenntnislage in der Mathematik*] (Weyl 1925) and in his essay in the “Handbook of Philosophy” (Weyl 1927), Sec. II A) he added the idea of “symbolic mathematics”. The place that Cassirer cites from the article in Math. Zeitschr. 10, 53 (Cassirer 1929, 433 f.) is insofar misleading as it stems from a flashback in the “semi-intuitionistic” position of 1918 that Weyl, a little later in the same text, *explicitly* rejects in favor of Brouwer’s intuitionism. The quotation [from Math. Zeitschr. 10, 54] in Cassirer (1929), 434, in which the peculiar concept of “abstracts of judgment” is applied to existential mathematical judgements, belongs to the second (intuitionistic) part of Weyl’s paper, whence the incongruity of the views that comes to light in the two quotations. I can hardly believe that there are essential differences between the *second* of Weyl’s positions (from 1921) and Brouwer’s position: both authors require the *effective* presentation of a construction by the proof of existence, that is, as far as sets are concerned, the “decidability [*Entscheidungsdefintheit*]”.

conceivable (imaginable in free phantasy) that an observer be magically carried in such inaccessible space or put back in those primitive times by a magician. This “conceivability” is essentially *more* than just consistency. For, it represents the possibility of reproducing something in imagination in all details.

In the realm of pure mathematics there is nothing similar. If one wishes, one can call a construction or a proof “possible” | if one can show that the *assumption* of its provability, or of its existence, are consistent with the other accepted premises. But, with this, it is not at all said how this “possible” construction or this “possible” proof should look like. While in the “empirical” case a fictional observer could perform fictional observations, in the “purely” mathematical case an imagined construction and a “real” construction would be immediately equivalent, they would not be distinguishable. Nor a mathematical proof loses its validity because “I found it in dreams” and not by methodic working in the awake state.¹³⁶ [38]

There is —this is the result of my previous considerations—*no* analogy between the *observabilité* of real objects in the sense of Leibniz and the “*constructabilité*” or “*démonstrabilité*” in the field of mathematics.

It seems not to be possible to give to the last two concepts a “*positive*” content, besides the “*privative*” one of *non-contradictoriness* [*Widerspruchslosigkeit*]. Indeed, the difference between the *positivity* of any even so fanciful “real” possibility or, still, of any just “conceivable” real impossibility, and the *privation* which lies in the mere freedom from contradiction [*Widerspruchsfreiheit*], is crucial. As far as “non-contradictoriness” is concerned, this is not only a *privative term*. It is *essential* for the recent consistency proofs of the Hilbert school that they *merely* prove the *impossibility* of any contradiction, the *impossibility* of the occurrence of the formula $0 \neq 0$, or of any other equivalent inconsistent statement, in a valid formal-mathematical deduction. Hilbert’s method makes nothing more. Neither it is able, nor it aims at all to provide “ideal” structures that would complete, in a somehow *formal-ontological suitable* way (say, as set-theoretical constructions) the already introduced “intuitionistic” structures. Therefore, | [Hilbert’s method] is fundamentally different from *all* previous methods for introducing “ideal” mathematical objects.¹³⁷ [39]

If one compares what we have said with the important observations Cassirer (*Ibid.*, 452 ff.) does on the question of the ideal elements in mathematics, one is confronted again with a result that it is not possible to endorse completely, even if one agrees with his general methodological approach. Indeed, Cassirer too tries

¹³⁶ It is not a matter of “chances” of finding a proof. To search for a proof of Fermat’s theorem is, after all, much more promising than trying to reach Venus on a space ship. Moreover, one has good chances to figure out the *kind* of that number-theoretical proof—not the proof itself!—, but nevertheless, in the sphere of pure mathematics, a modal intermediate stage between consistency and “factual” presentation of the proof is not conceivable.

¹³⁷ In particular [Hilbert’s method] is fundamentally different, as it must be here emphasized against Adolf Fraenkel, from Cauchy’s and related methods for introducing the imaginary, which all use certain set-theoretic constructions. For further information see my article in “Philos. Anzeiger”, vol. III, 369 ff.; especially 377, fn. 3 (Becker 1929).

to single out a *more positive* characterization of the sense and the justification of the addition of ideal elements, beyond the mere non-contradictoriness and thus, he also is eager to connote the concept of “possibility” of a “formalistic” construction in a *positive* way. Cassirer assumes that, for Hilbert, to add ideal (transfinite) propositions to the finitary propositions is justified by the consistency of the whole theory once the addition is made, and by a sort of permanence of the formal laws.¹³⁸ Cassirer himself feels the need to require more than the mere non-contradictoriness when considering the issue from the standpoint of the “philosophical critique of knowledge”. He intends to go beyond the idea of “a combination free from contradiction”, of the “merely formal compatibility”, towards “a guarantee for a true inner conjunction”, for a “homogenous logical structure of mathematics”. This can only be secured if the new added elements do not *just* occur *beside* the old ones, but “are a systematically necessary unfolding of the latter”. And this requires “a primary logical kinship” of the old and new elements, in virtue of which the implicit *sense* of the old elements is made explicit through the new ones. Not just the sphere of mathematical objects is expanded, but its foundations are more radically grounded. Every new “expansion” amounts here to a “logical intensification”. Even if the *ratio essendi* of the ideal constructs [*Gebilde*] must be sought in the realm of the old ones, still the *ratio cognoscendi* of the latter lies in the ideal elements. Cassirer thinks here, specifically, of the clarification of non-transparent relations in the field of real numbers, when considering these from the standpoint of complex numbers (fundamental theorem of algebra, circle of convergence, etc.) and similar things (ideal numbers, ideal and imaginary | constructs in geometry, etc.). From the standpoint of the logic of relations we are dealing with *new* systems of *relations* that have the *old* ones as a point of departure, with “systems of systems”, systems of higher order and richer structure, in which the old systems are embedded, and through which they become more comprehensible in their own structure. The ideal elements are composed “of no different logical stuff than these elementary objects”; the judgments we frame concerning the ideal elements can “*always* [!]”¹³⁹ be formulated in a way *that permits them to be transformed back into judgments concerning the first class of objects*, except that now it is not individual objects,

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¹³⁸ [Eds.] If one looks at Cassirer’s original version (Cassirer 1929, 455) and not only at Becker’s presentation thereof, it is quite clear that Cassirer interprets Hilbert’s position as follows. It is possible to expand a certain numerical field with ideal elements, if the theory of that field is *conservative* with respect to the old elements: “Hilbert—so Cassirer—has recently insisted on his view that number theory can never be truly worked out unless we decide to adjoin the ideal propositions of mathematics to its finitary propositions. The right to do so is adequately secured if the mathematician can show on the one hand that the new objects which he adopts obey the same formal laws of combination that had been established for the old ones, and if he can furthermore prove that the addition of the new, ideal elements can never give rise to contradictions in the old, restricted sphere—that the relations prevailing among the old elements when the ideal elements are eliminated are still valid.” Cassirer explicitly refers, at this place, to Hilbert’s famous talk of 1926 *Über das Unendliche* (Hilbert 1926, 174 ff., 179).

¹³⁹ [Eds.] Emphasis and exclamation mark by Oskar Becker.

but *groups* and *totalities* that function as the subjects of these judgments”¹⁴⁰ (real numbers as “cuts” or “segments” of rational numbers as well as complex numbers as ordered pairs of real numbers, and so on;—*in any case*, certain set-theoretical constructions!).

These considerations have been reported so extensively since—as we will show immediately below—they indirectly throw light upon the *relation between intuitionism and the logic of modalities*. Cassirer, in his interpretation of the ideal mathematical constructs, explicitly considers Hilbert’s “ideal propositions”. Indeed, he takes them as point of departure for his exposition of the whole question (*Ibid.*, 455). However, though his interpretation of the logical “value” of the *other* ideal elements hits the nail on his head, he is wrong, in our opinion, as far as the ideal *propositions* are concerned. Judgements upon Hilbert’s “ideal propositions” can *in no way* be transformed back into judgments concerning the “first class of objects”, i.e. into finitary propositions (which hold also for the *intuitionistic* logic). *There is no—even so complex—“collection” of “finitary” propositions that is able to represent a “transfinite” (“ideal”) proposition.* Thus, the parallel between the “ideality” of the transfinite propositions in Hilbert’s mathematical logic and the [ideality] of the other “ideal” constructs, known up to now, collapses at a crucial point!

Anyway, one could perhaps still take this failure as ultimately irrelevant. One could give up the concrete thesis | of “transforming back” judgements of the new field into judgments of the old field and still take as correct Cassirer’s fundamental thesis of the “logical intensification” through the addition [of ideal elements] as well as his belief that the *ratio cognoscendi* of the old constructs lies in the new ones—notwithstanding it may be difficult to understand this properly in single cases. But even so the “facts” (*sit venia verbo!*) are against Cassirer! For, Hilbert’s logic “extended” with the ideal propositions is nothing else than the “ordinary” logical calculus of Schröder, Peano, Frege and Russell-Whitehead. It is *much simpler* than the “intuitionistic” logic, as recently set up in detail by Heyting. By no way, Hilbert’s system is to Brouwer’s and Heyting’s system as complex algebra is to real algebra! For instance, in Heyting there are *three primitive* irreducible *modalities*, while in Hilbert there are only *two* primitive irreducible modalities (namely, the well known truth values: true and false).¹⁴¹ [41]

One cannot find the new and much more complex logical relations of intuitionistic logic enclosed as “special cases” in those of classical logic. On the contrary, it

¹⁴⁰ *Ibid.*, 460.

¹⁴¹ [Eds.] The three intuitionistic irreducible modalities Becker is alluding to are those expressed by “*p*”, “ $\neg p$ ” and “ $\neg\neg p$ ”; see also our comment in fn. 127. A formula containing exactly one propositional variable *p* may be seen as expressing a modality (acting on *p*). In classical propositional logic, up to logical equivalence, there are only two such irreducible modalities, *true* (*p*) and *false* ($\neg p$). Quite differently, as proved in Nishimura (1960), in intuitionistic logic there are *infinitely many* non equivalent formulas in one propositional variable, thus—using Becker’s terminology—*infinitely many primitive irreducible modalities; not only three*.

is classical logic that can be seen, in a sense, as a “part” of the new logic,¹⁴² just like Euclidean geometry can be seen as a part of Riemann’s geometry.¹⁴³ Moreover, it cannot be irrelevant that classical logic has been known since long ago, while intuitionistic logic has been just discovered, or rather, has not yet been completely discovered—in direct contradiction to the historical relation of the “old” and “new” ideal constructs *everywhere* else, as even the terminology shows.

Nevertheless, the relationship of intuitionistic logic as a whole to its “classical” part has something paradoxical | as long as the concept of the former is determined solely by the “*factum*” of the intuitionistic mathematics—this is also shown by Brouwer’s derivation of classical logic from the *finite* set theory, while intuitionistic logic corresponds to the *transfinite* set theory.¹⁴⁴ [42]

Here the question arises of determining the difference between the two “logics” by intrinsic logical, or better, *genuine logical means*. But it’s not just the conception of two “logics” an intolerable paradox? At this place MacColl’s idea¹⁴⁵ turns out to be useful to set up a many-valued, not bivalent formal logic that takes into account *all* modalities. Such “many-valued” logic seems to be the *legitimate expansion of classical logic* and it is rather *this* expansion that should be compared with that of real algebra to complex algebra. Of course such many-valued logic *is not in need of*

¹⁴² [Eds.] The set of classical tautologies is a proper subset of the set of intuitionistic tautologies: so, in this extensional sense, intuitionistic logic is a proper part of classical logic. Becker is considering here the relation between the two logics from a different, more deep perspective. According to Glivenko’s Theorem of 1929 (Glivenko 1929), which was known to Oskar Becker, classical *propositional* logic can be seen, “under a suitable translation”, as a proper “part” of intuitionistic *propositional* logic—as Becker is maintaining here: a propositional formula A is a classically valid if and only if its double negation $\neg\neg A$ is classically valid. Glivenko’s translation does not work at the quantificational level. A different translation of classical predicate logic into intuitionistic predicate logic, the so called “negative translation”, was discovered a few years later by K. Gödel (Gödel 1933b) (see also Gentzen 1936). Under the negative translation, atomic formulas P are translated to $\neg\neg P$, disjunctions $B \vee C$ to $\neg(\neg B \wedge \neg C)$ and existential formulas $\exists x B$ to $\neg\forall x\neg B$.

¹⁴³ Both in the case of the “non-Euclidean” geometries and in that of analogous geometries (“non-Archimedean”, “non-Pascalian”, “non-Desarguean” geometry, etc.) we are confronted with something very similar to the “non-Aristotelian” (intuitionistic) logic. The *rejection* of certain axioms leads to new fields of a much *richer* structure than the classically known discipline. However, also here, there is a significant difference between logic and geometry: the constructs of the “pathological” geometries can in most cases (perhaps always?) be represented through more complex constructs in the Euclidean system (straight lines, for instance, through families of circles, etc.), *in logic this is not possible!* The reason is to be found, in part, in the “nominalism” of (abstract) geometry. One can “call” families of circles or other constructs “straight lines”—without too many worries: but within the field of formal logic, which is in some way eternally *absolute* and non-conventional, one cannot call “ideal proposition” something that *is not* at all a proposition!

¹⁴⁴ Notice that here the roles that finite and transfinite play in Hilbert are reversed: in Hilbert the classical logic is the transfinite logic and it also grounds the transfinite set theory of the formalistic conception, that Brouwer rightly questions because of a too extensive analogy with the finite. (Hereto cp. the reproach of “existential absolutism” that Weyl levels against Cantor’s classical set theory!)

¹⁴⁵ See above p. 57.

ideal elements, of introducing “ideal” modalities beside the truth and the false, it just needs to reflect on the very “real”—i.e. immediately grounded in the most concrete logical “phenomena”—modalities of possibility, necessity, impossibility, and so on. If a complete reduction of “intuitionistic logic” to modal logic had already been achieved—a goal that has not been completely reached up to now—there would be nothing paradoxical in the “factum” of the intuitionistic logic, for this “factum”—or this at best “*contingent-a priori*” phenomenon—would have been proved to be a pure necessity.

Considered from the standpoint of modalities, also the problem of *mathematics and temporality* finally receives a strong, though unilateral, clarification. Here we will briefly go into this more in detail, since Cassirer sees in this point the decisive difference between “logical idealism” and “hermeneutic phenomenology”, as far as their application to the problem of mathematics is concerned.¹⁴⁶

The “logic” of modalities has an inner relation to temporality. One can even say that the philosophical “discovery” of the “*modi obliqui*”, *possibility* and *necessity*, which have acquired their ontological acknowledgement only after the “*modus rectus*”, the “*truth*” (reality) (and its opposite, the “falsity” or “unreality”), represent a first step towards the discovery of the “authentic” temporality. For, the characteristic “*modi*” of the latter, the *future* and the “*past*”, can be literally designated as “*modi obliqui temporales*” as opposed to the temporal “*modo recto*”, that is, the “*present*”.¹⁴⁷ The “future” is indeed the “possible” and so is the past, in so far as it is an “*is as already been* [Gewesenes]” or, to put it pointedly, an “*is in the process of having been* [Gewesendes]” (Heidegger). This holds for the “*historical*” temporality, that is, for the “authentic time” (Heidegger).¹⁴⁸ “Natural”

¹⁴⁶ I refer here not only to Cassirer’s observations mentioned above (see p. 97) but also to his critical remarks (*Ibid.*, 469, fn. 3) on my monograph “Mathematical Existence” (in vol. VIII of this Yearbook (Becker 1927)). I do this the more readily because these critical remarks, at a variance with some other criticism raised on my monograph, basically share the same “idealistic” position and, moreover, also show a very deep understanding of the philosophical perspective that I have tried to outline. However, I believe I will stay true to my “anthropological” standpoint, notwithstanding Cassirer’s critique.

[Eds.] Cassirer’s remarks are reproduced below, p. 112.

¹⁴⁷ The expressions “*modus rectus – obliquus*”, already used previously [see pp. 82, 93], are obviously taken from the Latin grammar; but they are not modeled on the names “*casus rectus – obliquus*”, but rather on the names of the “*verbal moods*”: the *indicative* mood is the “direct” mood, the *subjunctive* is the “indirect” (or “oblique”) mood. Also the distinction between “direct” and “indirect” speech (*oratio recta – obliqua*) is similar. Indeed, the *modi verbi* refer in principle to the actuality (*indicative*) or, respectively, to the possibility or the impossibility (*Potentialis, Irrealis*) of a *process* (active or passive), i.e. of a *temporal being* (*verb* = temporal word). The “*tempora*” of the verb, not its *modi*, seem to denote linguistically the temporal modalities. However, the “*tempora*” only denote the non-analyzed or already constituted temporality, that is a temporality that has already been subsumed under the schema of an objective process (*event*). The *origin* of time, the “*enactment-character*” of its peculiar “*temporalizing itself*” is rather rendered through the *modi verbi*.—Hereto cp. below, p. 111, my remark on F. Brentano’s linguistic use.

¹⁴⁸ Historical “repetition” means to make “possibilities” available again, or to put oneself before those possibilities again, that is, to make them into authentic possibilities, to make “futures” out

temporality, in which the same events recur over and over again, can be seen—from the standpoint of historical temporality—as determined through the modality of necessity.¹⁴⁹

From its own perspective—inasmuch as this is possible—pure “Nature” does not know the difference of the logical modi, in particular, she does not know the “*modi obliqui*”. The mere “reality” solely “is”, like the pure present, the “pure being-there”.¹⁵⁰ [44]

The Parmenidean ontology only accepts the Being as “presence”, as eternal present; at a variance, the Aristotelian ontology accomplishes, through the *modal* distinction potentiality—energy, the first tentative explanation of the phenomenon of the historical time. Aristotle however, as well as Hegel much later, when explicitly dealing with time (Physics, Book IV, c. 10–14) does not develop completely the new concept of δύναιμις, but rather remains tied to the Parmenidean and Platonic tradition at crucial places.¹⁵¹ However, if we restrict ourselves to consider Aristotle’s investigations on potentiality and actuality (above all, in *Metaphysics*, book Θ), we can say that the actual being of a “present” potential being appears, from the latter’s point of view, as a future being, but also as an “is as already been [*gewesenes*]” (think for instance, of a child, an adult and a forefather). Or, formulated more in accordance with what we have said previously: “the potential being” (δυνάμει ὄν)—the child—“will be (actual)” —as an adult—and, therefore, he is “in the time to come” (pregnant with future) but he is also an “already been [*gewesenes*]”—like

of them. (Nietzsche, *On the Advantage and Disadvantage of History for Life* (Nietzsche 1874); Heidegger, *Being and Time I*, (Heidegger 1927), Part II, Ch. 5.)

¹⁴⁹ The predicted solar eclipse, for instance, occurs “necessarily”; each tragic “fate” is necessary; the unavoidability of an instinct stimulus is necessary, or “fatal”; Nietzsche’s “eternal recurrence” is necessary.

¹⁵⁰ Already Aristotle has distinguished three essential meanings of necessity (ἀναγκάσιον):

1. that without which, as a condition, a thing cannot live [as breathing and food for an animal],
2. the “compulsory” and compulsion,
3. the “mere” necessary, which has nothing to do with the necessity in the first sense.

This third form of necessity is the proper one and the most important. (Met. Δ, 1015 b 11: ὥστε τὸ πρῶτον καὶ κυρίως ἀναγκάσιον τὸ ἀπλοῦν ἐστίν [Therefore the necessary in the primary and strict sense is the simple].) This is the form in which the “natural” is *simply* necessary and just because of this “simplicity” it stands out “by itself”, i.e., from the perspective of the natural, not at all as “indirect modality” of that of the simple being-there [*Da-Sein*]: Cp. in addition to Met. Δ 5 also the very meaningful remark by Aristotle in Met. Λ, 1072 b 11–13: τὸ γὰρ ἀναγκάσιον τοσαυταχῶς, τὸ μὲν βίαι ὅτι παρὰ τὴν ὁρμὴν (2), τὸ δὲ οὐ οὐκ ἔστιν (1), τὸ δὲ μὴ ἐνδεχόμενον ἄλλως ἀλλ’ ἀπλῶς (3). [For the necessary has all these senses—that which is necessary perforce because it is contrary to impulse [(2)], that without which the good is impossible [(1)], and that which cannot be otherwise but is absolutely necessary [(3)].

¹⁵¹ In Hegel, the idea of the “dialectical process” has an historical-temporal character. On the contrary, the interpretation of time in the part of the *Encyclopaedia* (Hegel 1817) (II, 1 A b, §§ 257–259) dedicated to the philosophy of nature is essentially a reproduction of the Aristotelian doctrin. (Cp. Heidegger, *Being and Time*, (Heidegger 1927) § 82a; Becker, *Mathematische Existenz* (Becker 1927, 668.))

his adult father—, and thus he is, at the same time, “in the process of having been [*Gewesendes*]” (since he “repeats” the existence of the father). (Met. Θ,8).¹⁵² |

We will not go into the “modal” differences between natural and historical temporality, at this place. What has been said is anyway sufficient to highlight the role of the so-called “logical” modalities in the analysis of the essence of temporality. Let us just add that also Heidegger’s analysis of being and time makes use, at a crucial place, of the concept of the *modus obliquus* “possibility”.¹⁵³ [45]

As a modal category of presence-at-hand, possibility signifies what is *not yet* actual and what is *not at any time* necessary. It characterizes the *merely* possible. Ontologically it is on a lower level than actuality and necessity. *On the other hand, possibility as an existentiale* [as a category] *is the most primordial and ultimate positive way in which the Being-there [Dasein] is characterized ontologically.*

One might perhaps briefly identify the fundamental trait of Heidegger’s ontology, compared to the Aristotelian one, through the similar and yet different roles attributed, each time, to the modalities possibility and actuality. The concept of dynamis, by Aristotle, is indeed the properly new and propulsive concept, when compared with the older concepts of energy (ἐργον) and entelechy (τέλος and ἔχειν).¹⁵⁴ However, it is those actuality-concepts that mark in Aristotle the authentic Being; the δυνάμει ὄν is indeed no longer a “non-being”, but yet, something of non-being still inheres in it: what *can* be may also not be.¹⁵⁵ But according to Heidegger, the *possible Being* (the potential Being [*das Seinkönnen*]) | is clearly primary; [46] it constitutes the proper *sense* of the Being [*Sein*] as well as of the Being-There [*Dasein*] (i.e. the “existence [*Existenz*]”). Only in this way the idea of dynamis has fully prevailed and fully overcome the “Eleatism”. The inner connection between

¹⁵² Cp. moreover the crucial distinctions between τὸ ἐνδεχόμενον καὶ ἄλλος ἔχειν [what is possible but not necessary] and its opposite, to which both the αἰεὶ (ἐξ ἀνάγκης) ὄν [what always (necessarily) is] and also the γεγονός [what has happened] belong, as well as the relation of the latter to the προαίρεσις and to the future in book VI of the *Nicomachean Ethics* (Aristotle 2014c), cp., for instance, Z. 2, 1139 b 5-9: οὐκ ἔστι δὲ προαιρετὸν οὐδὲν γεγονός . . . οὐ δὲ γὰρ βουλευέται περὶ τοῦ γεγονότος, ἀλλὰ περὶ τοῦ ἐσομένου καὶ ἐδεχομένου, τὸ δὲ γεγονός οὐκ ἐνδέχεται μὴ γενίσθαι [Choice is not concerned with what has happened already [. . .]; for neither does one deliberate about what has happened in the past, but about what still lies in the future and may happen or not; what has happened cannot be made not to have happened] (and of course, just as little, the αἰεὶ ὄν).

¹⁵³ *Being and Time* (Heidegger 1927), I, 143–144. [Eds.] Emphasis added by Oskar Becker in the last two lines.

¹⁵⁴ To avoid misunderstandings: it is not a matter of *words*. It is not meant that “δύναμις” is an archaic word, while “ἐνέργεια” and “ἐντελέχεια” are neologisms. It is a matter of the *ontological thoughts* expressed by those words. The Being, labelled as pure “ἐνεργεῖν”, is nothing else than the old, self-sufficient, “everlasting” (Heidegger) Being of Parmenides; yet, that the δυνάμει ὄν is an ὄν and not a μὴ ὄν, and that the “oblique” mode of being of the possibility is on a par with reality—even if reality maintains priority—is, even compared to Plato, decisively new and makes, among others, the whole positive doctrine of the movement (κίνησις) possible. [Eds.] Quotations marks added by the editors.

¹⁵⁵ Met. (Aristotle 2014b) Λ 6, 1071 b 19: ἐνδέχεται γὰρ τὸ δυνάμει ὄν μὴ εἶναι [that which is *potentially* may possibly not be], cp. Θ 3 (1047 a 35–37); Θ 8 passim.

the “hermeneutical” analysis of time and this change of the modal schema is evident: the “future as the *having-been coming* [*gewesende Zukunft*]”, the unity of the *modi obliqui temporales*, is the decisive “time-ecstasis [*Zeit-Ekstase*]”—even the term “ἐκ-στασις” should emphasize (quite in opposition to its use in Plotin) the “dynamic” against the static (στάσις), against the unmoved of the ἀεὶ ὄν.

These considerations seem to lead away from the issue of mathematics and time. However, the mathematical aspect comes immediately to the fore, as soon as one considers the crucial meaning of the modality, in particular again of the “oblique” possibility, for the ontological characteristic of the *infinite*. The ἄπειρον, as Aristotle has shown in analyses that are exemplary even today, is a δυνάμει ὄν, indeed with a specific modification that, already in Aristotle, connects it with time.¹⁵⁶ It is not like a matter in relation to its (separated) form (like the bronze and the statue), but it is ὡς ἡ ἡμέρα καὶ ὁ ἀγών,¹⁵⁷ as the day and the (olympic) play that, because of its regular recurrence, served the Greeks as a *chronometer*. It is characterized as τὸ ἐν τῷ γίνεσθαι τὸ εἶναι ἔχον (Simplicius)¹⁵⁸—it is in virtue of “the being changed and always changed” (τῷ ἀεὶ ἄλλο καὶ ἄλλο γίνεσθαι)—, as the never-ending (μηδέποτε ὑπόλειπον, as the ἀεὶ πάλιν καὶ πάλιν), as the eternal recurrence.¹⁵⁹ As shown elsewhere, this conception of the temporal origin and the ontic meaning [*Seinsinn*] of the ἄπειρον pertains to the Greek philosophy in general.

The reason why *number* and *time* belong together lies in the following: the *infinite* number series—in so far as it is *infinite*—is determined through the potentiality and hence through the “futuralness [*Zukünftigkeit*]”. It is the peculiar

¹⁵⁶ As to the exact references for the following quotations see *Mathematical Existence* (Becker 1927), 641 ff.

¹⁵⁷ [Eds.] Cp. Aristotle, *Physics* III, 206a, 20–29: “Being is spoken of in many ways, and we say that the infinite is in the sense in which we say it is day or it is the games [ὡσπερ ἡ ἡμέρα ἔστι καὶ ὁ ἀγών], because one thing after another is always coming into existence. For of these things too the distinction between potential and actual existence holds. We say that there are Olympic games, both in the sense that they may occur and that they are actually occurring. The infinite exhibits itself in different ways—in time, in the generations of man, and in the division of magnitudes. For generally the infinite has this mode of existence: one thing is always being taken after another, and each thing that is taken is always finite, but always different” (Aristotle 2014d, 776).

¹⁵⁸ [Eds.] It should be “τὸ ἐν τῷ γίνεσθαι τὸ εἶναι ἔχον” (what has its being in becoming), cp. Simplicius (1882), 492.28. We thank Christoph Helmig for the following comment: “Simplicius is often using the phrasing “τὸ ἐν τῷ γίνεσθαι τὸ εἶναι ἔχον” to characterize the particular kind of things which are “potentially” (δυνάμει ὄν), e.g. in *Physics* 492.28; 493.6–7; 493.25–26; 494.17; 494.27; cf. 494.34–35; 495.1; 495.16–17; 497.15–16” (personal communication).

¹⁵⁹ [Eds.] Simplicius (1882). Cp. the parallel place from Becker (1927), 642: “[...] Where is the ἄπειρον to be found? Primarily in *time* and next in *number*. [...] As the common and main trait is indicated τὸ ἐν τῇ νοήσει μὴ ὑπολείπειν (203 b24), the “never giving out in our thought”, that Simplicius closer explains as follows: “ἡ τῆς νοήσεως ἡτοι φαντασίας τῆς ἡμετέρας δύναμις αἰεὶ τι καὶ προστιθέναι καὶ ἀφαιρεῖν ἰσχύουσα καὶ μηδέποτε ἡττωμένη καὶ ὑπολείπουσα” (p. 467.6–8 Diels (Simplicius 1882)), that is, “The power of our thought or imagination (of our intuitive presenting) can *always* add something and take away something and is *never* defeated and exhausted.”

character of “*obliquitas*” that turns out to be decisive again here. Cassirer¹⁶⁰ says that the only “temporality” that comes into question in mathematical contexts is the “universal serial principle”, the “universal schema of the ‘order in progression’ (‘order in progression’ in the sense of W. Hamilton)”. This is perhaps correct for *finite* series, which are characterized by the fact that their terms are *commutable*, even in an arbitrary | or purely combinatorial, “lawless” way. But this is no longer true of infinite series. Already Kant, in “*The Antithetic of Pure Reason*”, leads back the infiniteness of *space* (and of the “series” in general) to the infiniteness of *time*.¹⁶¹ “Time is in itself a series (*and the formal condition of all series*)”¹⁶² But as for space, in it there is no difference between progress and regress, because it constitutes an *aggregate*, but *not* a *series*, since all its parts exist simultaneously. ... Yet the synthesis of the manifold parts of space, through which we apprehend it, is nevertheless successive, and thus occurs in time and contains a series. ... Thus regarding boundedness, the progression is also a regress, and the transcendental idea of the absolute totality of a synthesis in the series of conditions also applies to space ...”.¹⁶³ [47]

This may, or rather *must* be extended to all other kinds of infinities. For “infinity”, explained more explicitly, is always endless *perspective*, endless *horizon*, unlimited “and so on”. All these phenomena have in common (according to Husserl’s fundamental analyses, which as a matter-of-fact are totally in harmony with the Aristotelian ones) the fundamental phenomenon “I can always again”, the δύνάμις that refers to the “πάντα καὶ πάντα”. “I can again and again”, in this lies: “I *will* always be able to”, and this implies the inner unity of possibility and future. Thus *every* infinity is *as such* bound, at one and the same time, to the *modi obliqui* of “logical” and temporal kind.

Of course one should never lose sight which “time” is meant. One should hold firm that it is the *original* temporality, the “temporalizing itself” of the time that is meant. Since the failure of Mill’s empiricism and his theory of mathematics, it has become a kind of *communis opinio* that the general idea of the series and of the ordinal number should abstract from the “sensible-intuitive”, “psychological-anthropological” element of time, as well as from many other things. But this *communis opinio* is grounded on a naive (Heidegger says “popular”) concept of time. In most cases the “inner” psychical time (the time of experience) is hardly differentiated from the time of the “outside” world, in | which the “human being” lives. No further questions are asked on the “time of experience”, viewed as a *finished* [48]

¹⁶⁰ *Ibid.*, 470. [Eds.] Becker refers here to Cassirer’s critical remark in Cassirer (1929) on *Mathematical Existence*, already mentioned by us on p. 29 and entirely reproduced at the end of this chapter, p. 112 ff.

¹⁶¹ Cp. *Critique of Pure Reason* (Kant 1998), A 413; B 438–440. [Eds.] Becker writes “B 348”, but this is obviously a lapse.

¹⁶² My emphasis. One sees how unkantian is W. Hamilton’s view mentioned in the text, which Cassirer endorses.

¹⁶³ [Eds.] Kant (1998), A 411–413; B 438–440.

phenomenon; in the best case there is some complaint about its mysteriousness—since Augustine. Even some of Cassirer’s assertions are not entirely free from this popular opinion, which is hardly felt more than questionable. When he writes (*Ibid.*, see above p. 97): “Mathematical activity is a purely intellectual activity; it does not *occur* in time but *first makes possible* a fundamental factor underlying time itself, the factor of ordering”, one can completely agree with him (up to the predicate “purely intellectual”, which does not seem to be unobjectionable), provided that “time” means the completely constituted psychical or physical time. Indeed, the mathematical original phenomenon of the *infinite horizon* does not “*unfold within the time*” (in which time, then?!), the spatial-kinematic metaphor implies a ὕστερον πρότερον. Similarly, it appears to be justified to ask about the “fundamental factor” that *first makes* “time” and “ordering” *possible*; for this means nothing else than asking about the *origin* of the constituted phenomena “time” and “series” (number). But this fundamental factor is just the “infinite horizon” or its grounding in possibility and *futurality* [*Zukünftigkeit*].

The “possible Being [*Seinkönnen*]” of the existence [*Dasein*] and its “having-been coming future [*gewesende Zukunft*]”, insofar as it is “historical”, its simple necessity (being carried [*Getragenheit*]¹⁶⁴) and its “*second coming present* [*wiederkünftige Gegenwart*]”, insofar as it is “natural”—these are the roots from which the number series and the constituted time originate at once. The “existence [*Dasein*]” meant here is, of course, not the human being “*in*” the time (in the “world time” of astronomers and historiographical chronologists, or the “experiential time” of experimental psychologists), but the “human being” in so far as he temporalizes the time itself,—and thus, *indirectly*, also that “world time” or “experiential time”. The “anthropologism” defended here is no hidden renewal of any empirical “psychologism”. That “finiteness” of the human being, that *first gives sense to mathematics altogether*, is not something objective or a “factum” that can be in a way described objectively. It “precedes” the possibility of the constitution of any objectivity in general, since it is properly only a short expression for the “ecstatic” (“oblique”) temporality, which is not where the “human being lives”, but rather “is” the human existence [*Dasein*] itself, is what originally constitutes its mode of being [*Seinsweise*]. When Cassirer adds here the concept of the “pure subjectivity” of the ideal mathematician and that of the “pure objectivity” of the | mathematical element, one may object the following: a “pure” subject (of which *ex definitione* it is not known, whether it is finite or not—where it should also be asked: what does it mean that a subject is “not finite”?) cannot reasonably do mathematics, in so far as mathematics means *mastering the infinite*. One may add, for the sake of clarity, “with *finite* means”: at root, this is indeed superfluous! The sense of struggle involved in the concept of “mastery” is indeed only possible under the assumption of a primal tension, and the latter has just the finite as an essential counterpart of the in-finite. Still more: even the constitution of the ἀπειρον δύναμει ὄν sends back to the “possible Being [*Seinkönnen*]” and the *futurality* [*Zukünftigkeit*]; the concept

[49]

¹⁶⁴ [Eds.] Cp. Becker’s *Dasein und Dawesen* (Becker 1963), 34: “*Getragenheit*” = “being borne”.

of infinite in itself and taken “alone”—setting aside from its “mastery”—has its source, in its original sense, from those structures of being [*Sein*] that mark the finiteness of the human being.

One should completely free oneself from the *communis opinio* that mathematics is something “objective”, a kind of object, say an “objective system of relations”: an “*objective infiniteness*” is a *contradictio in adjecto*!

Nevertheless, mathematics is *a priori*, it is above the contingencies of the “ontic” facts. However, the “facticity” of the existence [*Dasein*] is itself not a “factum”, but an *a priori* fundamentally ontological structure.¹⁶⁵ “Mortality” is a pure structure of this “facticity”; “death”, in the sense relevant here, is not a “factum”. It happens also to the “eternal demon”.¹⁶⁶ Only for this reason he can “mathematize”, because he is, like the human being, a “Ζῶον ἐν τῇ γίγνεσθαι τὸ εἶναι ἔχον” (Simplicius)¹⁶⁷ and, because of this, he is *capable* of infinite.¹⁶⁸ |

Thus, the hint at Husserl’s “pure ego” and Kant’s “transcendental apperception” [50] is not decisive. For these have *at most* the “bad infinity” of a “demonic creature” (though this is especially by Kant very dubious!). Moreover, the only mathematics that we know is not at all “demonic”, but rather “human”.¹⁶⁹ Precisely who moves from the *factum* of human science, shall not neglect this. Indeed, mathematics as it has first been conceived by the Greeks and as we own it today, after Leibniz’s breakthrough towards the “*mathesis universalis*”, means “freeing” ourselves from a number of primitive restrictions. But mathematics can never serve to make the human being transcend himself and wrest himself from the indissoluble “pure

¹⁶⁵ Properly speaking, a “hyper-ontological”, both “ontological” and “para-ontological” (“para-existential”) structure. But we cannot undertake a closer examination of this here.

¹⁶⁶ Cp. here my paper in *Phil. Anzeiger III* (Becker 1929), 380 ff. as well as *Mathematical Existence* (Becker 1927), 751.

¹⁶⁷ [Eds.] The human being is “an animal which has its being in becoming”. Hereto cp. fn. 158.

¹⁶⁸ Husserl’s “pure ego” is mortal in the same sense of the “demon”; “pure consciousness”, were it be thought as a concrete “instance”, would be nothing else than such a demon; it is, in a sense, the species (*Eidos*) of such a demon.

And finally, the whole Kantian philosophy, as it is, is grounded on the fundamental thesis of the *finiteness of the philosophizing subject*. The antinomies, in particular, gain first of all from here a sense. Didn’t we encounter, right there, the connection between infinity and time? And, is it possible to deny Kant’s thesis that the number series is the schema of the temporal sequence, without unsettling his whole philosophy? Not to mention here Heidegger’s book on Kant (Heidegger 1929), which takes the idea of *finiteness* and *temporality* as the fundamental idea of Kant’s philosophy. The *finiteness* thesis seems to me to be sound, even if one may not completely agree with the second thesis as *historical interpretation*.

¹⁶⁹ This holds in particular for modern western mathematics. Even the widespread use, there (in contrast to ancient mathematics), of *abbreviating sign systems* makes sense only if the finite number of elements that can be intuited in a proper way is bounded by a certain limit (which is *de facto* very low). One sees that the “immanent power” of the mathematical “method” is at issue, and not an “arbitrary” introduction of extraneous subjective-psychological views or an “artificial” anthropological interpretation. Also cp. Hilbert’s essay *Axiomatic Thinking*, often quoted in analogous contexts in previous works (Math. Annalen, vol. 78, Hilbert 1918). Also see: *Mathematische Existenz* (Becker 1927), 772 ff.

structures” that constitute his essence as a human being. Still more: if one contrasts the natural primitiveness with the height of the historical (i.e., self-aware) Being [*Dasein*], then the “progress” towards the advanced civilization does *not* mean overcoming the finite, but rather *finitization* [*Ver-Endlichung*], distinct emergence of the finite traits in the image of the human being. This intrinsically neat evolution is however troubled by the effect of “archaisms” that frequently cross it: the principal one is the illusion of a completely “pure” “infinite” *Ratio* freed of any “anthropomorphic” bond,—an illusion whose “*transcendental appearance* [*transzendentaler Schein*]” was recognized by Kant, who strived to get rid of it. In this respect the idea of a “pure science” is dangerously ambiguous: it tends toward self-consciousness or | self-justifying *methodical* consciousness, *but* also toward the separation from the fundamental sense of any μέθοδος as a *way*: the very idea of a *way* includes the finiteness; “in the infinite” all “*places*” would be together; there would be no need of a *way*. Precisely because the “logical idealism” reflects again and again on the “*peculiarity of the method*”, it should always be mindful of this fundamental definition of the method as a *way*. And it is nothing other than the essential finiteness of this pure “method” what makes the most abstract and “freest” sciences of the *mathesis universalis* primarily and in principle determined through the concept of finiteness. [51]

Additional Remark:

On F. Brentano’s use of the terms “*modo recto*” and “*modo obliquo*”

(p. 104, fn. 147.)

Brentano has very often employed the expressions “*modo recto*” and “*modo obliquo*” in his *Psychology* (recently published in 3 volumes by O. Kraus, Leipzig 1924–1928 (Brentano 1924); cp., in particular, Vol. II, Appendixes I–IV; Vol. III, Section 1, Ch. 5). However, he means something completely different from what is meant in our text. For Brentano the *modo recto* and the *modo obliquo* are always modes of *presenting* [*vorstellen*] or of *sensing* [*empfinden*], they are not *logical* modalities. Precisely, presenting in *modo recto* is a *direct* or *primary* way [of referring to something], while presenting in *modo obliquo* is the *indirect* or *secondary* way (which occurs ἐν παρεργῶ, cp. Aristotle Met. (Aristotle 2014b) Δ 9, 1074b 35–36). Brentano writes (*Ibid.*, II, 134): “If I think of someone who loves flowers, then the person who loves flowers is the object I am thinking of *in recto*, but the flowers are what I am thinking of *in obliquo*. That, however, is similar to the case in which I am thinking of someone who is taller than Caius. The taller person is thought of *in recto*, Caius *in obliqua*.”¹⁷⁰ Even if there are many modes of indirect, “*oblique*” presenting by Brentano—indeed, the mentioned example gives

¹⁷⁰ [Eds] Becker is quoting from Brentano (1924), II, 134. Engl. transl.: II, *Supplementary Remarks Intended to Explain and Defend, as well as to Correct and Expand upon the Theory*, 212.

only a faint idea thereof—the distinction between actuality and possibility is never considered, the less so the *specific character* of such modal logical distinctions. Any object whatever that stands in a certain relation to another object, which is represented *directly* (*modo recto*), is represented *modo obliquo*, according to Brentano (*Ibid.*, III, 42).

In a sense Brentano also knows the “*modi obliqui temporales*”, cp. *Ibid.*, III, 39: “[...] we shall add at this point that we think of ourselves *in modo recto* in the *modus praesens*. Yet we think of ourselves as something which, in addition, thinks of other things *in modo obliquo*, whereby some of them are presented in the *modus praesens* while others, continually differing from one another, are thought of in a series of *modis praeteritis*. To summarize, I will say that I perceive myself *in modo recto* as existing in the present and simultaneously, or as existing | at a point of time more or less later with respect to certain external objects that are given to me in *modo obliquo* in different *modi praeteritis*”.¹⁷¹ Hence the past is also in Brentano characterized by a presenting in *modo obliquo*, even if, as the above place shows, the *modus praeteritus* is distinct from the *modus obliquus* and is not simply conceived as a special case of the latter. It seems to me therefore that O. Kraus (Archiv f. d. ges. Psychologie, Vol. 75 (1930) (Kraus 1930), 2) goes too far when he speaks of the “conception of the temporal *modi as modi obliqui*”. I can agree with him, on the contrary, when he says (*Franz Brentano*, München 1919 (Kraus 1919), 40) “that (Brentano’s) doctrine of the temporal *modi* is basically nothing other than an application of the theory of *modus rectus* and *modus obliquus*”.—Be that as it may, the view advocated in this essay does not share Brentano’s basic ideas about the distinction between direct and indirect presenting; the expressions “*modus rectus–obliquus*” are applied *primarily* to the logical modalities, and only *per analogiam* to the temporal *modi*; moreover, they are not taken over from Brentano—only subsequently I became aware of Brentano’s linguistic usage anew—but are borrowed from the well known terminology of the Latin grammar, from which it is likely that also Brentano derived them. [52]

Cassirer’s Remark on Oskar Becker’s Monograph *Mathematical Existence*¹⁷²

This chapter on mathematical concept formation was already completed in the autumn of 1927 when Oskar Becker’s *Mathematische Existenz* appeared in *Jahrbuch für philosophische und phänomenologische Forschung* 8 (1927), 441–809. It would not be doing justice to the significance of this work if I were to attempt, in a brief postscript and in passing, to make a criticism of it, for such a criticism could be provided only on another basis than that of our own systematic problem. I should

¹⁷¹ [Eds.] Brentano (1929), 29.

¹⁷² [Eds.] Cassirer (1929), 446–447; Cassirer (1957), 404–405.

like to make only a few remarks: I thoroughly grant Becker the starting point of his investigation, the principle which he calls the “principle of phenomenological approach.” “To all objectivity”—he formulates—“there is (in principle, i.e. technical difficulties aside) an approach. It is this that first characterizes all objectivity as a phenomenon and satisfies the absolutely universal claim of transcendental phenomenology (for which all being is synonymous with being constituted)” (p. 502 n.). I also agree fully with the consequence that Becker draws from this principle, namely his assertion of the primacy of the concept of number over the concept of quantity (cf. pp. 559 ff.). But what strikes me as not demonstrated by Becker’s argumentation, and moreover seems to me indemonstrable in the present state of mathematics, is the necessity and the justification for linking the universal “serial principle” underlying number with the phenomenon of *time*, as Becker does throughout his work. Even if we recognize “the decisive role of temporality for the character of mathematical objects,” this temporality is nothing other than the universal schema of “order in progression” (to use Hamilton’s term). This “objective” time of mathematics must not be confused with “historical” time, or with the “experiential” time of the mathematician (cf. Becker, pp. 657 ff.). As far as I can see, mathematics today offers no more justification than ever before for an attempt to anthropologize transcendental idealism. The “subject” to which the pure constructive principles of mathematics and hence the realm of mathematical objectivity must be referred, remains the “I think” of Kant’s “transcendental apperception,” hence the “pure ego,” the “I-pole,” which is also Husserl’s original starting point. Becker, however, seeks rather to refer the content and meaning of mathematics to a definite mode and trend of “factual life phenomena”: he sees mathematical “existence” as ultimately grounded in certain “modes of factual life” (pp. 621 ff.). Mathematics, I believe, will always have to resist any such grounding in a mere facticity, because of the fundamental postulate of “objectivity” that is inherent in it. In a recent work, *Über den sogenannten Anthropologismus in der Philosophie der Mathematik*, Philosophischer Anzeiger, 3 (1929) (Becker 1929), 369 ff., which became accessible to me only while I was correcting the present volume, Becker says that the Marburg Neo-Kantians had rightly attempted to approach the problem of the philosophical foundations of mathematics from an “idealistic” point of view but goes on to say that they failed, because “they formulated the idea of the subject of knowledge too indeterminately, or rather, they did not even bring the already achieved determinacy of subjectivity to bear in their formulation of the problem” (p. 381). If this reproach of “indeterminacy” means merely that logical idealism rejects a mixture of *pure* subjectivity with determinations of *man’s* subjectivity, I believe that it is based on a *petitio principii*. Logical idealism starts from an analysis of mathematical “objects,” and seeks to apprehend the peculiar determinacy of these objects by explaining them through the peculiarity of the mathematical “method,” the mathematical concept formation and formulation of its problem. As to what this method itself “is”—this the logical idealists derive solely from its immanent achievement. And in this achievement subjectivity, insofar as it is understood as the concrete subjectivity of the mathematician, is not present; as a constitutive factor, rather, this subjectivity is consciously eliminated. When Becker attempts to deny

this state of affairs—when he puts forward the thesis that the mere *definition* of mathematics (as the science which strives to master the infinite with finite means) points immediately and necessarily to the mathematician himself (p. 379), such a conclusion, it seems to me, is nowhere justified by the “content” of mathematical knowledge, but can stem only from an artificial interpretation of it. For even in the cited definition of mathematics the emphasis is not on the fact that mathematics uses “finite means,” but on the fact that through these finite means it masters the infinite. If we start from the problem of this mastery and go on to ask how it is “possible,” we are led to an entirely different “time form” than that in which the “existence of man,” as a necessarily finite existence, moves. As far as I can see, no path leads either from the object of mathematics nor from its method to the “very definite and concrete structures” (death, historicity, “freedom,” “guilt”) which Becker’s analyses seek to disclose: it seems to me that the “essential and evident relation between the structure of mathematical meaning” as such and the “existential meaning of the finite being, man” (cf. p. 383), can be found no more in the present stage of mathematics than in any previous stage.

Chapter 5

Appendix



This Appendix is addressed in particular to the readers who are also interested in “technical” logical details. It is organized as follows.

Sections 5.1–5.8 are intended as a self-contained and comprehensive survey of the main syntactical and semantical results concerning Lewis’s system **S3**—Becker’s main object of investigation and the basis of the extended systems he introduced in the first part of *On the Logic of Modalities*.

Thus armed, in the final Sect. 5.9 we refine with technical details the evaluation of Becker’s logical accomplishments and failures we hinted at in Chap. 2, Sect. 2.9, and in the Editors’ footnotes to the translation of Becker’s text in Chap. 4.

5.1 Lewis’s Axiomatizations of S3

Lewis’s original axiomatization of the *Survey System S3* (Lewis 1918), indeed a systematically reworked version of the calculus he introduced in the earlier paper Lewis (1914), is based—as we already know from Becker’s presentation—on a language containing:

- propositional atoms: $p, q, \dots$
- four primitive logical constants: the two *material* connectives “ $\neg$ ” (Boolean negation) and “ $\times$ ” (Boolean conjunction), and the two *strict* connectives “ $\sim$ ” (impossibility) and “ $=$ ” (equivalence).

Lewis does not employ special letters to denote arbitrary, not necessarily atomic, formulas; also, he systematically replaces the symbol $\times$ for conjunction with the juxtaposition of the two conjuncts (so “ $(p q)$ ” abbreviates “ $(p \times q)$ ”), and Becker follows suit. At a variance, we will use the letters $A, B, \dots$ as metavariables for formulas in this language, and will not replace $\times$ with juxtaposition.

Other logical operators are introduced by Lewis through appropriate *definitions*. Notice however that, differently from the now current practice, such definitions are not, so to speak, external to the calculus, that is, they are not to be intended as *metalinguistic* abbreviations. Rather, one might better say that the *axioms* of Lewis's calculus come in two groups, namely *Definitions* (which have always the form $\bullet p = B$ or $p \bullet q = B$, where $\bullet$ is a possibly non primitive¹ unary or binary logical operator) and *Postulates*, as follows.

Definitions

- D1. $p \circ q = - \sim (p \times q)$ (*consistency*: $p \circ q$ says that p and q are jointly possible)
- D2. $p \supset q = \sim (p \times -q)$ (*strict implication*: $p \supset q$ says that it is impossible that p is true and q is false)²
- D3. $p \subset q = -(p \times -q)$ (*material implication*)
- D4. $p \wedge q = \sim (-p \times -q)$ (*strict disjunction*)³
- D5. $p + q = -(-p \times -q)$ (*material disjunction*)
- D6. $(p = q) = (p \supset q) \times (q \supset p)$ (*strict equivalence*)
- D7. $(p \equiv q) = (p \supset q) \times (q \supset p)$ (*material equivalence*)

Postulates⁴

- P1. $p \times q \supset q \times p$
- P2. $q \times p \supset p$
- P3. $p \supset p \times p$
- P4. $(p \times (q \times r)) \supset (q \times (p \times r))$
- P5. $p \supset - -p$
- P6. $(p \supset q) \times (q \supset r) \supset (p \supset r)$
- P7. $\sim p \supset - p$
- P8. $(p \supset q) \supset (\sim q \supset \sim p)$
- P9. $(\sim q \supset \sim p) \supset (p \supset q)$

The last two axioms, P8 and P9, can be equivalently replaced with the single axiom

$$P8^*. (p \supset q) = (\sim q \supset \sim p).$$

Finally, there are four **inference rules**—which in Becker's text are (strangely enough) not specified at all:

- R1. “*Substitution*”, 1. This is the usual substitution rule, needed because the axioms are not given in schematic form:

$$\frac{A}{A'}$$

¹ Indeed, D6 is a definition of the strict equivalence operator $=$, which is primitive!

² In Becker (1930), as we have seen, the symbol “ $\supset$ ” is written “ $<$ ”.

³ Notice the peculiar use of the symbol “ $\wedge$ ”, which will become, much later, the standard symbol for conjunction. Notice also that the symbol for material implication is “ $\subset$ ” and not the russellian “ $\supset$ ”.

⁴ Convention: the connective “ $\times$ ” binds stronger than the connective “ $\supset$ ”.

where A' is any substitution instance of A .

R2. “*Substitution*”, 2. This is the rule allowing the substitution of strictly equivalent formulas:

$$\frac{A = B \quad C[\dots A \dots]}{C[\dots B \dots]}$$

R3. “*Inference*”. This is the rule of strict detachment:

$$\frac{A \supset B \quad A}{B}$$

R4. “*Production*”. This is the adjunction rule:

$$\frac{A \quad B}{A \times B}$$

The above 1918 version of the calculus, as we already know, collapses into classical logic. Emil Post indeed discovered that the formula $\sim p = -p$ is a theorem, and so that each strict logical notion becomes equivalent to the corresponding material one. Here is a short proof of the collapse formula:

- | | | |
|----|---|--------------------------|
| 1. | $(\sim p \supset \sim -p) \supset (-p \supset p)$ | by P9, R1 |
| 2. | $\sim (\sim p \times -\sim -p) \supset \sim (-p \times -p)$ | by 1, D2, R1, R2 |
| 3. | $(-p \times -p) \supset (\sim p \times -\sim -p)$ | by 2, P9, R1, R3 |
| 4. | $-p \supset (\sim p \times -\sim -p)$ | by 3, P3, P6, R1, R3 |
| 5. | $-p \supset \sim p$ | by 4, P1, P2, P6, R1, R3 |
| 6. | $-p = \sim p$ | by 5, P5, R4, D6, R2 |

The axiom P9, which is apparently the cause of the modal collapse, is thus dropped in the *emended* 1920 version (Lewis 1920), where the *Survey system* takes its definitive form as resulting from D1–D7, P1–P8 and the rules R1–R4. This is the system we find, under the name **S3**, in the *Appendix II* of Lewis and Langford (1932).⁵

⁵ Apart from two unessential differences: (i) “ $\Diamond$ ” (possibility) is taken as primitive in place of “ $\sim$ ” (impossibility), and (ii) the symbol “ $-$ ” for negation is replaced by the symbol “ $\sim$ ”.

As was proved by Lewis, in **S3** it is still possible to derive all classical tautologies.⁶ On the other side, the emended system *does not collapse* into the system of material implication.⁷

5.2 Lemmon-Style Axiomatization of S3

Lewis-style axiomatizations of modal systems are not constructed as extensions of the classical propositional calculus **CPC** and have been in force during the early development of modal logic up until the 1940s. The now familiar axiomatic presentations, adopting the much more “user-friendly” format “**CPC** + specific modal rules and axioms”, were anticipated by Gödel in 1933⁸ and have become standard since the 1950s, both for the normal and—thanks in particular to the work of Lemmon (Lemmon 1957)⁹—for the non normal modal systems.

As far as our **S3** is concerned, Lemmon’s axiomatization (to which we will henceforth refer) is as follows:

Language $\mathcal{L}^\square$

The language $\mathcal{L}^\square$ is determined by the infinite set $AT = \{p_0, p_1, p_2 \dots\}$ of atoms (propositional variables) and the primitive logical operators $\neg$ and $\rightarrow$ (Boolean connectives), together with the modal operator $\square$. The other Boolean connectives $\wedge$, $\vee$, $\leftrightarrow$ and the modal operator $\diamond$ are defined as usual. We denote by $FOR^\square$ the set of all formulas of $\mathcal{L}^\square$.

Axioms and Axiom Schemas

- A1. all **CPC** tautologies
- A2. $\square A \rightarrow A$ (schema *T*)
- A3. $\square(A \rightarrow B) \rightarrow \square(\square A \rightarrow \square B)$ (schema *K*⁺)

Inference Rules

$$\frac{A \rightarrow B \quad A}{B} \text{MP} \quad (\textit{modus ponens})$$

$$\frac{A}{\square A} \text{RN}^- \quad \text{provided } A \text{ is an (instance of an) axiom (schema)}$$

⁶ That all classical tautologies are theorems of the unamended system is already proved in Lewis (1918), Chapter V, Section III. The proof depends however on the axiom P9, and in Lewis (1920), 301–2, it is shown how the proof can be otherwise effected without making use of P9.

⁷ This is proved in Lewis and Langford (1932), 493ff., by using the many-valued matrixes method.

⁸ Gödel (1933a). Gödel’s system $\mathfrak{S}$ coincides, up to the use of ‘ $\mathbf{B}$ ’ in place of ‘ $\square$ ’, with Lewis’s system **S4** recasted in such a format.

⁹ See also Feys (1950) and Prior (1955).

The necessitation rule (RN^-) comes in a *restricted* form.¹⁰ Indeed, it turns out that **S3** is not closed under the unrestricted necessitation rule (RN), hence *it is not a normal modal system*. For instance, denoting by “ $\top$ ” any classical tautology, say $p \rightarrow p$, $\Box\top$ is a theorem of **S3**, while by contrast $\Box\Box\top$ is not a theorem (see Sect. 5.4). Indeed, one can prove that **S3** extended with the axiom $\Box\Box\top$ is equivalent to the *normal* system **S4**. Consider indeed the following theorems and eliminable inference rules of **S3** (which will also be also useful in Sect. 5.5).

Lemma 1

1. $\vdash_{\mathbf{S3}} \Box A \rightarrow \Box(\Box\top \rightarrow \Box A)$
2. $\vdash_{\mathbf{S3}} \Box(\Box A \rightarrow \Box\top)$
3. $\vdash_{\mathbf{S3}} \Box(\Box A \wedge \Box B \leftrightarrow \Box(A \wedge B))$
4. *any axiomatic extension of **S3** is closed under the rules*

$$\frac{A}{\Box(\Box\top \rightarrow A)} \text{ R.1} \qquad \frac{A \rightarrow B}{\Box(A \rightarrow (\Box\top \rightarrow B))} \text{ R.2}$$

Proof

(1)

- | | | |
|----|--|-------------------|
| 1. | $A \rightarrow (\top \rightarrow A)$ | A1 |
| 2. | $\Box(A \rightarrow (\top \rightarrow A))$ | $RN^-: 1$ |
| 3. | $\Box(A \rightarrow (\top \rightarrow A)) \rightarrow \Box(\Box A \rightarrow \Box(\top \rightarrow A))$ | A3 |
| 4. | $\Box(\Box A \rightarrow \Box(\top \rightarrow A))$ | MP: 2,3 |
| 5. | $\Box A \rightarrow \Box(\top \rightarrow A)$ | MP: A2, 4 |
| 6. | $\Box(\top \rightarrow A) \rightarrow \Box(\Box\top \rightarrow \Box A)$ | A3 |
| 7. | $\Box A \rightarrow \Box(\Box\top \rightarrow \Box A)$ | transitivity: 5,6 |

(2)

- | | | |
|----|---|------------|
| 1. | $\top \rightarrow (A \rightarrow \top)$ | A1 |
| 2. | $\Box(\top \rightarrow (A \rightarrow \top))$ | $RN^-: 1$ |
| 3. | $\Box\top \rightarrow \Box(A \rightarrow \top)$ | 1, A3, A2 |
| 4. | $\Box\top$ | A1, RN^- |
| 5. | $\Box(A \rightarrow \top)$ | MP: 3, 4 |
| 6. | $\Box(\Box A \rightarrow \Box\top)$ | 5, A3 |

¹⁰ Notice that an equivalent axiomatization in which MP is the unique inference rule can be easily obtained: just replace A1–A3 with the corresponding boxed-versions $A1^\Box$ – $A3^\Box$ ($\Box A$, where A is a classical tautology, etc.), add A2 and delete RN^- .

(3) straightforward.

(4) Let $\mathbf{S}$ be an axiomatic extension of $\mathbf{S3}$. First, we prove that $\mathbf{S}$ is closed under R.1, by induction on the derivation of A .

– A is an axiom of $\mathbf{S}$.

1.	A	Axiom
2.	$\Box A$	$\text{RN}^-: 1$
3.	$A \rightarrow (\Box \top \rightarrow A)$	A1
4.	$\Box(A \rightarrow (\Box \top \rightarrow A))$	$\text{RN}^-: 3$
5.	$\Box A \rightarrow \Box(\Box \top \rightarrow A)$	4, A3 and A2
6.	$\Box(\Box \top \rightarrow A)$	MP: 2, 5

– A follows from B and $B \rightarrow A$ by MP. Then by the I.H.:

$$(*) \quad \vdash \Box(\Box \top \rightarrow B) \quad \text{and} \quad \vdash \Box(\Box \top \rightarrow (B \rightarrow A))$$

Now $(\Box \top \rightarrow (B \rightarrow A)) \rightarrow ((\Box \top \rightarrow B) \rightarrow (\Box \top \rightarrow A))$ is a tautology, so by RN^- , A3, A2 and MP, together with $(*)$, we get $\vdash \Box(\Box \top \rightarrow A)$.

– A follows from B by RN^- . Then $A \equiv \Box B$ and B is an axiom. Thus:

1.	B	Axiom
2.	$\Box B$	$\text{RN}^-: 1$
3.	$B \rightarrow (\top \rightarrow B)$	A1
4.	$\Box(B \rightarrow (\top \rightarrow B))$	$\text{RN}^-: 3$
5.	$\Box B \rightarrow \Box(\top \rightarrow B)$	4, A3 and A2
6.	$\Box(\top \rightarrow B)$	MP: 2, 5
7.	$\Box(\Box \top \rightarrow \Box B)$	6, A3

Notice that the I.H. is not used in this case!

The closure of $\mathbf{S}$ under R.2 now follows from R.1 and the $\mathbf{S3}$ theorem

$$\Box(\Box \top \rightarrow (A \rightarrow B)) \rightarrow \Box(A \rightarrow (\Box \top \rightarrow B)),$$

which is easily derivable using A1, RN^- , A2 and A3.

Theorem 1 $\mathbf{S4} \equiv \mathbf{S3} + (\Box \Box \top) \equiv \mathbf{S3} + (\Box A \rightarrow \Box \Box A)$.

Proof By $\vdash_{\mathbf{S3}} \Box A \rightarrow \Box(\Box \top \rightarrow \Box A)$ (Lemma 1) and $\vdash_{\mathbf{S3}} \Box(\Box \top \rightarrow \Box A) \rightarrow (\Box \Box \top \rightarrow \Box \Box A)$ (A3 and A2) we get, by transitivity, $\vdash_{\mathbf{S3}} \Box A \rightarrow (\Box \Box \top \rightarrow \Box \Box A)$ and then $\vdash_{\mathbf{S3}} \Box \Box \top \rightarrow (\Box A \rightarrow \Box \Box A)$.

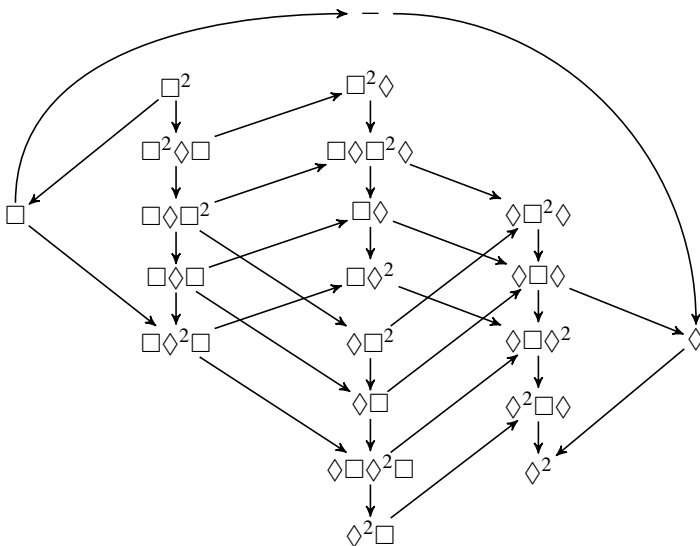
Thus $\mathbf{S3} + \Box A \rightarrow \Box\Box A \subseteq \mathbf{S3} + \Box\Box\top$. The converse inclusion follows immediately from $\vdash_{\mathbf{S3}} \Box\top$.

To conclude that $\mathbf{S4} \equiv \mathbf{S3} + \Box A \rightarrow \Box\Box A$, it is of course sufficient to prove that $\mathbf{S} := \mathbf{S3} + \Box A \rightarrow \Box\Box A$ is closed under the unrestricted necessitation rule RN. To this aim, suppose $\vdash_{\mathbf{S}} A$. Then by the closure of $\mathbf{S}$ under the rule R.1 of Lemma 1, $\vdash_{\mathbf{S}} \Box(\Box\top \rightarrow A)$ and so also (by using A3 and A2) $\vdash_{\mathbf{S}} \Box\Box\top \rightarrow \Box A$. As we have proved above, $\vdash_{\mathbf{S3}} \Box\Box\top$, so, by MP, $\vdash_{\mathbf{S}} \Box A$.

5.3 Irreducible Modalities in S3

As we know, Becker was working under the assumption that the number of irreducible modalities in Lewis's system $\mathbf{S3}$ was infinite.¹¹ In 1939, W. Parry (Parry 1939) proved instead that the number of irreducible positive and negative modalities in $\mathbf{S3}$ is exactly 42—including the two “trivial” modalities “—” (*it is true, that ...*) and “¬” (*it is false, that ...*).

The 21 *positive* irreducible modalities are displayed in the diagram below, together with their mutual logical-strength relations: for any two modalities Λ and Π , there is an arrow going from Λ to Π if, and only if, $\vdash_{\mathbf{S3}} \Lambda A \rightarrow \Pi A$ for any A .



¹¹ Cp. Becker (1930), 16 (74).

The 21 *negative* irreducible modalities as well as the corresponding diagram are of course obtained by putting a negation in front of each of the 21 positive modalities, and by reversing the arrows.

We do not present Parry's proof here. It may be (patiently) reconstructed by using syntactical tools (see Sect. 5.2) to show that any modality is **S3**-equivalent to one of Parry's 42, and the Kripke-style semantical characterization of **S3** (see below Sects. 5.4 and 5.5) to show that they are indeed irreducible.

5.4 Kripke-Style Semantics for S3

Possible-worlds (or *relational*) semantics for *non normal* modal logics, including **S3**, was introduced by Saul Kripke in Kripke (1965) where, in order to deal with modal systems that are not closed under the necessitation rule, he suitably adapted the possible-words semantics for *normal* modal logics he had presented some years earlier in Kripke (1963). The key ideas of such an adaptation are:

- (i) to divide the worlds of a relational model $\langle W, R, v \rangle$ into *normal* and *non normal* ones: the rules for evaluating a compound formula A at a world $x \in W$ are the usual ones when x is normal, whereas they are suitably modified (for non-Boolean compounds only) when x is not normal, to the effect that at any non normal world every formula of the form $\Box B$ is evaluated false, and every formula of the form $\Diamond B$ is evaluated true;¹²
- (ii) to define the notion of a formula being *true in a model* as truth of that formula at all the *normal* worlds of the model;
- (iii) to identify the adequate notion of validity (as *truth in all models*) for the various non normal modal systems under consideration by imposing suitable structural conditions on the accessibility relation R and on the set N of normal worlds of the models.

Let us see how all of this works in the case of our system **S3**. Actually, one can find in the literature a number of non-equivalent definitions of what a Kripke model for **S3** is—two of them can indeed be found already in Kripke's already mentioned paper. Let us start with the most general type of **S3**-model.

Definition 1 (Mod₁ Class of Models)

1. A *type-1 S3*-model is a quadruple

$$\mathcal{M} = \langle W, N, R, v \rangle$$

¹² In a sense, non normal worlds behave in the opposite way as dead-ends—worlds x such that $\neg \exists y (x R y)$ —behave in a *normal* model, in so far as all formulas of the form $\Box A$ ($\Diamond A$) are evaluated true (false) at a dead-end.

where:

- W is a non-empty set (the set of all *worlds*)
- N is a non-empty subset of W (the set of all *normal worlds*)
- R is a binary relation on W , which is *reflexive* on N and *transitive* on W
- $v : W \times AT \longrightarrow \{0, 1\}$ is (as usual) the valuation function for atoms.

Henceforth, we conveniently denote by Q the set $W \setminus N$ of all the non normal worlds of a model. Also, we denote by $\mathbf{Mod}_1$ the class of all type-1 **S3**-models.

2. The relation

$$x \Vdash_{\mathcal{M}} A$$

where $x \in W$ and $A \in FOR^{\square}$, is defined inductively as follows:

- $x \Vdash_{\mathcal{M}} p$ iff $v(x, p) = 1$
- $x \Vdash_{\mathcal{M}} \neg B$ iff $x \not\Vdash_{\mathcal{M}} B$
- $x \Vdash_{\mathcal{M}} B \wedge C$ iff $x \Vdash_{\mathcal{M}} B$ and $x \Vdash_{\mathcal{M}} C$
- $x \Vdash_{\mathcal{M}} B \vee C$ iff $x \Vdash_{\mathcal{M}} B$ or $x \Vdash_{\mathcal{M}} C$
- $x \Vdash_{\mathcal{M}} B \rightarrow C$ iff $x \not\Vdash_{\mathcal{M}} B$ or $x \Vdash_{\mathcal{M}} C$
- $x \Vdash_{\mathcal{M}} \square B$ iff $x \in N$ and, for all $y \in W$ such that xRy , $y \Vdash_{\mathcal{M}} B$
- $x \Vdash_{\mathcal{M}} \diamond B$ iff $x \in Q$, or $y \Vdash_{\mathcal{M}} B$ for some $y \in W$ such that xRy

3. A formula A is true in $\mathcal{M}$, in symbols $\models_{\mathcal{M}} A$, iff $x \Vdash_{\mathcal{M}} A$ for all $x \in N$.

Before considering a number of narrower classes of **S3**-models, let us verify that every **S3**-theorem is true in each model $\mathcal{M} \in \mathbf{Mod}_1$.

Theorem 2 (Soundness of S3 with Respect to $\mathbf{Mod}_1$) *For every formula A in $FOR^{\square}$ and every model $\mathcal{M} \in \mathbf{Mod}_1$, $\models_{\mathcal{M}} A$.*

Proof Let $\mathcal{M} = \langle W, N, R, v \rangle \in \mathbf{Mod}_1$. Since $\mathcal{M}$ -truth is trivially closed under *modus ponens*, to conclude it is sufficient to verify that any (instance of an) axiom (schema) of **S3** is true in any world $x \in W$ (be it normal or not!). This indeed also implies the closure of $\mathcal{M}$ -truth under RN^- . Now:

- AX1. Obviously any **CPC** tautology is true at any $x \in W$.
- AX2. Suppose $x \Vdash_{\mathcal{M}} \square B$; then $x \in N$ and $y \Vdash_{\mathcal{M}} B$ for all y such that xRy . Since R is reflexive on N , $x \Vdash_{\mathcal{M}} B$ follows. Thus $x \Vdash_{\mathcal{M}} \square B \rightarrow B$.
- AX3. Suppose $x \not\Vdash_{\mathcal{M}} \square(A \rightarrow B) \rightarrow \square(\square A \rightarrow \square B)$; then

$$(i) \ x \Vdash_{\mathcal{M}} \square(A \rightarrow B), \text{ and so } x \in N, \quad (ii) \ x \not\Vdash_{\mathcal{M}} \square(\square A \rightarrow \square B)$$

By (ii) and $x \in N$ it follows that there exists a $y \in W$ such that xRy and

$$(iii) \ y \Vdash_{\mathcal{M}} \square A, \text{ and so } y \in N, \quad (iv) \ y \not\Vdash_{\mathcal{M}} \square B$$

By (iv) and $y \in N$ it follows that there exists a $z \in W$ such that yRz and

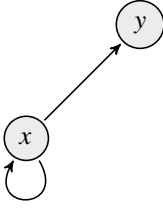
$$(v) z \not\models_{\mathcal{M}} B$$

On the other side, by (i), xRy , yRz and the transitivity of R we obtain

$$z \models_{\mathcal{M}} A \rightarrow B$$

while by (iii) and yRz we get $z \models_{\mathcal{M}} A$; thus $z \models_{\mathcal{M}} B$, in contradiction with (v).

Here is a very simple countermodel $\mathcal{M} \in \mathbf{Mod}_1$ showing (thanks to the Soundness Theorem) that $\not\models_{\mathbf{S3}} \Box\Box\top$.



where $N = \{x\}$ and the valuation v is arbitrary.

Indeed, $\not\models_{\mathcal{M}} \Box\Box\top$ since $x \in N$ and $x \not\models \Box\Box\top$, due to the fact that $y \not\models \Box\top$ (since $y \in Q$) and xRy .

As one can easily realize from the proof of the completeness theorem given in the next section, the (possible) presence of non normal worlds in a model is necessary in order to provide counterexamples to formulas which *are not* theorems of **S3**; on the other side, non normal worlds do not have any *active* role in the validation of **S3**-theorems. Thus—while keeping (2) and (3) of Definition 1 unchanged—a number of additional conditions can be imposed on the structural part $\langle W, N, R \rangle$ of a type-1 model (in particular on the set Q of non normal worlds) that, without of course affecting the validity of the soundness theorem, allow to prove *finer* versions of the completeness theorem—as we shall see.

Definition 2 (Mod₁^{*} Class of Models) A type-1* **S3**-model is a type-1 model $\mathcal{M} = \langle W, N, R, v \rangle$ where, additionally:

- (i) $\forall x \in W \forall y \in W (xRy \rightarrow x = x)$ (*quasi-reflexivity*)
- (ii) $\forall x \in W (x = x \rightarrow x \in N)$

Type-1* models are the **S3**-models originally introduced in Kripke (1965) together with the more general type-1 models.¹³ It is easy to verify that in any type-1* model all the non normal worlds are irreflexive, and indeed that a world x is non normal if and only if it is a dead-end.

¹³ To be precise, Kripke's definition of what we call here a type-1* model is slightly different, in so far as N is not taken as a primitive notion: a model is a triple $\langle W, R, v \rangle$ where R is a quasi-reflexive and transitive relation on W , and $N =_{df} \{x \in W \mid xRx\}$ is non-empty.

Quite to the opposite, one can restrict the notion of a type-1 model by asking that R is reflexive not only on N but on the set W of *all* worlds; even, one may ask that R is antisymmetric, so that $\langle W, R \rangle$ becomes a *partial ordering*.¹⁴

Definition 3 (Mod₂ and Mod₃ Classes of Models)

1. A *type-2 S3-model* is a type-1 model $\mathcal{M} = \langle W, N, R, v \rangle$ where, additionally, R is reflexive on W .
2. A *type-3 S3-model* is a type-1 model $\mathcal{M} = \langle W, N, R, v \rangle$ where, additionally, R is reflexive and antisymmetric on W . Since R turns out to be a *partial order* relation, in this case it will be often convenient to use the symbol $\leq$ in place of R .

Finally, one may impose to type-2 and type-3 models further restrictions, that make it so that any non normal world x is always seen by a normal world,¹⁵ and sees only itself.

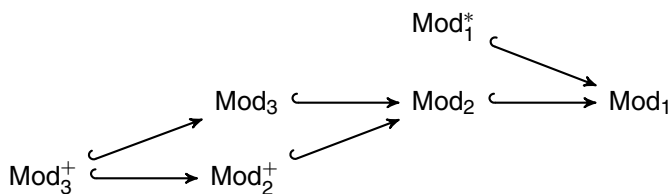
Definition 4 (Mod₂⁺ and Mod₃⁺ Classes of Models) Consider the conditions:

$$(q_1) \quad \forall y(y \in Q \rightarrow \exists x(x \in N \wedge xRy))$$

$$(q_2) \quad \forall x\forall y(x \in Q \wedge xRy \rightarrow x = y)$$

A *type-2⁺ (type-3⁺) S3-model* is a type-2 (type-3) model additionally satisfying the above conditions (q_1) and (q_2) .

The inclusion relations between the classes of models introduced above (which can be easily deduced from the definitions together with simple appropriate counterexamples) are summarized in the picture below, where an arrow ($\hookrightarrow$) denotes proper inclusion:



In order to conclude that each of these classes of models characterizes **S3** semantically, it will be thus sufficient to prove—given the above Theorem 2, stating that **S3** is sound with respect to the class **Mod₁** of models—that **S3** is complete with respect to **Mod₁^{*}** as well as with respect to **Mod₃⁺**.

¹⁴ These variants are considered in Oakes (1999).

¹⁵ Cp. Hughes and Cresswell (1996), 201.

5.5 Completeness Theorem(s) for S3

The completeness theorem for **S3** (actually, for a number of non normal systems including **S3**) with respect to the relational semantics has been proved for the first time by Kripke himself (Kripke 1965), making use of the Beth–Hintikka method of *tableaux* already employed by him for normal modal systems in his Kripke (1963). More elegant and simple proofs, using instead the *canonical model* method (that is, Lindenbaums’s Lemma and maximal sets of formulas) were provided a few years later by Makinson (Makinson 1966) and Routley (Routley 1970). We follow here, with some adaptations and improvements, the completeness proof given by Cresswell (Cresswell 1982), which has the advantage of not making reference to Lemmon’s E-systems¹⁶ and, therefore, of being self-contained in our present context.

Preliminarily, let us briefly recall some basic notions and results (we omit the proofs) concerning maximal sets of formulas.

Definition 5 Let **L** be any extension of **CPC** (here, typically, a modal extension thereof), and let Γ be a possibly infinite set of formulas in the language of **L** (here $\mathcal{L}^\square$). We write “ $\Gamma \vdash_{\mathbf{L}} A$ ” as an abbreviation for “ $\vdash_{\mathbf{L}} B_1 \wedge \dots \wedge B_n \rightarrow A$ for some $B_1, \dots, B_n \in \Gamma$ ($n \geq 0$)”.

The set Γ is said to be:

1. **L-consistent**, provided $\Gamma \not\vdash_{\mathbf{L}} \perp$ (where $\perp$ is an arbitrarily fixed **CPC**-contradiction, e.g. $\neg(p \rightarrow p)$);
2. **L-maximal**, provided Γ is **L-consistent** and any proper extension Δ of Γ is **L-inconsistent**.

Proposition 1 (Properties of L-Maximal Sets) *Let Γ be a **L-maximal** set of formulas. Then, for any formulas A, B :*

1. $\Gamma \vdash_{\mathbf{L}} A \Rightarrow A \in \Gamma$ (in particular, any **L-theorem** belongs to Γ);
2. either $A \in \Gamma$ or $\neg A \in \Gamma$, not both;
3. $A \in \Gamma$ and $A \rightarrow B \in \Gamma \Rightarrow B \in \Gamma$;
4. $A \wedge B \in \Gamma \Leftrightarrow A \in \Gamma$ and $B \in \Gamma$.

Proposition 2 (Extension Lemma—Lindenbaum) *Any **L-consistent** set Γ of formulas can be extended to a **L-maximal** set of formulas $\Delta \supseteq \Gamma$.*

Definition 6 Given a set $\Gamma \subseteq FOR^\square$, we set $\sqrt{\Gamma} := \{A \mid \Box A \in \Gamma\}$.

¹⁶ Lemmon’s E-systems E1–E5 are epistemic counterparts of Lewis’s system **S1–S5**, and have no theorems of the form $\Box A$. Cp. Lemmon (1957), 181ff.

5.5.1 Completeness of S3 with Respect to Mod_2^+ and Mod_1^*

We start by proving that **S3** is complete with respect to the class of type-2⁺ **S3**-models.

Definition 7 (Canonical Model for S3) The canonical model $\mathcal{M}^c = \langle W, N, R, v \rangle$ is made up as follows:

- (a) $N := \{\Gamma \mid \Gamma \text{ is } \mathbf{S3}\text{-maximal}\}$
- (b) $Q := \{\Gamma \mid \Gamma \text{ is } \mathbf{CPC}\text{-maximal, } \Box\top \notin \Gamma, \text{ and } \sqrt{\Delta} \subseteq \Gamma \text{ for some } \Delta \in N\}$
- (c) $W = N \cup Q$
- (d) for $\Gamma, \Delta \in W$: $\Gamma R \Delta$ iff either $\Gamma \in N$ and $\sqrt{\Gamma} \subseteq \Delta$, or $\Gamma \in Q$ and $\Gamma = \Delta$
- (e) for $\Gamma \in W$ and $p \in AT$: $v(\Gamma, p) = 1$ if $p \in \Gamma$, and $= 0$ otherwise

Notice that N and Q are disjoint, since $\Box\top \in \Gamma$ for every **S3**-maximal set Γ .

Lemma 2 $\mathcal{M}^c$ is a type-2⁺ **S3**-model.

Proof $N \subseteq W$ by definition, and N is non-empty by the Extension Lemma (e.g., it contains a **S3**-maximal extension of $\{\Box\top\}$, which is **S3**-consistent because $\vdash_{\mathbf{S3}} \Box\top$).

R is reflexive on W : indeed, if $\Gamma \in W$ then either $\Gamma \in N$ or $\Gamma \in Q$. In the first case, since $\vdash_{\mathbf{S3}} \Box A \rightarrow A$ for all A , it holds $\sqrt{\Gamma} \subseteq \Gamma$ and so $\Gamma R \Gamma$ does hold; in the second case $\Gamma R \Gamma$ holds trivially by the definition of R .

R is transitive on W : indeed, suppose $\Gamma R \Delta$ and $\Delta R \Sigma$. If $\Gamma \in Q$ then $\Gamma = \Delta = \Sigma$, so trivially $\Gamma R \Sigma$. If $\Gamma \in N$ and $\Delta \in Q$, then $\Sigma = \Delta$, so trivially again $\Gamma R \Sigma$. In the remaining case, $\Gamma, \Delta \in N$, suppose $A \in \sqrt{\Gamma}$. Since

$$\vdash_{\mathbf{S3}} \Box A \rightarrow \Box(\Box\top \rightarrow \Box A)$$

(Lemma 1, point 1) and $\Box A \in \Gamma$, we have $\Box(\Box\top \rightarrow \Box A) \in \Gamma$ and so, by $\Gamma R \Delta$, $\Box\top \rightarrow \Box A \in \Delta$. Since also $\Delta \in N$, we have $\Box\top \in \Delta$, hence $\Box A \in \Delta$ and, by $\sqrt{\Delta} \subseteq \Sigma$, $A \in \Sigma$.

Finally, it is easy to verify, on the basis of the definitions of Q and of R , that $\mathcal{M}^c$ satisfies the additional conditions (q_1) and (q_2) of Definition 4.

Lemma 3 For all $A \in \mathcal{L}^\Box$, $\Gamma \Vdash_{\mathcal{M}^c} A \Leftrightarrow A \in \Gamma$.

Proof By induction on the construction of A . The case $A \in AT$ is immediate by the definition of v . The cases $A \equiv \neg B$ and $A \equiv B \circ C$ ($\circ \in \{\wedge, \vee, \rightarrow\}$) are straightforward by I.H. and the properties of maximal sets. For $A \equiv \Box B$ we reason as follows.

[$\Leftarrow$] Suppose $\Box B \in \Gamma$. If $\Gamma \in Q$ then $\Box\top \notin \Gamma$ and $\sqrt{\Delta} \subseteq \Gamma$ for some $\Delta \in N$; thus by point 2 of Lemma 1 $\Box(\Box B \rightarrow \Box\top) \in \Delta$, hence $\Box B \rightarrow \Box\top \in \sqrt{\Delta} \subseteq \Gamma$, and so also, by **CPC**-maximality of Γ , $\Box\top \in \Gamma$: contradiction. Therefore $\Gamma \in N$, and then $\Gamma \Vdash_{\mathcal{M}^c} \Box B$ since for all Δ s.t. $\Gamma R \Delta$ we have $B \in \sqrt{\Gamma} \subseteq \Delta$, and so $\Delta \Vdash_{\mathcal{M}^c} B$ by the I.H.

[$\Rightarrow$] Suppose $(*) \Box B \notin \Gamma$. If $\Gamma \in Q$ then, trivially, $\Gamma \not\vdash_{\mathcal{M}^c} \Box B$. If instead $\Gamma \in N$, then consider the set

$$\Theta := \sqrt{\Gamma} \cup \{\neg B\}$$

and distinguish the following two cases.

Case 1 Θ is **S3**-consistent. Then by the Extension Lemma there is a $\Delta \in N$ s.t. $\Theta \subseteq \Delta$. Thus $\Gamma R \Delta$, $B \notin \Delta$ and, by the I.H., $\Delta \not\vdash_{\mathcal{M}^c} B$. Therefore $\Gamma \not\vdash_{\mathcal{M}^c} \Box B$.

Case 2 Θ is **S3**-inconsistent. Then there exist formulas $\Box C_1, \dots, \Box C_n \in \Gamma$ such that

$$\vdash_{\mathbf{S3}} C_1 \wedge \dots \wedge C_n \rightarrow B$$

It follows by the admissible rule R.2 (Lemma 1) that

$$\vdash_{\mathbf{S3}} \Box(C_1 \wedge \dots \wedge C_n \rightarrow (\Box T \rightarrow B))$$

and also, using A3, A2 and (3) of Lemma 1,

$$\vdash_{\mathbf{S3}} \Box C_1 \wedge \dots \wedge \Box C_n \rightarrow \Box(\Box T \rightarrow B)$$

Thus, using $\Box C_1, \dots, \Box C_n \in \Gamma$,

$$\Box(\Box T \rightarrow B) \in \Gamma \tag{5.1}$$

And we can now show that

$$\mathcal{E} := \sqrt{\Gamma} \cup \{\neg B, \neg \Box T\} \text{ is } \mathbf{CPC}\text{-consistent} \tag{5.2}$$

Indeed, suppose $\mathcal{E}$ is **CPC**-inconsistent. Then there are formulas $\Box D_1, \dots, \Box D_m \in \Gamma$ such that

$$\vdash_{\mathbf{CPC}} D_1 \wedge \dots \wedge D_m \rightarrow \Box T \vee B$$

whence by RN^-

$$\vdash_{\mathbf{S3}} \Box(D_1 \wedge \dots \wedge D_m \rightarrow \Box T \vee B)$$

and so, using A3, A2 and (3) of Lemma 1,

$$\vdash_{\mathbf{S3}} \Box D_1 \wedge \dots \wedge \Box D_m \rightarrow \Box(\Box T \vee B)$$

Now, being $\Box D_1, \dots, \Box D_m \in \Gamma$, we get

$$\Box(\Box T \vee B) \in \Gamma \tag{5.3}$$

By (5.1) and (5.3), together with $\vdash_{\mathbf{S3}} \Box(\Box\top \rightarrow B) \rightarrow (\Box(\Box\top \vee B) \rightarrow \Box B)$, we get $\Box B \in \Gamma$, against the initial assumption (*).

Thus, by (5.2) and the Extension Lemma, there exists a **CPC**-maximal set Σ such that $\mathcal{E} \subseteq \Sigma$. Therefore $\sqrt{\Gamma} \subseteq \Sigma$ and $\Box\top \notin \Sigma$, whence $\Sigma \in Q$ and $\Gamma R \Sigma$ follows, as well as $B \notin \Sigma$, whence $\Sigma \not\models_{\mathcal{M}^c} B$. Thus we conclude that, also in Case 2, $\Gamma \not\models_{\mathcal{M}^c} \Box B$.

Theorem 3 *S3 is sound and complete w.r. to the class $\mathbf{Mod}_2^+$ of models.*

Proof Soundness follows from Theorem 2 and the inclusion $\mathbf{Mod}_2^+ \subseteq \mathbf{Mod}_1$. As to completeness, suppose $\not\models_{\mathbf{S3}} A$. Then $\{\neg A\}$ is **S3**-consistent and so by the Extension Lemma there is a **S3**-maximal set Γ such that $A \notin \Gamma$. Thus $\Gamma \not\models_{\mathcal{M}^c} A$ by Lemma 3, and $\Gamma \in N$. Hence A is not true in the type-2⁺ model $\mathcal{M}^c$.

By slightly modifying the above canonical model construction, one can also prove

Theorem 4 *S3 is sound and complete w.r. to the class $\mathbf{Mod}_1^*$ of models.*

Proof Just replace the definition of the accessibility relation R in the canonical model $\mathcal{M}^c$ (Definition 7, item (d)) with

(d') for $\Gamma, \Delta \in W$: $\Gamma R \Delta$ iff $\Gamma \in N$ and $\sqrt{\Gamma} \subseteq \Delta$.

It is then easy to verify that $\mathcal{M}^c \in \mathbf{Mod}_1^*$, and that Lemma 3 still goes through.

5.5.2 Completeness of S3 with Respect to $\mathbf{Mod}_3^+$

We now show that the addition of the *antisymmetry* of R to the structural conditions of type-2⁺ models does not destroy the completeness result proved above, and so that **S3** is complete also with respect to the narrower class $\mathbf{Mod}_3^+$ of models. To this aim, we make use of the *unfolding* technique.

Definition 8 Let $\mathcal{M} = \langle W, N, R, v \rangle$ be a type-2 model. The associated model

$$\mathcal{M}^t = \langle W^t, N^t, \preceq, v^t \rangle$$

is defined as follows.

- W^t is the set of all finite sequences $\sigma \equiv \langle x_1, \dots, x_n \rangle$ of elements of W ($n \geq 1$) satisfying:
 - $n = 1$ and $x_1 \in N$, or $n > 1$ and at least $x_1, \dots, x_{n-1} \in N$
 - $\forall i (1 \leq i < n \rightarrow x_i R x_{i+1})$.

Given an element $\sigma \equiv \langle x_1, \dots, x_n \rangle$ of W^t and an element y of W , we denote:

- by $\hat{\sigma}$ the end-component x_n of σ
- by $\sigma * y$ the extension $\langle x_1, \dots, x_n, y \rangle$ of σ . This is an element of W^t provided that (i) $\hat{\sigma} \in N$ and (ii) $y \in W$ is such that $\hat{\sigma} R y$.

- $N^t := \{\sigma \in W^t \mid \hat{\sigma} \in N\}$
- for $\sigma, \tau \in W^t$, $\sigma \preceq \tau$ iff σ is a (possibly improper) initial segment of τ .
- for $\sigma \in W^t$ and $p \in AT$: $v^t(\sigma, p) = v(\hat{\sigma}, p)$

Lemma 4 *If $\mathcal{M} \in \text{Mod}_2^+$ then $\mathcal{M}^t \in \text{Mod}_3^+$.*

Proof Clearly, $\langle W^t, \preceq \rangle$ is a partial ordering, actually a tree ordering; so $\mathcal{M}^t \in \text{Mod}_3$. To conclude, one has to verify that also the two conditions (q_1) and (q_2) of Definition 4 are satisfied.

Indeed, if $\sigma \in Q^t$ then $\sigma \equiv \langle x_1, \dots, x_n \rangle$ and $n > 1$, $x_1, \dots, x_{n-1} \in N$, $x_n \in Q$, so $\tau \equiv \langle x_1, \dots, x_{n-1} \rangle \in N^t$ and $\tau \preceq \sigma$. Moreover, if $\sigma, \tau \in Q^t$ and $\sigma \preceq \tau$ then σ and τ must coincide, otherwise τ would contain two components belonging to Q , against the definition of W^t .

Lemma 5 *For all A and $\sigma \in W^t$, $\sigma \Vdash_{\mathcal{M}^t} A \Leftrightarrow \hat{\sigma} \Vdash_{\mathcal{M}} A$*

Proof By induction on A . Let us discuss only the nontrivial case $A \equiv \Box B$.

[$\Rightarrow$] Assume $\sigma \Vdash_{\mathcal{M}^t} \Box B$. Then $\sigma \in N^t$, so $\hat{\sigma} \in N$ and, for all $y \in W$ such that $\hat{\sigma} R y$, $\sigma * y \in W^t$ and $\sigma \preceq \sigma * y$. Therefore $\sigma * y \Vdash_{\mathcal{M}^t} B$ and, by I.H., $\widehat{\sigma * y} = y \Vdash_{\mathcal{M}} B$. Thus $\hat{\sigma} \Vdash_{\mathcal{M}} \Box B$.

[$\Leftarrow$] Assume $\hat{\sigma} \Vdash_{\mathcal{M}} \Box B$. Then $\hat{\sigma} \in N$, so $\sigma \in N^t$ and for all $\tau \in W^t$ such that $\sigma \preceq \tau$ we have, by transitivity of R , $\hat{\sigma} R \hat{\tau}$, hence $\hat{\tau} \Vdash_{\mathcal{M}} B$ and, by I.H., $\tau \Vdash_{\mathcal{M}^t} B$. Thus $\sigma \Vdash_{\mathcal{M}^t} \Box B$.

Theorem 5 *S3 is sound and complete w.r. to the class Mod_3^+ of models.*

Proof By Theorem 3 and Lemmas 4 and 5.

A natural question to ask at this point is whether **S3** is semantically characterized also by the class of all finite type-3 or type-3⁺ models.

Although, as we will see in the next section, **S3** enjoys the *finite model property* with respect to the class Mod_2^+ of models, the answer to the above question is *negative*. Consider indeed the following variant of Grzegorczyk's schema:¹⁷

$$(GRZ') \quad \Box((\Box\Box\top \rightarrow \Box(A \rightarrow \Box A)) \rightarrow A) \rightarrow A$$

We have:

Theorem 6

1. *The schema (GRZ') is true in all finite models $\mathcal{M} \in \text{Mod}_3$*
2. $\not\models_{\text{S3}} (GRZ')$

¹⁷ $(GRZ) := \Box(\Box(A \rightarrow \Box A) \rightarrow A) \rightarrow A$, known as *Grzegorczyk's schema* (cp. Grzegorczyk 1967), is the characteristic axiom schema of the normal modal system **GRZ** := **K**+(GRZ). **GRZ** is a proper extension of **S4**, and is not included in **S5**.

Proof

- (1) Let $\mathcal{M} = \langle W, N, \leq, v \rangle$ be a finite, partially ordered type-3 model and, in view of a contradiction, suppose there is a normal world $x_0 \in W$ such that

$$x_0 \not\models_{\mathcal{M}} \Box((\Box\Box\top \rightarrow \Box(A \rightarrow \Box A)) \rightarrow A) \rightarrow A \quad (5.4)$$

for some formula A . We inductively define a set $\{x_n\}_{n \geq 1}$ of normal worlds, satisfying:

- (i) $x_0 \leq x_1 \leq x_2 \leq x_3 \dots$
- (ii) if n is even, $x_n \not\models_{\mathcal{M}} A$
- (iii) if n is odd, $x_n \not\models_{\mathcal{M}} \Box A$ and $x_n \Vdash_{\mathcal{M}} A$

For $x_0 \in N$ we have:

$$(a) \ x_0 \Vdash_{\mathcal{M}} \Box((\Box\Box\top \rightarrow \Box(A \rightarrow \Box A)) \rightarrow A), \quad (b) \ x_0 \not\models_{\mathcal{M}} A \quad (5.5)$$

Suppose now $x_0, \dots, x_k$ ($k \geq 1$) satisfy (i)–(iii), and pick x_{k+1} as follows.

Case 1: k is even. Then by (5.5)(a) together with reflexivity (if $k = 0$) and transitivity (if $k > 0$) of $\leq$, we have $x_k \Vdash_{\mathcal{M}} (\Box\Box\top \rightarrow \Box(A \rightarrow \Box A)) \rightarrow A$, so by (ii) $x_k \not\models_{\mathcal{M}} \Box\Box\top \rightarrow \Box(A \rightarrow \Box A)$. Thus

$$(a) \ x_k \Vdash_{\mathcal{M}} \Box\Box\top \quad (b) \ x_k \not\models_{\mathcal{M}} \Box(A \rightarrow \Box A) \quad (5.6)$$

By (5.6)(b) and $x_k \in N$ we can choose a world $x_{k+1} \geq x_k$ such that

$$x_{k+1} \not\models_{\mathcal{M}} A \rightarrow \Box A.$$

Then by (5.6)(a) $x_{k+1} \Vdash_{\mathcal{M}} \Box\top$ and so

$$x_{k+1} \in N, \quad x_{k+1} \Vdash_{\mathcal{M}} A, \quad x_{k+1} \not\models_{\mathcal{M}} \Box A$$

Case 2: k is odd. Then by (iii) and $x_k \in N$ we can choose a world $x_{k+1} \geq x_k$ such that $x_{k+1} \not\models_{\mathcal{M}} A$. Since by (5.5)(a) and transitivity

$$x_{k+1} \Vdash_{\mathcal{M}} (\Box\Box\top \rightarrow \Box(A \rightarrow \Box A)) \rightarrow A,$$

it follows that $x_{k+1} \not\models_{\mathcal{M}} \Box\Box\top \rightarrow \Box(A \rightarrow \Box A)$, so in particular

$$x_{k+1} \Vdash_{\mathcal{M}} \Box\Box\top$$

whence $x_{k+1} \in N$.

So, in both cases, $x_0, \dots, x_{k+1}$ satisfy (i)–(iii), and we are done.

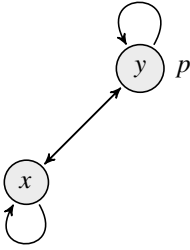
Now the desired contradiction is at hand: since by assumption W is finite, two numbers $n, m \in \mathbb{N}$ must exist, such that $m < n$ and $x_m = x_n$. By (ii) and (iii) above, either both n and m are even, or both n and m are odd.

In the first case, $m+1$ is odd and $m+1 < n$; so by (i) we have $x_m \leq x_{m+1}$ and also (using the transitivity of $\leq$), $x_{m+1} \leq x_n = x_m$; therefore, by antisymmetry, $x_m = x_{m+1}$, which is impossible by (ii) and (iii). We reason analogously in the second case.

(2) It is easily seen that the atomic instance

$$\Box((\Box\Box\top \rightarrow \Box(p \rightarrow \Box p)) \rightarrow p) \rightarrow p$$

of (GRZ') is false in the following model $\mathcal{M} \in \mathbf{Mod}_1$ (actually, a **S5** model):



where $N = \{x, y\}$, $v(x, p) = 0$, $v(y, p) = 1$.

5.6 Finite Model Property and Decidability of S3

We prove that **S3** is complete w.r. to the class of *finite type-2⁺ models*, and indeed that any formula A such that $\not\models_{\mathbf{S3}} A$ has a type-2⁺ countermodel with at most n_A worlds, where n_A functionally depends on the complexity of A . The proof employs the *filtration* method.

Let $\mathcal{M} = \langle W, N, R, v \rangle$ be a type-2⁺ model, and let Φ be a set of formulas closed under subformulas and containing $\Box\top$. Define:

- for $x, y \in W$, $x \sim_\Phi y \Leftrightarrow$ for all $B \in \Phi$, $x \Vdash_{\mathcal{M}} B$ iff $y \Vdash_{\mathcal{M}} B$.

Clearly, $\sim_\Phi$ (in the following we drop the subscript Φ whenever possible) is an equivalence relation.

- $W^\Phi := \{[x]_\sim \mid x \in W\}$, the quotient of W modulo the equivalence relation $\sim_\Phi$ (where $[x]_\sim := \{y \in W \mid x \sim_\Phi y\}$).
- $N^\Phi := \{[x]_\sim \mid x \in N\}$.

Notice that N^Φ is well defined: indeed, by $\Box\top \in \Phi$ we have that $x \sim y$ implies $x \in N$ iff $y \in N$.

- for $[x], [y] \in W^\Phi$, $[x]R^\Phi[y]$ iff either

- $x \in N$,

- for all $\Box B \in \Phi$, if $x \Vdash_{\mathcal{M}} \Box B$ then $y \Vdash_{\mathcal{M}} B$,
- and, if $y \in N$, then for all $\Box B \in \Phi$, if $x \Vdash_{\mathcal{M}} \Box B$ then $y \Vdash_{\mathcal{M}} \Box B$

or $x, y \in Q$ and $x \sim y$.

- $v^\Phi([x], p) = v(x, p)$, if $p \in \Phi$, and $= 0$ otherwise.

Of course, one has to verify that the definitions of N^Φ , R^Φ and v^Φ are correct, that is, they are independent of the representatives used for the equivalence classes. This is obvious for v^Φ and straightforward for N^Φ and R^Φ once one recalls (see above) that $x \sim y$ implies $x \in N$ iff $y \in N$.

Lemma 6 *For all $x, y \in W$ and all formulas $\Box B \in \Phi$:*

1. if xRy then $[x]R^\Phi[y]$
2. if $[x]R^\Phi[y]$ and $x \Vdash_{\mathcal{M}} \Box B$ then $y \Vdash_{\mathcal{M}} B$.

Proof

- (1) Let xRy . If $x \in N$, suppose $x \Vdash_{\mathcal{M}} \Box B$. Then $y \Vdash_{\mathcal{M}} B$ and, in case $y \in N$, also $y \Vdash_{\mathcal{M}} \Box B$ —since $x \Vdash_{\mathcal{M}} \Box B \rightarrow (\Box \top \rightarrow \Box B)$, which is a theorem of **S3**, and $y \Vdash_{\mathcal{M}} \Box \top$. Thus $[x]R^\Phi[y]$. If instead $x \in Q$, then $x = y$ because $\mathcal{M}$ is a type-2⁺ model, so $x \sim y$ and $[x]R^\Phi[y]$ by definition of R^Φ .
- (2) is immediate from the definition of R^Φ , considering that $x \Vdash_{\mathcal{M}} \Box B$ is always false in case $x \notin N$.

Lemma 7 *The Φ -filtration $\mathcal{M}^\Phi$ of a type-2⁺ model $\mathcal{M}$ is itself a type-2⁺ model, and $\mathcal{M}^\Phi$ is finite if $\mathcal{M}$ is finite.*

Proof Obviously, $\emptyset \neq N^\Phi \subseteq W^\Phi$. Next:

- (a) R^Φ is reflexive. Let $[x] \in W^\Phi$, and suppose first that $[x] \in N^\Phi$, $\Box B \in \Phi$, $x \Vdash_{\mathcal{M}} \Box B$: by the soundness of **S3** w.r. to type-2⁺ models, $x \Vdash_{\mathcal{M}} \Box B \rightarrow B$ and $x \Vdash_{\mathcal{M}} \Box B \rightarrow (\Box \top \rightarrow \Box B)$, which are theorems of **S3**. Thus $x \Vdash_{\mathcal{M}} B$ and also $x \Vdash_{\mathcal{M}} \Box B$, because $x \Vdash_{\mathcal{M}} \Box \top$ by $x \in N$. Thus $[x]R^\Phi[x]$ follows by the definition of R^Φ . Next, suppose $[x] \in Q^\Phi$: then $x \in Q$ and $[x]R^\Phi[x]$ trivially holds.
- (b) R^Φ is transitive. Suppose $[x]R^\Phi[y]$ and $[y]R^\Phi[z]$, and distinguish the following four cases.
 - (i) $x \in Q$: then $x \sim y \sim z$ and $[x]R^\Phi[z]$ holds trivially.
 - (ii) $x \in N$ and $y \in Q$: then $z \in Q$ and $y \sim z$, thus $x \Vdash_{\mathcal{M}} \Box B$ (where $\Box B \in \Phi$) implies $y \Vdash_{\mathcal{M}} B$, which in turn implies $z \Vdash_{\mathcal{M}} B$. Hence $[x]R^\Phi[z]$.
 - (iii) $x, y \in N$ and $z \in Q$: then $x \Vdash_{\mathcal{M}} \Box B$ (where $\Box B \in \Phi$) implies $y \Vdash_{\mathcal{M}} \Box B$, which in turn implies $z \Vdash_{\mathcal{M}} B$. Hence $[x]R^\Phi[z]$.
 - (iv) $x, y, z \in N$: then $x \Vdash_{\mathcal{M}} \Box B$ (where $\Box B \in \Phi$) implies $y \Vdash_{\mathcal{M}} \Box B$, which in turn implies $z \Vdash_{\mathcal{M}} \Box B$ and $z \Vdash_{\mathcal{M}} B$. Hence $[x]R^\Phi[z]$.

- (c) Every non normal world of $\mathcal{M}^\Phi$ is seen from a normal one. Indeed, if $[y] \notin N^\Phi$ then $y \in Q$, so there is $x \in N$ such that xRy because $\mathcal{M}$ is a type-2⁺ model. But then $[x] \in N^\Phi$ and, by (1) of Lemma 6, $[x]R^\Phi[y]$.
- (d) Every non normal world of $\mathcal{M}^\Phi$ sees only itself. Indeed, if $[x] \notin N^\Phi$ and $[x]R^\Phi[y]$ then, by definition of R^Φ , $x, y \in Q$ and $x \sim y$, so $[x] = [y]$.

Lemma 8 (Filtration Lemma) *For all formulas $A \in \Phi$ and every world x of the model $\mathcal{M}$,*

$$x \Vdash_{\mathcal{M}} A \quad \Leftrightarrow \quad [x] \Vdash_{\mathcal{M}^\Phi} B$$

Proof By induction on the construction of $A \in \Phi$. The use of (1) and (2) of Lemma 6 makes the verification of the only non trivial case, $A \equiv \Box B$, straightforward.

Theorem 7 (Finite Model Property of S3) *S3 is sound and complete w.r. to the class of finite, type-2⁺ models.*

Proof Suppose $\not\models_{S3} A$. By Theorem 3 there are a type-2⁺ model $\mathcal{M}$ and a normal world x in it, such that $x \not\Vdash_{\mathcal{M}} A$. Let Φ_A be the following finite set of formulas:

$$\{B \mid B \text{ is a subformula of } A\} \cup \{\Box \top\}.$$

By Lemma 7 the filtrated model $\mathcal{M}^{\Phi_A}$ is a finite type-2⁺ model, $[x]$ is a normal world in it, and by Lemma 7 $[x] \not\Vdash_{\mathcal{M}^{\Phi_A}} A$.

Notice that the cardinality of $\mathcal{M}^\Phi$ functionally depends on the complexity of A . More precisely, let $s(A) \in \mathbb{N}$ be the number of occurrences of atoms and primitive logical operators in A . It is easily seen that A has at most $s(A)$ subformulas, so $\|\Phi_A\| \leq s(A) + 1$ and $\|\mathcal{M}^\Phi\| \leq 2^{s(A)+1}$.

Theorem 8 (Decidability of S3) *S3 is decidable.*

Proof By the observation concluding the *proof* of Theorem 7, it follows that in order to decide whether a formula A is or not a theorem of S3, it is sufficient to effectively explore the finite space containing all type-2⁺ models $\mathcal{M} = \langle W, N, R, v \rangle$ such that $W \leq 2^{s(A)+1}$ and $v(x, p) = 0$ for every $x \in W$ and every atom p not occurring in A .

5.7 Modal Translation of IPC in S3 and Embeddings of S4 in S3

As we already said, Becker's idea of a modal interpretation of the intuitionistic propositional calculus **IPC** was realized in 1933 by Kurt Gödel, who did the “future work” Becker intended to pursue after the failure of his own attempts in *On the Logic of Modalities*.

To be precise, Becker was asking for a translation of **IPC** into **S3**, and Gödel found a translation of **IPC** into the stronger system **S4**. In 1963 Ian Hacking (Hacking 1963) proved that the McKinsey-Tarski translation provides a sound and faithful translation of **IPC** *already into Lewis's S3*. Hacking's proof-theoretical demonstration, which makes use of a normalization theorem for a suitable natural deduction presentation of **S3**, is rather convoluted. We present here a much simpler and entirely semantical proof, obtained by exploiting the ideas and a conjecture contained in Oakes (1999). Let us first recall how Gödel's translation $(\dots)^G$ and the McKinsey-Tarski translation $(\dots)^*$ are defined.

Definition 9 The mappings $(\dots)^G$ and $(\dots)^*$ from FOR_I into $FOR^\square$ are inductively defined as follows

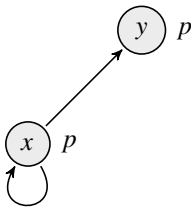
1. Gödel's translation $(\dots)^G$:

- $p^G := p$, where p is a propositional atom
- $(\neg\beta)^G := \neg\Box\beta^G$
- $(\beta \wedge \gamma)^G := \beta^G \wedge \gamma^G$
- $(\beta \vee \gamma)^G := \Box\beta^G \vee \Box\gamma^G$
- $(\beta \rightarrow \gamma)^G := \Box\beta^G \rightarrow \Box\gamma^G$

2. McKinsey-Tarski translation $(\dots)^*$:¹⁸

- $p^* := \Box p$, where p is a propositional atom
- $(\neg\beta)^* := \Box\neg\beta^*$
- $(\beta \wedge \gamma)^* := \beta^* \wedge \gamma^*$
- $(\beta \vee \gamma)^* := \beta^* \vee \gamma^*$
- $(\beta \rightarrow \gamma)^* := \Box(\beta^* \rightarrow \gamma^*)$

The $(\dots)^G$ translation turns out *not to be sound* with respect to **S3**. For instance, $\vdash_{\mathbf{IPC}} p \rightarrow p \vee q$, but the $(\dots)^G$ translation of this formula, $\Box p \rightarrow \Box(\Box p \vee \Box q)$, is not a theorem of **S3**. In the very simple type-1 countermodel $\mathcal{M}$ below we have indeed $x \not\models_{\mathcal{M}} \Box p \rightarrow \Box(\Box p \vee \Box q)$:



where $N = \{x\}$, $v(x, p) = v(y, p) = 1$.

Let us now prove that the McKinsey-Tarski translation does instead work, also for **S3**.

¹⁸ McKinsey and Tarski (1948).

Lemma 9 *Let $\mathcal{M} = \langle W, N, \leq, v \rangle$ be a type-3 model. Then, for every $A \in \mathcal{L}_I$ and for every $x \in W$:*

$$x \Vdash_{\mathcal{M}} A^* \Rightarrow x \in N$$

Proof By induction on the construction of A .

- $A \equiv p \in AT$. Then $p^* \equiv \Box p$, and obviously $x \in N$ if $x \Vdash_{\mathcal{M}} \Box p$.
- $A \equiv B \wedge C, B \vee C$. Immediate by the induction hypothesis.
- $A \equiv \neg B, B \rightarrow C$. Then, by the definition of the $*$ -translation, A^* is a boxed formula and therefore $x \Vdash_{\mathcal{M}} A^*$ implies that x is normal.

Definition 10 Let $\mathcal{M} = \langle W, N, \leq, v \rangle$ be a type-3 model. We canonically associate to $\mathcal{M}$ the **IPC** model

$$\mathcal{M}_N = \langle N, \leq| N, v_N \rangle$$

where, for every $p \in AT$, $v_N(p) = \{x \in W \mid x \Vdash_{\mathcal{M}} \Box p\}$.

Clearly, $v_N(p) \subseteq N$. In order to conclude that $\mathcal{M}_N$ is an intuitionistic model, it is sufficient to verify that $v_N(p)$ is closed under $\leq$. Suppose $x, y \in N$ and $x \in v_N(p)$. Then $x \Vdash_{\mathcal{M}} \Box p$ and for all $z \in W$, if $y \leq z$ then $x \leq z$ by transitivity, and so $z \Vdash_{\mathcal{M}} p$. Hence $y \Vdash_{\mathcal{M}} \Box p$ and $y \in v_N(p)$.

Lemma 10 *Let $\mathcal{M} = \langle W, N, \leq, v \rangle$ be a type-3 model, and $\mathcal{M}_N = \langle N, \leq| N, v_N \rangle$ be its associated **IPC** model. Then for every formula $A \in \mathcal{L}_I$ and every $x \in N$:*

$$x \Vdash_{\mathcal{M}_N} A \Leftrightarrow x \Vdash_{\mathcal{M}} A^*$$

Proof By induction on the construction of $A \in \mathcal{L}_I$.

- $A \equiv p \in AT$. Then $p^* \equiv \Box p$ and, by construction, $x \Vdash_{\mathcal{M}_N} p$ iff $x \Vdash_{\mathcal{M}} \Box p$.
- $A \equiv B \wedge C, B \vee C$. Immediate by the induction hypothesis.
- $A \equiv B \rightarrow C$. Then $A^* \equiv \Box(B^* \rightarrow C^*)$.

Suppose $x \not\Vdash_{\mathcal{M}_N} B \rightarrow C$. Then there is $y \geq x, y \in N$, such that $y \Vdash_{\mathcal{M}_N} B$ and $y \not\Vdash_{\mathcal{M}_N} C$. Then by I.H. $y \Vdash_{\mathcal{M}} B^*$ and $y \not\Vdash_{\mathcal{M}} C^*$, so $y \not\Vdash_{\mathcal{M}} B^* \rightarrow C^*$ and $x \not\Vdash_{\mathcal{M}} \Box(B^* \rightarrow C^*)$.

Conversely, suppose $x \not\Vdash_{\mathcal{M}} \Box(B^* \rightarrow C^*)$. Since $x \in N$, there is $y \geq x, y \in W$, such that $y \Vdash_{\mathcal{M}} B^*$ and $y \not\Vdash_{\mathcal{M}} C^*$. We can now apply Lemma 9 to $y \Vdash_{\mathcal{M}} B^*$ and infer $y \in N$, so that we may apply the I.H. obtaining $y \Vdash_{\mathcal{M}_N} B$ and $y \not\Vdash_{\mathcal{M}_N} C$, whence $x \not\Vdash_{\mathcal{M}_N} B \rightarrow C$ follows.

- $A \equiv \neg B$. Then $A^* \equiv \Box \neg B^*$, and we reason as in the previous case, making use of Lemma 9.

Theorem 9 (Modal Translation of IPC in S3) *For any formula $A \in \mathcal{L}_I$:*

$$\vdash_{\text{IPC}} A \Leftrightarrow \vdash_{\text{S3}} A^*$$

Proof $[\Rightarrow]$ Suppose $\not\vdash_{\mathbf{S3}} A^*$; then by Theorem 5 there are a type-3 model $\mathcal{M}$ and a normal world x in $\mathcal{M}$, such that $x \not\Vdash_{\mathcal{M}} A^*$. Then, in the associated intuitionistic model $\mathcal{M}_N$, we have that $x \not\Vdash_{\mathcal{M}_N} A$ by Lemma 10. Therefore $\not\vdash_{\mathbf{IPC}} A$ by the soundness theorem for **IPC**.

$[\Leftarrow]$ This direction, of course, follows immediately from the faithfulness of the *-translation of **IPC** into **S4**¹⁹ and **S3** $\subseteq$ **S4**. But there is also a very simple and direct semantic proof. Suppose indeed that $\not\vdash_{\mathbf{IPC}} A$; then by the completeness theorem for **IPC** there exist an intuitionistic model

$$\mathcal{M} = \langle W, \leq, v \rangle$$

and a world $x \in W$, such that $x \not\Vdash_{\mathcal{M}} A$. Now, $\mathcal{M}$ can be seen as a **S3** type-3 model in which every world is normal. More precisely, let $\mathcal{M}'$ be the type-3 model $\langle W, W, \leq, v \rangle$: then it is easily verified that for every formula $B \in \mathcal{L}_I$ and every $x \in W$,

$$x \Vdash_{\mathcal{M}} B \Leftrightarrow x \Vdash_{\mathcal{M}'} B^*$$

Thus $x \not\Vdash_{\mathcal{M}} A$ yields $x \not\Vdash_{\mathcal{M}'} A^*$ and so, by the soundness theorem for **S3**, $\not\vdash_{\mathbf{S3}} A^*$.

Other sound and faithful translations of **IPC** into **S3** can be obtained by composing the McKinsey-Tarski translation with following *compositional* embedding of **S4** into **S3**.

Definition 11 The mapping $\tau : FOR^{\square} \longrightarrow FOR^{\square}$ is inductively defined as follows:

- $p^{\tau} := p$, where p is a propositional atom
- $(\neg B)^{\tau} := \neg B^{\tau}$
- $(B \circ C)^{\tau} := B^{\tau} \circ C^{\tau}$ ($\circ \in \{\wedge, \vee, \rightarrow\}$)
- $(\Box B)^{\tau} := \Box(\Box \top \rightarrow B^{\tau})$

Theorem 10 For every $A \in \mathcal{L}^{\square}$:

$$\vdash_{\mathbf{S4}} A \Leftrightarrow \vdash_{\mathbf{S3}} A^{\tau}$$

Proof $[\Rightarrow]$ Given a type-2 model $\mathcal{M} = \langle W, N, R, v \rangle$, let $\mathcal{M}'$ be the **S4** model

$$\langle N, R \upharpoonright N, v \upharpoonright (N \times AT) \rangle.$$

It holds that for all $x \in N$ and all $A \in FOR^{\square}$:

$$(1) \quad x \Vdash_{\mathcal{M}'} A \Leftrightarrow x \Vdash_{\mathcal{M}} A^{\tau}.$$

¹⁹ McKinsey and Tarski (1948): $\vdash_{\mathbf{S4}} A^* \Rightarrow \vdash_{\mathbf{IPC}} A$ for any formula $A \in \mathcal{L}_I$.

The proof is by induction on A and the only non trivial case is $A \equiv \Box B$.

Suppose $x \in N$ and $x \not\models_{\mathcal{M}'} \Box B$. Then there exists a world y of $\mathcal{M}'$, hence a $y \in N$, such that xRy and $y \not\models_{\mathcal{M}'} B$. Then, by the induction hypothesis, $y \not\models_{\mathcal{M}} B^\tau$. On the other side, by $y \in N$ we have $y \Vdash_{\mathcal{M}} \top$, so $y \not\models_{\mathcal{M}} \Box \top \rightarrow B^\tau$ and $x \not\models_{\mathcal{M}} \Box(\Box \top \rightarrow B^\tau)$, that is $x \not\models_{\mathcal{M}} (\Box B)^\tau$.

Conversely, suppose $x \in N$ and $x \not\models_{\mathcal{M}} (\Box B)^\tau$. Then there exists $y \in W$ such that xRy , $y \Vdash_{\mathcal{M}} \Box \top$, so $y \in N$, and $y \not\models_{\mathcal{M}} B^\tau$. Then the I.H. applies, and $y \not\models_{\mathcal{M}'} B$, whence $x \not\models_{\mathcal{M}'} \Box B$ follows.

Now, assume $\not\models_{\mathbf{S3}} A^\tau$. By the completeness theorem for **S3**, there exist a type-2 model $\mathcal{M}$ and a normal world x , such that $x \not\models_{\mathcal{M}} A^\tau$. Then, in the associated **S4** model $\mathcal{M}'$ we have, by (1) above, $x \not\models_{\mathcal{M}'} A$. Hence $\not\models_{\mathbf{S4}} A^\tau$.

[$\Leftarrow$]. It is straightforward to verify inductively that, for every $A \in \mathcal{L}^\Box$,

$$\vdash_{\mathbf{S4}} A \leftrightarrow A^\tau.$$

The conclusion then follows by the inclusion $\mathbf{S3} \subseteq \mathbf{S4}$.

We conclude by observing that there is also a very simple *non-compositional* embedding of **S4** into **S3**.

Theorem 11 *For every $A \in \mathcal{L}^\Box$:*

$$\vdash_{\mathbf{S4}} A \quad \Leftrightarrow \quad \vdash_{\mathbf{S3}} \Box \Box \top \rightarrow A$$

Proof The right-to-left direction is trivial. Conversely, suppose $\not\models_{\mathbf{S3}} \Box \Box \top \rightarrow A$. Then, by the completeness theorem for **S3**, there exists a type-2 model $\mathcal{M} = \langle W, N, R, v \rangle$ and a world $x \in N$, such that $x \not\models_{\mathcal{M}} \Box \Box \top \rightarrow A$, and so $x \Vdash_{\mathcal{M}} \Box \Box \top$ and $x \not\models_{\mathcal{M}} A$. Now, the accessibility relation R of $\mathcal{M}$ is reflexive and transitive, and we can assume, without limitations, that xRy for all worlds $y \in W$. This, by $x \Vdash_{\mathcal{M}} \Box \Box \top$, implies that $y \Vdash_{\mathcal{M}} \Box \top$ for all worlds y of $\mathcal{M}$, or, otherwise said, that $W = N$. Thus $\mathcal{M}$ is a **S4** countermodel for A , and $\not\models_{\mathbf{S4}} A$.

5.8 Lewis's System S2

The “System 2”, **S2**, of strict implication—which we present here for future reference in the next Sect. 5.9—was introduced by Lewis in Chapter VI of *Symbolic Logic* (Lewis and Langford 1932), together with the weaker system **S1**. Among the two systems, and indeed among all the systems of strict implication, **S2** was Lewis's preferred system, by his own acknowledgment.²⁰

²⁰ “Although System 1 is more economical in its assumptions, and although there is no usable principle of inference in System 2 which is not also contained (in more complex form perhaps) in System 1, nevertheless we regard System 2 as superior on account of its slightly greater simplicity.

The Lewis-style axiomatization of **S2** is obtained from that of **S3** (see Sect. 5.1 above) by modifying the postulates as follows (all the rest remains unchanged):²¹

- (P2) $q \times p \rightarrow p$ is replaced by:
 (P2') $p \times q \rightarrow p$
- (P4) $(p \times (q \times r)) \rightarrow (q \times (p \times r))$ is replaced by:
 (P4') $((p \times q) \times r) \rightarrow (p \times (q \times r))$
- (P7) $\sim p \rightarrow \sim p$ is replaced by:
 (P7') $p \times (p \rightarrow q) \rightarrow q$
- (P8) $(p \rightarrow q) \rightarrow (\sim q \rightarrow \sim p)$ is replaced by the *consistency postulate*:
 (P8') $\sim \sim (p \times q) \rightarrow \sim \sim p$

The Lemmon-style axiomatization of **S2** is obtained from the corresponding one of **S3** (see Sect. 5.2 above) as follows:

- the axiom schema (A.3) $\Box(A \rightarrow B) \rightarrow \Box(\Box A \rightarrow \Box B)$ is replaced by:
 (A.3') $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ (schema *K*)
- A third inference rule is added:

$$\frac{\Box(A \rightarrow B)}{\Box(\Box A \rightarrow \Box B)} \text{ RK}^+$$

We know from Theorem 1 that **S3** is the “non normal companion” of the normal system **S4**, in the precise sense that **S4** is equivalent to **S3** + $\Box\Box\top$. Analogously, **S2** is the “non normal companion” of the normal system **T** (see Sect. 2.3):

Proposition 3 *T is equivalent to S2 + $\Box\Box\top$.*

This is easily proved after having checked that items (2), (3) and (4) of Lemma 1, which we stated for **S3**, do hold already for the weaker system **S2** (while item (1) does not hold in **S2**, see below).

Like its non normal companion **T**, and unlike **S3**, the system **S2** has an *infinite* number of irreducible modalities. This was first proved by J.C.C. McKinsey (McKinsey 1940), about 25 years before the invention of the possible-worlds semantics—given the characterization of **S2** by means of such semantics, the result is of course more or less straightforward.

A sound and complete Kripke-style semantic characterization for **S2** (originally devised in Kripke 1965) can be easily described as follows, given the semantics for **S3** presented in Sect. 5.4 above: just modify the notion of *type-1* or *type-2* **S3**-model, by dropping the requirement of *transitivity* on the accessibility relation *R*. **S2** is easily proved to be sound with respect to such modified models, while

It is this System 2 which we desire to indicate as the System of Strict Implication.” Lewis and Langford (1932), 177.

²¹ As mentioned in footnote 5, the primitive symbols adopted in Lewis and Langford (1932) are different from those in Lewis (1918). For ease of comparison with the axiomatization of the *Survey* system **S3**, we keep using the primitive symbols of the former work.

the corresponding completeness theorem can be demonstrated along the lines of the proof of the completeness theorem for **S3** given in Sect. 5.5. Indeed, a careful inspection of the latter shows that there is only one **S3** tautology which is used in an essential way and which is not a **S2** tautology, namely

$$\Box A \rightarrow \Box(\Box \top \rightarrow \Box A)^{22}$$

however, this tautology is needed *only* in order to show that the accessibility relation in the canonical model is transitive.

5.9 Becker's “Experiments”: A Technical Evaluation

5.9.1 The “Variant” Six Modalities System **S3***

Recall that **S3*** is obtained from **S3** through the addition of the axiom schema

$$(\Box E^*) \quad \Box(\Box \Diamond A \rightarrow \Box A)$$

As we anticipated in Chap. 2, W. Parry (Parry 1939) proved that **S3*** collapses. Let us see a detailed proof thereof.

First of all, we have:

$$\vdash_{\mathbf{S3}} \Box \Diamond(A \rightarrow \Box A) \quad (5.7)$$

We can benefit of the completeness theorem for **S3** to give a quick semantical proof of (5.7). Suppose indeed, in view of a contradiction, that

$$\not\models_{\mathbf{S3}} \Box \Diamond(A \rightarrow \Box A)$$

Then by the completeness theorem(s) proved in Sect. 5.5, there exist a **S3**-model (say, a type-2 model) $\mathcal{M} = \langle W, N, R, v \rangle$ and a *normal* world $x \in W$ such that

$$x \not\models_{\mathcal{M}} \Box \Diamond(A \rightarrow \Box A)$$

Therefore, since $x \in N$, there is a world $y \in W$ such that

$$x R y \quad \text{and} \quad y \not\models_{\mathcal{M}} \Diamond(A \rightarrow \Box A) \quad (5.8)$$

The world y is normal, since non normal worlds force all formulas of the form $\Diamond B$, and

$$z \not\models_{\mathcal{M}} A \rightarrow \Box A \quad \text{for all } z \in W \text{ such that } y R z \quad (5.9)$$

²² See Lemma 1, item (1). A type-2 **S2**-countermodel for $\Box p \rightarrow \Box(\Box \top \rightarrow \Box p)$ is obtained by taking $W = \{x, y, z\}$, $N = \{x, y\}$, $R = \{\langle x, x \rangle, \langle y, y \rangle, \langle x, y \rangle, \langle y, z \rangle\}$, and $v(x, p) = v(y, p) = 1$, $v(z, p) = 0$.

Hence, by the reflexivity of R ,

$$y \not\models_{\mathcal{M}} A \rightarrow \Box A, \quad \text{that is (i) } y \models_{\mathcal{M}} A \quad \text{and (ii) } y \not\models_{\mathcal{M}} \Box A \quad (5.10)$$

Since $y \in N$, by (ii) of (5.10) there is a world u such that

$$yRu \quad \text{and} \quad u \not\models_{\mathcal{M}} A \quad (5.11)$$

But then $u \models_{\mathcal{M}} A \rightarrow \Box A$, and we are in contradiction with (5.9).

It now easily follows that $\mathbf{S3}^*$ collapses, that is

$$(\text{COLL}) \quad \vdash_{\mathbf{S3}^*} \Box(A \leftrightarrow \Box A)$$

Indeed, $\vdash_{\mathbf{S3}^*} \Box(A \rightarrow \Box A)$:

1. $\Box(\Box\Diamond(A \rightarrow \Box A) \rightarrow \Box(A \rightarrow \Box A))$ instance of $(\Box E^*)$
2. $\Box\Diamond(A \rightarrow \Box A) \rightarrow \Box(A \rightarrow \Box A)$ MP: A2, 1
3. $\Box(A \rightarrow \Box A)$ MP: 2, (5.7)

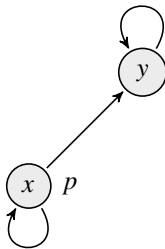
which, together with $\vdash_{\mathbf{S3}} \Box(\Box A \rightarrow A)$ (by A2 and RN^-) yields (COLL).

5.9.2 The Ten Modalities System $\mathbf{S3}''$

Recall that $\mathbf{S3}''$ is obtained from $\mathbf{S3}$ through the addition of the axiom schemas

$$(\Box B) \quad \Box(A \rightarrow \Diamond\Box A) \quad \text{and} \quad (\Box 4) \quad \Box(\Box A \rightarrow \Box\Box A)$$

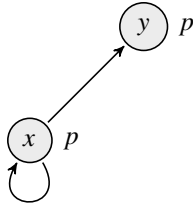
Becker is of course assuming that the schemas $(\Box B)$ and $(\Box 4)$ are not provable in $\mathbf{S3}$ —and he is right, indeed. Actually, as is well known to us, the normal modal system $\mathbf{S4}$ does not prove (B) and its boxed version $(\Box B)$ —just consider the *normal*, reflexive and transitive Kripke model



$$\text{where } W = N = \{x, y\}, v(x, p) = 1, v(y, p) = 0$$

in which, as it is easily verified, $x \not\models_{\mathcal{M}} (\Box)(p \rightarrow \Box\Diamond p)$. Thus $\mathbf{S3}$, which is a subsystem of $\mathbf{S4}$, does not prove (B) and $(\Box B)$.

On the other side, here is a countermodel $\mathcal{M} \in \mathbf{Mod}_1$ showing (thanks to the Soundness Theorem for **S3**) that $\not\models_{\mathbf{S3}} \Box(\Box p \rightarrow \Box\Box p)$.



where $N = \{x\}$, $v(x, p) = v(y, p) = 1$.

Indeed, $x \Vdash_{\mathcal{M}} \Box p$ since $x \in N$, $x \Vdash_{\mathcal{M}} p$, $y \Vdash_{\mathcal{M}} p$, but $x \not\Vdash_{\mathcal{M}} \Box\Box p$ since xRy and, by $y \in Q$, $y \not\Vdash_{\mathcal{M}} \Box p$.

Becker is also pretty sure, see Chap. 4, fn. 75, that $(\Box B)$ and $(\Box 4)$ are *independent* on the basis of Lewis's system **S3**. Here, unfortunately, he is wrong. More exactly, it turns out that:

Lemma 11

1. $(\Box B)$ is not a theorem of **S3** + $(\Box 4)$
2. $\vdash_{\mathbf{S2}+(\Box B)} \Box\Box\top$
3. $(\Box 4)$ is a theorem of **S3** + $(\Box B)$

Proof As to (1.), the claim immediately follows from **S3** + $(\Box 4) \equiv \mathbf{S4}$, see Theorem 1, and from the above observations about the unprovability of (B) and $(\Box B)$ in **S4**.

As to (2.), which concerns the weaker system **S2** (see Sect. 5.8) and is included here for future use, the following is a proof of $\Box\Box\top$ in the system **S2** extended with the additional axiom schema $(\Box B)$:

- | | | |
|-----|---|------------------------|
| 1. | $\Box(\top \rightarrow \Box\Diamond\top)$ | instance of $(\Box B)$ |
| 2. | $\Box(\Box\top \rightarrow \Box\Box\Diamond\top)$ | RK^+ , 1 |
| 3. | $\Box\top \rightarrow \Box\Box\Diamond\top$ | MP: A2, 2 |
| 4. | $\Box\top$ | RN^- : A1 |
| 5. | $\Box\Box\Diamond\top$ | MP: 3, 4 |
| 6. | $\Diamond\top \rightarrow \top$ | A1 |
| 7. | $\Box(\Diamond\top \rightarrow \top)$ | RN^- : 6 |
| 8. | $\Box(\Box\Diamond\top \rightarrow \Box\top)$ | RK^+ , 7 |
| 9. | $\Box(\Box\Box\Diamond\top \rightarrow \Box\Box\top)$ | RK^+ , 8 |
| 10. | $\Box\Box\Diamond\top \rightarrow \Box\Box\top$ | MP: A2, 9 |
| 11. | $\Box\Box\top$ | MP: 5, 10 |

Finally, (3.) follows from (2.), $\mathbf{S2} \subseteq \mathbf{S3}$ and $\mathbf{S3} + (\Box\Box\top) \equiv \mathbf{S4}$, which holds by Theorem 1.

5.9.3 The System SM

A preliminary clarification about the "definition" of Becker's system **SM** is necessary. Indeed, while the other systems introduced by Becker (**S3'**, **S3'**^{*}, **S3''**) are *explicitly* formulated by him as axiomatic extensions of **S3**, the same cannot be said of **SM**.

What clearly and explicitly results from Becker's text is that:

1. a *modality* ($\Lambda, \Pi, \dots$) is a finite, possibly empty string of $\Box$'s and $\neg$'s (" $\Diamond$ " being also used as an abbreviation of " $\neg\Box\neg$ "); a modality containing an even (odd) number of negations is said to be *positive* (*negative*);²³
2. given any modal system **S** and modalities Λ, Π , by writing

$$\Lambda \rightarrow \Pi \quad (\text{or, pedantically, } \Lambda \rightarrow_{\mathbf{S}} \Pi)$$

it is meant that

$$\vdash_{\mathbf{S}} \Box(\Lambda A \rightarrow \Pi A) \quad \text{for all formulas } A$$

(equivalently: $\vdash_{\mathbf{S}} \Box(\Lambda p \rightarrow \Pi p)$ for an atom p);²⁴

and that **SM** should obey the following conditions:

- (SM.1) **SM** contains all classical tautologies and is closed under the (meta)rule of transitivity for the relation $\rightarrow_{\mathbf{SM}}$;²⁵
- (SM.2) for any modalities Λ and Λ' , if $\Lambda \rightarrow_{\mathbf{SM}} \Lambda'$ then also²⁶

²³ Cp. Becker (1930), 25–6 (85 f).

²⁴ With *exactly this meaning*, the "arrow" notation between modalities is systematically employed in Becker (1930) starting from p. 17 (74).

²⁵ The second requirement is already implicit in Becker's request that the system **SM** should allow a *linear ordering* of positive (resp. negative) modalities, thus in particular that $\Lambda \rightarrow \Sigma$ follows from $\Lambda \rightarrow \Pi$ and $\Pi \rightarrow \Sigma$, cp. Becker (1930), 28–9 (89). The transitivity of the arrow relation $\rightarrow$ is also explicitly stated by Becker, cp. *Ibid.*, 28 (89). A sufficient condition for this is e.g. that **SM** be closed under the transitivity rule for strict implication, that is: if $\vdash_{\mathbf{SM}} \Box(A \rightarrow B)$ and $\vdash_{\mathbf{SM}} \Box(B \rightarrow C)$ then $\vdash_{\mathbf{SM}} \Box(A \rightarrow C)$.

²⁶ These three closure conditions correspond to Becker's *Rule II* and *Rule III* in Becker (1930), 26 (86 f). More precisely, (R.1) corresponds to Rule II, while (R.2) and (R.3) correspond to the two parts (concerning positive, resp. negative modalities) of Rule III. We do not consider Becker's *Rule I* (the juxtaposition of modalities is associative and not commutative) since it is implicit in the formalism we are using.

- $\Lambda\Pi \rightarrow_{\mathbf{SM}} \Lambda'\Pi$ for any modality Π (R.1)²⁷
- $\Pi\Lambda \rightarrow_{\mathbf{SM}} \Pi\Lambda'$ for any positive modality Π (R.2)
- $\Pi\Lambda' \rightarrow_{\mathbf{SM}} \Pi\Lambda$ for any negative modality Π (R.3)

Concerning these *closure conditions*, it is very important to stress (as already anticipated in Chap. 2) that the inference rules

$$\frac{\Box(A \rightarrow B)}{\Box(\Box A \rightarrow \Box B)} BR_{\Box} \qquad \frac{\Box(A \rightarrow B)}{\Box(\Diamond A \rightarrow \Diamond B)} BR_{\Diamond}$$

known since Churchman (1938) as Becker’s rules (the primitive inference rule RK^+ of **S2** is exactly $BR_{\Box}$, see Sect. 5.8), *are not special cases of (R2) above*, as erroneously assumed by Churchman. More precisely: a modal system **S** which is closed under $(BR_{\Box})$ and $(BR_{\Diamond})$ is indeed closed under (R.1)–(R.3), as it is immediate to verify; but from the fact that a modal system **S** satisfies Becker’s explicit conditions on **SM**, here (SM.1)–(SM.4), it is not possible to infer, in general, that **S** is closed under $(BR_{\Box})$ and $(BR_{\Diamond})$. See Centrone and Minari (2022) for a detailed analysis and discussion.

- (SM.3) (i) $\Box \rightarrow_{\mathbf{SM}} W$ and (ii) $W \rightarrow_{\mathbf{SM}} \Diamond$
 (where by “ W ” we denote *here* the empty modality, “it is true, that”);²⁸
 (SM.4) **SM** contains the “generalized Brouwer’s axioms” B_n , for all $n \geq 1$.²⁹

$$\vdash_{\mathbf{SM}} \Box(A \rightarrow \overbrace{\Box \dots \Box}^n \Diamond A)$$

As *cleverly and correctly* demonstrated by Becker (see our explicatory footnotes to Becker 1930, 29–30 (90–91)), the conditions (SM.1)–(SM.4), in particular the schemas (B_n) , allow to *linearly order* all the positive (and, separately, all the negative) modalities. The problem is, however, whether all these infinite modalities are *irreducible*, as Becker seems to claim. And this leads to a second problem: the notion of irreducibility is of course dependent on the modal system under consideration, but *the conditions* (SM.1)–(SM.4) *do not allow to single out one and only one modal system*.

In particular, both the systems **S2** and **S3** satisfy (SM.1)–(SM.3); actually:

Lemma 12 *Any axiomatic extension of **S2** satisfies (SM.1)–(SM.3).*

²⁷ This rule is, in a sense, trivial: for any **S**, if $\vdash_{\mathbf{S}} \Box(\Lambda A \rightarrow \Lambda' A)$ for all formulas A then, in particular, this holds for all formulas A of the form ΠB , where B is arbitrary; therefore $\Lambda\Pi \rightarrow_{\mathbf{S}} \Lambda'\Pi$.

²⁸ “So, initially $N \rightarrow M$ holds, as is well known; [...] One can get one step further if one takes into account that $N \rightarrow W \rightarrow M$.” Becker (1930), 28 (88).

²⁹ Becker (1930), 28–9 (89).

Proof Let **S** be an axiomatic extension of **S2**. Then, **S** satisfies (SM.1) and (SM.3) because $\vdash_{\mathbf{S2}} \Box(A \rightarrow B) \wedge \Box(B \rightarrow C) \rightarrow \Box(A \rightarrow C)$ (a verification using the Kripke semantics for **S2** is immediate) and $\vdash_{\mathbf{S2}} \Box(\Box A \rightarrow A)$ (this is the axiom schema (A.2) of **S2**, see Sect. 5.8). As to (SM.2), the closure conditions (R.1)–(R.3) hold, as we said above, in any system closed under *Becker's rules* and so in any axiomatic extension of **S2**, because the latter contains the primitive inference rule RK^+ .

Thus, the axiomatic extensions of **S2** and **S3** with the generalized Brouwer's axioms, that is

- **S2sm** := **S2** + $\{(B_n) \mid n \geq 1\}$
- **S3sm** := **S3** + $\{(B_n) \mid n \geq 1\}$

are “**SM**-systems”. *Both these systems, however, boil down to S5*, although not for exactly the same reasons.

As for **S3sm**, already the addition of the schema (B_1) to **S3** gives **S5**. Indeed, (B_1) coincides with the schema $(\Box B)$, so it is an immediate consequence of Theorem 1 together with (3.) of Lemma 11 that **S3** + (B_1) is equivalent to **S4** + $(A \rightarrow \Box \Diamond A)$ which, as is well known, is nothing but **S5**.

As for **S2sm**, on the contrary, the addition to **S2** of the sole schema (B_1) is not sufficient to get **S5**. Indeed, by Proposition 3 together with (2.) of Lemma 11, we have that

$$\mathbf{S2} + (B_1) \text{ is equivalent to } \mathbf{T} + (A \rightarrow \Box \Diamond A) \quad (5.12)$$

where the latter is the normal proper subsystem of **S5** known in the literature as system **B** (see Sect. 2.3), that is characterized in Kripke's semantics by the class of all reflexive and symmetric frames.

However, once *both the schemas* (B_1) *and* (B_2) *are added to S2* (**S2sm** contains all the (B_n) 's!) we get by (5.12) the system **B** + (B_2) , but in turn the latter proves $\Box A \rightarrow \Box \Box A$ (schema 4) and so boils down to **S5**:

- | | | |
|----|--|--|
| 1. | $\Box(\Box A \rightarrow \Box \Box \Diamond \Box A)$ | instance of (B_2) |
| 2. | $\Box A \rightarrow \Box \Box \Diamond \Box A$ | MP: A2, 1 |
| 3. | $\Diamond \Box A \rightarrow A$ | theorem of B $\equiv$ S2 + (B_1) |
| 4. | $\Box \Box \Diamond \Box A \rightarrow \Box \Box A$ | RN and $K : 3$ |
| 5. | $\Box \Box A \rightarrow A$ | transitivity: 2, 4 |

Summing up: it is not at all clear whether Becker intended his experimental system **SM** as being “based” on Lewis's system **S3**, which is the starting point of his investigations, or maybe on some weaker system comparable in force with **S2** (remind that at the time he was writing the latter system had not yet been introduced). If this is the case, then also this ingenious experiment of him actually

ends up in a further axiomatization of **S5**. If this is not the case (as in our opinion is not), it would be interesting to find (if any) a *non normal* modal system **S** complying the conditions (SM.1)–(SM.4) and having an infinite number of irreducible positive modalities linearly ordered as Becker expected.

We conclude with a comment on a variant of the schemas (B_n) alluded to by Becker in a footnote (Becker (1930), 29 (89), fn. 109): “The choice of the implicative relation $MN \rightarrow W \rightarrow NM$ [that is, schema $(\Box B) = (B_1)$ and its dual form] in place of $NM \rightarrow W \rightarrow MN$ [...] would bring along drastic modifications of the ranking relations in the whole schema. Altogether, it can be questioned whether a linear rank order of *all* modalities is still possible.”

The question Becker is asking is: what happens if the schemas (B_n) , which generalize Brouwer’s axiom $A \rightarrow \Box \Diamond A$, are replaced by the analogous generalizations (B'_n) of $(B'_1) = \Box \Diamond A \rightarrow A$, that is $\Box(\Box \Diamond A \rightarrow A)$? Here is the answer: *the system collapses*; more precisely, already **S2** + (B'_1) proves the collapse formula $\Box(\Box A \leftrightarrow A)$. Indeed, in Sect. 5.9.1 we have shown that $\vdash_{\mathbf{S3}} \Box \Diamond(A \rightarrow \Box A)$, but the semantic proof we have given there does not make use of the transitivity of the accessibility relation, and so (given the characterization of **S2** in Kripke’s semantics, see Sect. 5.8), it holds:

$$\vdash_{\mathbf{S2}} \Box \Diamond(A \rightarrow \Box A) \quad (5.13)$$

It follows that

$$\vdash_{\mathbf{S2}+(B'_1)} A \rightarrow \Box A \quad (5.14)$$

- | | | |
|----|--|----------------------|
| 1. | $\Box(\Box \Diamond(A \rightarrow \Box A) \rightarrow (A \rightarrow \Box A))$ | instance of (B'_1) |
| 2. | $\Box \Diamond(A \rightarrow \Box A) \rightarrow (A \rightarrow \Box A)$ | MP: A2, 1 |
| 3. | $\Box \Diamond(A \rightarrow \Box A)$ | (5.13) |
| 4. | $A \rightarrow \Box A$ | MP: 2, 3 |

By instantiating A with $\Box \top$ in (5.14) and using $\vdash_{\mathbf{S2}} \Box \top$, we get:

$$\vdash_{\mathbf{S2}+(B'_1)} \Box \Box \top \quad (5.15)$$

It follows by (5.15) that **S2** + (B'_1) is closed under the necessitation rule. Therefore, by necessitation of (5.14), $\vdash_{\mathbf{S2}+(B'_1)} \Box(A \rightarrow \Box A)$ which, together with the schema (A.2) of **S2**, gives the collapse: $\vdash_{\mathbf{S2}+(B'_1)} \Box(A \leftrightarrow \Box A)$.

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